

Michaels: Lifetime Census and Closing Consequences

Complete record, compressed interactions, exact prime residual, and fictional closure audit

Christophe Michaels research programme

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Reading guide to this consolidated edition

This edition preserves the earlier census and both prior continuation rounds in sections 1-51. Sections 52-67 document the compression, positive allowance, exact residual, and fictional closing experiment. Sections 68-73 now carry out the direct joint transfer, with exact prime jumps, a finite expansion, and an unconditional allowance for a complete insertion range. Every step is labeled as an identity, proved estimate, conditional deduction, hypothetical law, or finite diagnostic.

For the current calculation, read sections 68-73. The small-insertion allowance is proved in section 71, the even-coefficient and prime-square credits in section 72, and the precise surviving signed comparison in section 73. Sections 59-60 define the original whole residual. The compression proof and historical means are in sections 53-54; the fixed-cutoff allowance is in sections 56-57. Section 61's growth law is explicitly fictional and is not assumed in sections 68-73. Section 62 supplies the conditional analytic closure. Sections 67 and 73 map the two latest continuations to their proofs and status.

For the earlier reductions, product-range and common-factor estimates are in sections 37-38, the complete squarefree transfer in sections 45-46, and the averaged alternatives in sections 40 and 47-48. The first version of the analytic closing argument remains in section 26. Later estimates supersede earlier cutoffs only where explicitly stated. Their hypotheses and inherited terms are retained.

The fictional law in section 61 has not been proved for the primes. Its conditional consequence is proved. The exact residual remains the original one-sided arithmetic comparison, and the hypothetical square budget is only one sufficient mechanism for it.

This is the lifetime-census companion. It is separate from the 12-page summary, the continuing proof manuscript, and the large research master. The source package preserves the previous edition and all material used to rebuild this one.

1. What would close, and the object being estimated

An all-scale subpower bound for the authentic remaining lifetime energy completes the selected RH deduction. The nested multiplicative identities are a proposed arithmetic mechanism for that estimate. This note proves a finite birth-scale reformulation and an equivalent counting target; it does not presume the counting target. No new independent condition is added after the existing historical-energy premise.

Let μ be the ordinary Möbius function, $M(x) = \sum_{n \leq x} \mu(n)$, and

$$I_M(N) = \int_1^N \frac{M(t)^2}{t^2} dt.$$

Each upcrossing of $|M|$ to an occupied integer depth h_u at an integer u creates a level. Pair it with the first subsequent downcrossing from h_u to $h_u - 1$, at v_u ; use $v_u = \infty$ if this never occurs. Put

$$w_u = 2h_u - 1, \quad d_u(N) = (\min(v_u, N, h_u^2) - u)_+.$$

Thus $d_u(N)$ is only the observed portion before the level's own height-square time. The already established reduction is

$$\mathcal{E}_*(N) = \sum_{w_u d_u(N) > u} \frac{w_u d_u(N)}{u[u + d_u(N)]}, \quad \boxed{\mathcal{E}_*(N) \leq I_M(N) \leq 2 \log N + \mathcal{E}_*(N).} \quad (1)$$

The sum is over actual level births. Its weights already include opening/closing cancellation. Large and unfinished lifetimes remain included.

One term has the exact shape

$$\boxed{c(u, h, d) = \frac{2h - 1}{u} \frac{d/u}{1 + d/u}.} \quad (2)$$

Feature	Mathematical role
Birth u	Later births reduce the cost through the reciprocal weight
Depth h	The odd weight $2h - 1$ accounts for the increment in a square
Early duration d	Its cost fraction $d/(u + d)$ rises toward one
Number of such lifetimes	Determines the aggregate cost; a bound on one term is not a bound on the sum

A remaining term must satisfy $h^2 > u$, $d > u/(2h - 1)$, and $d \leq h^2 - u$. Its full original lifetime may be arbitrarily long. The later portion is already paid by the logarithmic term in (1).

2. The geometric shape

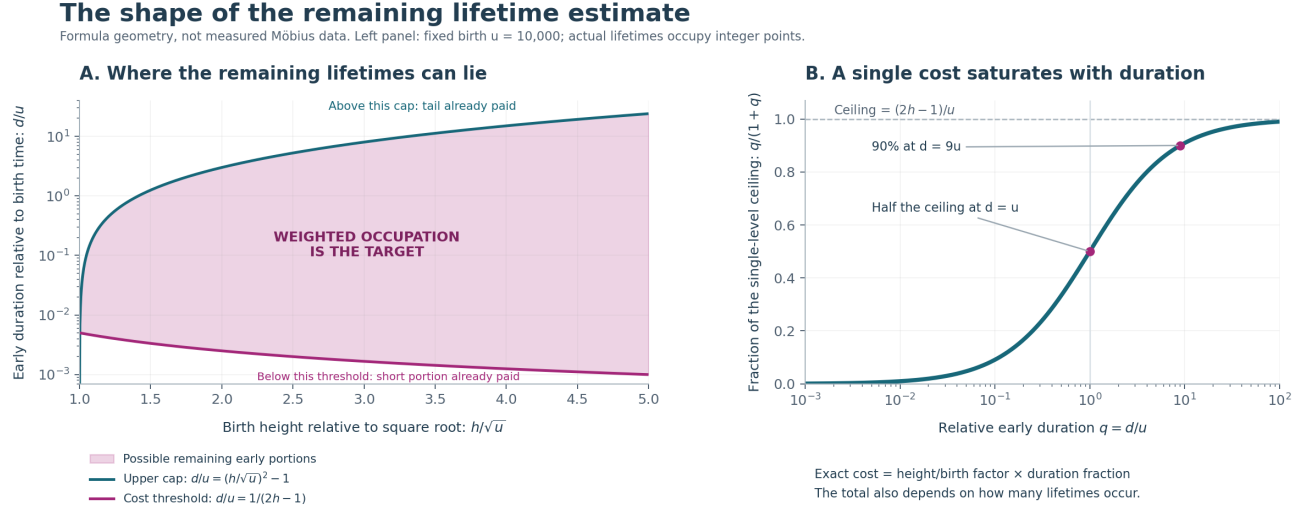


Figure 1: The exact eligibility region and duration response. No measured Möbius data are plotted.

Fixing a birth time u gives a two-dimensional slice through the three coordinates (u, h, d) . In panel A, write $r = h/\sqrt{u}$ and $q = d/u$. Remaining points can occupy only

$$\frac{1}{2r\sqrt{u} - 1} < q \leq r^2 - 1. \quad (3)$$

The shaded region is an eligibility region, not a claim that actual lifetimes fill it. Actual arithmetic produces discrete points, each with cost (2). The desired theorem controls their weighted occupation over all birth scales.

Panel B shows a useful distinction: duration does not create an indefinitely linear cost. For a fixed birth and depth,

$$0 \leq c(u, h, d) < \frac{2h-1}{u}, \quad \frac{\partial c}{\partial d} = \frac{2h-1}{(u+d)^2}. \quad (4)$$

The marginal cost decays. This does not by itself control how many expensive births occur. It tells us exactly how their frequency and duration must enter a collective estimate.

The multiplicative structure acts before this census is formed. For a finite set S of odd primes, $P = \prod_{p \in S} p$, define

$$A_S(x) = \sum_{(m, 2P)=1} \mu(m) \mathbf{1}_{[m, 2m)}(x), \quad D_p f(x) = f(x/p).$$

The saved exact identity is

$$M = \prod_{p \in S} (I - D_p) A_S. \quad (5)$$

It connects every color to the same signed count. The proposed use of these dependencies is to restrict the authentic points that can occupy the shaded region, with the weights in (2) retained. No favorable sign for each separate interaction is assumed.

3. A finite birth-scale census: proved reformulation

Define the settled early duration

$$d_u^\infty = (\min(v_u, h_u^2) - u)_+,$$

and for dyadic $X = 2^k \geq 1$ let

$$K(X) = \sum_{\substack{X \leq u < 2X \\ w_u d_u^\infty > u}} \frac{w_u d_u^\infty}{u(u + d_u^\infty)}. \quad (6)$$

Finite horizon. All the terms of $K(X)$ are determined by the Möbius walk through $4X^2$. Indeed, $h_u \leq u < 2X$, so $h_u^2 < 4X^2$. Thus $d_u(4X^2) = d_u^\infty$, whether or not the original lifetime ever closes.

Each summand, with its threshold indicator, is nondecreasing as its observed duration grows. Consequently

$$K(X) \leq \mathcal{E}_*(4X^2), \quad \mathcal{E}_*(N) \leq \sum_{\substack{X \leq N \\ X \text{ dyadic}}} K(X). \quad (7)$$

It follows that $\mathcal{E}_*(N) = O_\varepsilon(N^\varepsilon)$ for every $\varepsilon > 0$ is equivalent to $K(X) = O_\varepsilon(X^\varepsilon)$ for every $\varepsilon > 0$. For the forward direction use exponent $\varepsilon/2$ at $4X^2$. For the reverse direction sum the geometric sequence X^ε over dyadic $X \leq N$.

Now let $C(X, H, D)$ count the births in (6) with

$$H \leq h_u < 2H, \quad D \leq d_u^\infty < 2D,$$

where H, D are powers of two. If $K(X; H, D)$ denotes their total cost, then

$$\frac{1}{4} \frac{HD}{X(X+D)} C(X, H, D) \leq K(X; H, D) \leq 8 \frac{HD}{X(X+D)} C(X, H, D). \quad (8)$$

To check the constants, use $H \leq 2h_u - 1 < 4H$, $D \leq d_u^\infty < 2D$, and $X \leq u < 2X$ in the numerator and denominator.

There are $O((1 + \log X)^2)$ height/duration bins: $h_u < 2X$ and $d_u^\infty < 4X^2$. Therefore the desired subpower estimate is equivalent to the following counting statement:

$$C(X, H, D) \leq A_\varepsilon X^\varepsilon \frac{X(X+D)}{HD} \quad \text{for every } \varepsilon > 0. \quad (9)$$

One constant must work for all bins and all dyadic X . Necessity follows from positivity and (8). For sufficiency use (9) with $\varepsilon/2$, sum the bins, and absorb the two logarithms. Equation (9) is the precise arithmetic target supplied by this reformulation, not an additional estimate assumed to have been proved.

4. What degree is needed, and why it reaches the critical line

For a representative range $H \asymp D \asymp X^{1/2+\delta}$, $0 < \delta < 1/2$, the cost of one term is of order $X^{-1+2\delta}$. Equation (9) requires

$$C(X, H, D) \ll_{\varepsilon} X^{1-2\delta+\varepsilon}. \quad (10)$$

For example, heights and durations near $X^{0.6}$ require a count at most of order $X^{0.8+\varepsilon}$. This is a statement about a scale range, not a measured count. It allows numerous expensive lifetimes while prescribing the needed rarity. If $D \geq X$, the allowance in (9) is instead of order $X^{1+\varepsilon}/H$: the duration cost has saturated.

The needed growth condition is equivalently

$$\boxed{\frac{\log(1 + \mathcal{E}_{\star}(N))}{\log N} \rightarrow 0.} \quad (11)$$

It permits unbounded, irregular growth. A single bound by $N^{0.01}$ supplies only a fixed exponent; (11) asks for every positive exponent, with a different finite constant allowed for each.

Here is the complete analytic connection. Fix $\sigma > 1/2$ and choose $0 < \eta < 2\sigma - 1$. Subpower energy and Cauchy-Schwarz give

$$\int_X^{2X} |M(x)| x^{-\sigma-1} dx \leq I_M(2X)^{1/2} \left(\int_X^{2X} x^{-2\sigma} dx \right)^{1/2} \ll_{\sigma, \eta} X^{(1+\eta)/2-\sigma}. \quad (12)$$

Summing dyadic intervals converges, locally uniformly with spare decay in every compact subset of $\Re s > 1/2$. Thus

$$F(s) = s \int_1^{\infty} M(x) x^{-s-1} dx$$

is holomorphic there. For $\Re s > 1$, partial summation and the Euler product give $F = 1/\zeta$. The holomorphic identity $(s-1)\zeta(s)F(s) = s-1$ extends throughout $\Re s > 1/2$. At any zero ρ there, its left side would be zero and its right side nonzero. Reflection under the functional equation then puts every nontrivial zero on $\Re s = 1/2$.

More quantitatively, $I_M(N) = O(N^{\delta})$ with $0 < \delta < 1$ yields the same argument for $\Re s > (1+\delta)/2$, hence confines nontrivial zeros to

$$|\Re \rho - \tfrac{1}{2}| \leq \tfrac{\delta}{2}. \quad (13)$$

Every positive exponent collapses these strips to the critical line. The implication uses no independent terminal, prime-transfer, or spectral-tail estimate.

5. What changes for primes, zeros, and the next arithmetic step

The same energy has a particularly simple shape in logarithmic coordinates. Put $Y = \log N$ and $U(y) = e^{-y/2} M(e^y)$. Direct substitution gives

$$I_M(e^Y) = \int_0^Y |U(y)|^2 dy, \quad F(s) = s \int_0^\infty U(y) e^{-(s-1/2)y} dy. \quad (14)$$

The target is subexponential accumulated squared amplitude in logarithmic time. Then every positive damping rate $\Re s - 1/2$ controls the transform. This is another coordinate description of the same estimate.

Once the subpower estimate is proved, standard RH consequences apply:

$$\psi(x) = x + O(\sqrt{x} \log^2 x), \quad \pi(x) = \text{Li}(x) + O(\sqrt{x} \log x). \quad (15)$$

Here $\psi(x) = \sum_{p^k \leq x} \log p$ counts prime powers with logarithmic weights, and $\pi(x)$ counts primes. These give quantitative limits on departures from their smooth approximations. They do not prescribe an identical local spacing.

For a zero $\rho = \beta + i\gamma$, an individual term in the explicit formula has shape

$$\frac{x^\rho}{\rho} = \frac{x^\beta}{\rho} e^{i\gamma \log x}. \quad (16)$$

The real part β sets the amplitude exponent; γ sets an oscillation frequency in $\log x$. The proved estimate would force $\beta = 1/2$ at every height. Individual zero ordinates and individual prime locations still require their own evaluation. Equation (15) controls the collective fluctuation.

Next arithmetic step. Use the exact multiplicative identities to prove (9), or a directly summed bound for (6). The target is frequency weighted by depth and saturated duration. Identities for separate colors must be recombined before their costs are counted. The existing cancellation is already built into the authentic lifetimes and must not be credited twice.

What this note establishes. Equations (6)-(10) give a proved finite-horizon reformulation of the remaining estimate. Equations (11)-(14) give its exact growth degree and analytic consequences. The all-scale counting bound (9) itself remains the estimate to establish. This distinction locates the arithmetic work while keeping the closing deduction fixed.

Sources and placement. The established lifetime reduction and nested identities are in the active running proof, sections 23-25 of its prime-carry continuation and its final nested-lifetime chapter; see the reorganized master, pages 221-225, and source NC, pages 265-267. This is a companion research note; the master has not been reissued in this turn.

External checks of standard consequences:

1. E. Bombieri, Problems of the Millennium: the Riemann Hypothesis, Clay Mathematics Institute, pp. 1 and 4. [Official description](#).
2. NIST DLME, [section 25.16](#), explicit formula and prime-counting equivalence; [section 25.10](#), zero symmetry.
3. E. S. Lee and P. Nosal, Sharper bounds for the error in the prime number theorem assuming the Riemann Hypothesis, [arXiv:2312.05628](#). Its stronger RH-conditional estimate implies the first bound in (15).

6. Direct attempt from the complete multiplicative dependencies

The requested next step is to derive (9), not assume it. This continuation performs the finite reconstruction, translates the surviving lifetimes into exact barrier tests, and applies the available population bounds. The reconstruction is a two-term instance of the finite convolution method already present in the research, including the six-channel reconstruction in Discounted Signed Energy, section 42. Its classical form is recorded by Helfgott and Thompson [4].

Fix the birth scale X and put $L = 2X$. Write $\mathbf{1}(n) = 1$, let δ_1 be the identity for Dirichlet convolution, and put

$$f(n) = \mu(n)\mathbf{1}_{n \leq L}, \quad g = \mu - f.$$

The identity $\mu * \mathbf{1} = \delta_1$ gives

$$g * g * \mathbf{1} = (\mu - f) * (\mu - f) * \mathbf{1} = \mu - 2f + f * f * \mathbf{1}.$$

Both nonzero factors in $g * g$ have indices at least $L+1$, so the left side vanishes through L^2 . Therefore

$$\boxed{\mu(n) = 2f(n) - (f * f * \mathbf{1})(n), \quad n \leq L^2.} \quad (17)$$

Summing this finite identity proves, for every real $1 \leq t \leq L^2$,

$$\boxed{M(t) = 2M(\min(t, L)) - \sum_{a,b \leq L} \mu(a)\mu(b) \left\lfloor \frac{t}{ab} \right\rfloor.} \quad (18)$$

Every ordered pair (a, b) is retained. No independent future coefficients, color independence, or dropped cross terms enter (18).

Connection to the square horizon. Section 3 proved that every settled early duration for births $X \leq u < 2X$ is determined by time $4X^2 = L^2$. Equation (18) now shows that all these duration tests can be written in terms of the single authentic prefix $\mu(1), \dots, \mu(2X)$. This is a precise form of the multiplicative reachback through a square horizon; the index L need not be prime.

Define $B_L(t)$ to be the full right side of (18). For a birth at u , write $h = |M(u)|$ and $\sigma = \text{sgn } M(u)$. For a positive integer D ,

$$\boxed{d_u^\infty \geq D \iff h^2 \geq u + D \text{ and } \sigma B_L(n) \geq h \text{ for all integers } u \leq n < u + D.} \quad (19)$$

If $u + D > L^2$, the first condition is impossible for a birth in this shell. Before the first downcrossing, the sign cannot change, since that would pass below level h . Thus (19) is exactly the survival condition, including an unfinished lifetime.

A census bin adds the birth test, the height range, the upper duration test $d_u^\infty < 2D$, and the original expensive-cost test. The complete multiplicative dependencies have therefore been substituted into the population problem, without replacing its actual conditions by unrelated independent events.

7. The population allowance that can be proved directly

Let

$$Q_X = \sum_{X \leq n < 2X} \mu(n)^2.$$

For the authentic census defined in section 3, including its costly and duration restrictions,

$$C(X, H, D) \leq \min \left\{ \left\lfloor \frac{Q_X + H}{2} \right\rfloor, H \left\lceil \frac{X}{D+1} \right\rceil \right\}. \quad (20)$$

Duration packing. Fix an integer depth $h \in [H, 2H)$. A counted birth at u cannot close before $u + D$. The next birth at that same depth must follow its closure by at least one counting step. Thus successive counted births at that depth differ by at least $D + 1$. There are at most $\lceil X/(D + 1) \rceil$ in the X integer positions of this shell. Summing over its H depths proves the second bound. This also covers a level that remains open.

Authentic squarefree support. Put

$$V(r) = \min\{H, \max(0, r - H + 1)\}.$$

Every unit upward change of $V(|M(n)|)$ is a birth in the height band. If U and A are the upward and downward change counts over $X \leq n < 2X$, then

$$U - A = V(|M(2X - 1)|) - V(|M(X - 1)|) \leq H, \quad U + A \leq Q_X.$$

The latter holds because a change requires $\mu(n) \neq 0$. Hence $C \leq U \leq (Q_X + H)/2$, giving the first bound. This retains the zero steps forced by repeated prime factors; it does not substitute a walk with arbitrary zero positions.

The elementary identity $\mu(n)^2 = \sum_{d^2|n} \mu(d)$ and absolute convergence give

$$Q_X = \frac{6}{\pi^2} X + O(\sqrt{X}). \quad (21)$$

This refines a counting constant but supplies no new exponent. Combining (20), $Q_X \leq X$, and the weight comparison (8) gives

$$K(X; H, D) \ll \min \left\{ \frac{HD}{X + D}, \frac{H^2}{X} \right\}. \quad (22)$$

For nonempty bins $H < 2X$; the ceiling in (20) is bounded by $1 + X/D$. Those observations make the constants in (22) absolute.

Direct comparison with the requested rate. At $H \asymp D \asymp X^{1/2+\delta}$, $0 < \delta < 1/2$, (22) gives $O(X^{2\delta})$. To establish (9) in that range we need a further population saving of size $X^{-2\delta+\varepsilon}$ relative to the $O(X)$ allowance. For $\delta = 0.1$, the direct count allows $O(X)$ births and the required count is $O_\varepsilon(X^{0.8+\varepsilon})$.

These are upper allowances, not lower estimates for the actual population. This derivation does not show that authentic Möbius lifetimes fill the permitted range. The saving must use their signed arithmetic restrictions beyond nesting and squarefree support.

8. Keep the centered bilinear form intact

The finite-prefix formula gives a concrete expression on which to attempt that signed saving. Define

$$b_L = \sum_{a \leq L} \frac{\mu(a)}{a}, \quad Z_L(t) = \sum_{a, b \leq L} \mu(a)\mu(b) \left\{ \frac{t}{ab} \right\},$$

where $\{x\} = x - \lfloor x \rfloor$. For $L \leq t \leq L^2$, (18) becomes

$$M(t) = 2M(L) - tb_L^2 + Z_L(t). \quad (23)$$

This is the same complete expression; Z_L is not assumed to be a small independent error. In particular, the main term and the fractional-part form must be estimated together.

For two times $L \leq t_0 \leq t \leq L^2$, subtracting gives

$$M(t) - M(t_0) = Z_L(t) - Z_L(t_0) - (t - t_0)b_L^2. \quad (24)$$

The inherited value $M(t_0)$ survives. Controlling changes alone would not bound how long a previously high level persists. For survival beyond L , the exact condition is

$$\sigma[Z_L(n) - Z_L(L) - (n - L)b_L^2] \geq h - \sigma M(L) \quad (25)$$

for every integer observation time n in the relevant part of the lifetime. The quantities on both sides are determined by the same prefix.

Attempted absolute estimate. Since every fractional part lies in $[0, 1)$,

$$|Z_L(t)| \leq \left(\sum_{a \leq L} |\mu(a)| \right)^2 \leq L^2. \quad (26)$$

Together with $|b_L| \leq 1 + \log L$, this supplies only a large power-sized bound in (23). It is weaker than the elementary $|M(t)| \leq t$ in much of the range. Thus it cannot furnish the saving in (9). Keeping the factors signed is necessary; asserting that their signs average independently would be an additional assumption.

Why generic positivity does not replace the missing estimate. The symmetric fractional-part matrix is not always positive semidefinite. At $L = 2, t = 3$, its $(a, b) \in \{1, 2\}^2$ block is

$$\begin{pmatrix} 0 & 1/2 \\ 1/2 & 3/4 \end{pmatrix}, \quad \det = -\frac{1}{4}.$$

Its sign cannot be assigned from entrywise nonnegativity. This does not rule out a bound for the authentic Möbius vector; it identifies why a matrix-sign shortcut would be invalid.

Outcome of this branch. The dependencies are now an explicit bilinear formula and a set of exact barrier tests. The bounds actually proved are (20) and (22), not the stronger all-scale allowance (9). A successful continuation must control the frequency of the actual barrier events (19), retaining the centered expression (23)-(25) and its inherited height.

9. Independent full-prime check and the precise stopping point

The prime-power recursion provides a second exact check of the same arithmetic information. From

$$\mu(n) \log n = - \sum_{d|n} \Lambda(d) \mu(n/d)$$

and partial summation,

$$M(x) \log x = \int_1^x \frac{M(y)}{y} dy - \sum_{d \leq x} \Lambda(d) M(x/d). \quad (27)$$

Every prime power is retained. Put $f(t) = e^{-t/2} M(e^t)$, zero for $t < 0$. The normalized identity is

$$t f(t) = \int_0^t e^{-(t-s)/2} f(s) ds - \sum_{d \leq e^t} \frac{\Lambda(d)}{\sqrt{d}} f(t - \log d). \quad (28)$$

Define $E(T) = \int_0^T f(t)^2 dt = I_M(e^T)$ and

$$J(T) = \int_0^T t f(t)^2 dt, \quad C_d(T) = \int_{\log d}^T f(t) f(t - \log d) dt,$$

$$A(T) = \int_0^T f(t) \int_0^t e^{-(t-s)/2} f(s) ds dt.$$

Multiplying (28) by $f(t)$ and integrating gives

$$J(T) = A(T) - \sum_{d \leq e^T} \frac{\Lambda(d)}{\sqrt{d}} C_d(T), \quad |A(T)| \leq 2E(T). \quad (29)$$

The bound uses the L^1 norm 2 of $e^{-t/2} \mathbf{1}_{t \geq 0}$ and the convolution inequality. Equation (29) keeps the common denominator and all inherited correlations. It does not assign a favorable sign to the prime correlations or prove a growth bound for their aggregate. Assuming such a sign would not be a consequence of (27).

What was attempted and what was obtained. The full multiplicative reconstruction was substituted into the finite population question, the lifetime nesting and authentic squarefree support were bounded, the resulting exponent was compared with the exact target, and absolute/positivity shortcuts for the remaining quadratic form were tested. The attempt proves (17)-(29), with their stated domains. It does not derive (9).

The next needed inequality is an arithmetic saving for the count of barrier events of the specific bilinear field B_L , sufficient to improve (20) to (9). That is the existing weighted-population objective expressed with all multiplicative dependencies visible. It introduces no additional requirement after the subpower estimate and does not undo the earlier completed-block, long-record, or cancellation results.

Research placement. The two-term reconstruction here is consistent with the archived six-channel reconstruction and its exact shared-prefix cancellation (Discounted Signed Energy, sections 42-43; active running proof, Part XV). The new application in this companion is the explicit connection to the finite lifetime census and its barrier events. No novelty claim is made for the classical convolution identity.

4. H. A. Helfgott and L. Thompson, Summing $\mu(n)$: a faster elementary algorithm, Research in Number Theory 9, article 6 (2023), sections 1.1 and 2, equations (2.1)-(2.2). [Published primary source](#). The combinatorial identity is used analytically here; no computation is treated as proof of a growth rate.

10. Change the census: occupation above the square-root curve

A collective alternative is to charge the total height present at each time, rather than track every level to its closure. This changes the observation horizon without imposing a maximum lifetime. Define

$$\mathcal{W}(t) = \frac{|M(t)|}{\sqrt{t}}, \quad \mathcal{H}(N) = \int_1^N (\mathcal{W}(t)^2 - 1)_+ \frac{dt}{t}.$$

The elementary decomposition $z = \min(z, 1) + (z - 1)_+$ for $z \geq 0$ proves

$$\boxed{\mathcal{H}(N) \leq I_M(N) \leq \log N + \mathcal{H}(N).} \quad (30)$$

Thus only the integrated excess above the square-root curve needs arithmetic control. A very high excursion is allowed if its weighted duration is sufficiently small; long and unfinished excursions are included automatically.

Exact connection to the existing level accounting. At time t , the occupied depths are $1, \dots, |M(t)|$. Their odd weights telescope. Those whose early portion is still active satisfy $h^2 > t$. Consequently the sum of all early level costs, before the expensive-term restriction, is

$$\mathcal{A}(N) = \int_1^N \frac{(M(t)^2 - \lfloor \sqrt{t} \rfloor^2)_+}{t^2} dt.$$

Since $0 \leq t - \lfloor \sqrt{t} \rfloor^2 \leq 2\sqrt{t}$,

$$\boxed{0 \leq \mathcal{A}(N) - \mathcal{H}(N) \leq 4.} \quad (31)$$

This is the same early energy, to a bounded correction, with simultaneously occupied levels combined into their actual total square. It does not credit their cancellation twice.

For dyadic $X \geq 2$, measure the logarithmic duration above normalized height r by

$$\nu_X(r) = \int_X^{2X} \mathbf{1}_{\{|M(t)| > r\sqrt{t}\}} \frac{dt}{t}, \quad r \geq 1.$$

Layer cake, using $|M(t)| \leq t$, gives

$$\boxed{\int_X^{2X} (\mathcal{W}(t)^2 - 1)_+ \frac{dt}{t} = 2 \int_1^{\sqrt{2X}} r \nu_X(r) dr.} \quad (32)$$

It follows that the existing subpower energy target is equivalent to

$$\boxed{\nu_X(r) \ll_{\varepsilon} \frac{X^{\varepsilon}}{r^2} \quad \text{for all dyadic } X \geq 2, r \geq 1, \varepsilon > 0.} \quad (33)$$

Necessity follows from $r^2 \nu_X(r) \leq \int_X^{2X} M(t)^2 t^{-2} dt$. For sufficiency, apply (33) with exponent $\varepsilon/2$ in (32), absorb $\log(2X)$ into $X^{\varepsilon/2}$, and sum dyadic shells in (30).

What changes. These are observation shells, not birth shells. Their tests end at $2X$, rather than the birth census's $4X^2$. An excursion crossing a shell boundary is simply charged for the portion inside each shell. Equation (33) is an equivalent form of the existing objective, not an additional gate after it.

11. A square-root prefix pays the inherited contribution

The shorter horizon admits a smaller exact multiplicative reconstruction. Fix dyadic $X \geq 2$ and now choose

$$L = \lfloor \sqrt{2X} \rfloor, \quad \mathcal{B}_X(t) = \sum_{a,b \leq L} \mu(a)\mu(b) \left\lfloor \frac{t}{ab} \right\rfloor. \quad (34)$$

The symbol $\mathcal{B}_X$ denotes only this floor sum; B_L in section 6 denoted the entire reconstructed value of M . This distinction preserves both definitions.

With the truncated coefficients f and remainder g from section 6, the first possible nonzero index of $g * g * \mathbf{1}$ is $(L+1)^2$. Since $(L+1)^2 > 2X$, the same convolution proof gives $\mu = 2f - f * f * \mathbf{1}$ at every integer through $2X$. Also $L \leq X$. Therefore throughout this entire shell,

$$\boxed{M(t) + \mathcal{B}_X(t) = 2M(L), \quad X \leq t \leq 2X.} \quad (35)$$

This slightly sharper support observation justifies the floor cutoff even when $2X > L^2$. Every pair and every common product denominator is retained.

Put

$$\mathcal{E}_X = \int_X^{2X} \frac{M(t)^2}{t^2} dt, \quad \mathcal{P}_X = \int_X^{2X} \frac{\mathcal{B}_X(t)^2}{t^2} dt.$$

The inherited constant has a bounded squared norm, using only the elementary bound $|M(L)| \leq L$:

$$\int_X^{2X} \frac{[2M(L)]^2}{t^2} dt = \frac{2M(L)^2}{X} \leq \frac{2L^2}{X} \leq 4. \quad (36)$$

Apply the reverse triangle inequality in $L^2([X, 2X], dt/t^2)$ to (35):

$$\boxed{|\sqrt{\mathcal{E}_X} - \sqrt{\mathcal{P}_X}| \leq 2.} \quad (37)$$

In particular,

$$\boxed{\mathcal{E}_X \leq 2\mathcal{P}_X + 8, \quad \mathcal{P}_X \leq 2\mathcal{E}_X + 8.} \quad (38)$$

The cross interaction with the prefix has not been discarded. The norm inequality controls it together with the complete floor sum. The inherited term itself costs at most 4 per shell, unconditionally.

Consequently

$$\boxed{I_M(N) = O_\varepsilon(N^\varepsilon) \text{ for every } \varepsilon > 0 \iff \mathcal{P}_X = O_\varepsilon(X^\varepsilon) \text{ for every } \varepsilon > 0.} \quad (39)$$

The forward direction uses (38) and $\mathcal{E}_X \leq I_M(2X)$. The reverse direction sums (38) over dyadic shells covering $[2, N]$; the constant payments contribute $O(\log N)$, and $[1, 2]$ costs $1/2$.

Proved simplification. The coefficient prefix has length at most $\sqrt{2X}$, and inherited centering has a fixed shell budget. This does not estimate $\mathcal{P}_X$ by itself. The outstanding bound is now a second-moment estimate for one complete signed floor sum, with no separate closure or prefix-height requirement.

12. Exact dictionary for product denominators

The pair sum in (34) can be grouped without losing signs. For each product k , let

$$R_L(k) = \#\{(a, b) : ab = k, a, b \leq L, a, b \text{ squarefree}\}.$$

Every nonzero pair has $\mu(a)\mu(b) = (-1)^{\Omega(k)}$, where Ω counts prime factors with multiplicity. Hence

$$\mathcal{B}_X(t) = \sum_{k \leq L^2} c_L(k) \left\lfloor \frac{t}{k} \right\rfloor, \quad c_L(k) = (-1)^{\Omega(k)} R_L(k). \quad (40)$$

The coefficient vanishes unless k is cube-free. Write such a product uniquely as $k = rs^2$, with r, s squarefree and coprime. Then

$$R_L(rs^2) = \#\{d \mid r : sd \leq L, sr/d \leq L\}, \quad c_L(rs^2) = (-1)^{\omega(r)} R_L(rs^2). \quad (41)$$

Indeed, each squared prime must occur in both factors; each prime occurring once is assigned to exactly one factor. Thus $(a, b) = (sd, sr/d)$ gives all ordered squarefree factorizations. Here $\omega(r)$ counts distinct prime factors; for squarefree r it equals $\Omega(r)$.

Research language	Exact object in this formulation
Color or prime-factor parity	The sign $(-1)^{\omega(r)}$
Multiple channels with the same denominator	The multiplicity $R_L(rs^2)$
Reachback response	$\lfloor t/k \rfloor$ at product denominator k
Inherited prefix	$2M(L)$, with squared shell cost at most 4
Complete surviving energy	$\mathcal{P}_X$, with all cross terms retained

This dictionary has a definite implication: **pairs with an identical product denominator reinforce rather than cancel**. Cancellation needed for a smaller complete response must occur across distinct denominators carrying opposite parity signs. Merely regrouping identical denominators does not produce a saving.

Define the exact positive semidefinite Gram kernel

$$\mathcal{G}_X(k, \ell) = \int_X^{2X} \left\lfloor \frac{t}{k} \right\rfloor \left\lfloor \frac{t}{\ell} \right\rfloor \frac{dt}{t^2}.$$

The complete norm is

$$\mathcal{P}_X = \sum_{k, \ell \leq L^2} (-1)^{\Omega(k) + \Omega(\ell)} R_L(k) R_L(\ell) \mathcal{G}_X(k, \ell). \quad (42)$$

Its positivity is automatic because it is a square integral; smallness is a different claim. The coefficients are the authentic cutoff factorization counts, not independent random signs. All denominators use the same measure dt/t^2 .

Current result and next estimate. Equations (30)-(42) are proved reductions and bounds. They replace the longer lifetime observation horizon by $[X, 2X]$ and pay the inherited prefix with an unconditional constant. The remaining arithmetic task is a subpower bound for (42), equivalently the occupation allowance (33). Such a bound feeds directly into section 4's already completed analytic deduction. The present argument supplies no exponent improvement for (42) and makes no claim that its final bound has been established.

13. Pay the small-product rounding errors as well

A further elementary bound removes one part of the complete field without any arithmetic cancellation assumption. With the same L and coefficients as in (34) and (40), put

$$\beta_L = \sum_{k \leq L} \frac{c_L(k)}{k}, \quad Z_{\text{low}}(t) = \sum_{k \leq L} c_L(k) \left\{ \frac{t}{k} \right\},$$

$$\mathcal{T}_X(t) = t\beta_L + \sum_{L < k \leq L^2} c_L(k) \left\lfloor \frac{t}{k} \right\rfloor.$$

Splitting off the fractional parts of the small denominators gives exactly

$$\boxed{\mathcal{B}_X(t) = \mathcal{T}_X(t) - Z_{\text{low}}(t).} \quad (43)$$

For $k \leq L$ all ordered squarefree factorizations fit the cutoff. Since $|c_L(k)| = R_L(k)$,

$$|Z_{\text{low}}(t)| \leq \sum_{k \leq L} R_L(k) = \sum_{ab \leq L} \mu(a)^2 \mu(b)^2 \leq \sum_{a \leq L} \left\lfloor \frac{L}{a} \right\rfloor \leq L(1 + \log L). \quad (44)$$

Here an absolute bound is sufficient because these products lie below the square-root scale. Since $L^2 \leq 2X$,

$$\boxed{\int_X^{2X} \frac{Z_{\text{low}}(t)^2}{t^2} dt \leq (1 + \log L)^2.} \quad (45)$$

Let $\mathcal{Q}_X = \int_X^{2X} \mathcal{T}_X(t)^2 t^{-2} dt$. Combining (37), (43), and (45) yields

$$\boxed{|\sqrt{\mathcal{E}_X} - \sqrt{\mathcal{Q}_X}| \leq 3 + \log L.} \quad (46)$$

Therefore the existing arithmetic target is also equivalent to

$$\boxed{\int_X^{2X} \left[\beta_L + \frac{1}{t} \sum_{L < k \leq L^2} c_L(k) \left\lfloor \frac{t}{k} \right\rfloor \right]^2 dt = O_\varepsilon(X^\varepsilon) \quad \text{for every } \varepsilon > 0.} \quad (47)$$

Necessity and sufficiency follow by squaring (46) with the inequality $(a + b)^2 \leq 2a^2 + 2b^2$, then summing shells as in section 11. The added squared logarithms sum to $O((\log N)^3)$, which is subpower.

Keep the centering inside the square. The smooth drift $t\beta_L$ has not been proved small separately. It remains combined with the high-product floor sum in $\mathcal{T}_X$. Bounding those two pieces separately could lose their essential signed interaction. The two pieces actually paid here are the inherited prefix and the low-product rounding errors.

Net outcome of this continuation. The remaining bound is localized to observation windows, depends on a square-root-length authentic prefix, and has no unpaid prefix or low-product rounding term. The signed high-product response relative to its smooth center remains the arithmetic estimate to establish. Equation (47) is an equivalent target, not a new assumption placed after an earlier successful bound. No numerical experiment or fictional decay law was used in these derivations.

14. Calibrated shadows of the complete centered response

The proposed indirect measurement can be made exact. The archive already proves the weighted mean/variance decomposition and pays fine fluctuations of M (active running proof, Part 0, sections 3-5, equations C1-C9). Here that result is transferred to the complete field $\mathcal{T}_X$ from section 13. It is not presented as a new discovery of the earlier decomposition.

For $X = 2^j \geq 2$, choose $h = 2^{\lfloor j/2 \rfloor} \leq \sqrt{X}$ and partition $[X, 2X]$ into integer-aligned cells $J = (a, b]$ of width h . Put

$$W_J = \frac{1}{a} - \frac{1}{b}, \quad \tau_J = \frac{1}{W_J} \int_J \mathcal{T}_X(t) \frac{dt}{t^2}, \quad \mathcal{S}_X = \sum_J W_J \tau_J^2. \quad (48)$$

Each τ_J is a signed linear measurement, or shadow. Let P be weighted orthogonal projection onto cell-constant functions in $L^2([X, 2X], dt/t^2)$. Pythagoras gives

$$\boxed{\mathcal{Q}_X = \mathcal{S}_X + F_X, \quad F_X = \|(I - P)\mathcal{T}_X\|^2 \geq 0.} \quad (49)$$

The question is whether F_X has already been paid. It has, as follows.

On each cell the essential range of M has length at most h , since every integer increment has magnitude at most one. A real variable in an interval of length d has variance at most $d^2/4$: expand $\mathbb{E}[(Y - u)(u + d - Y)] \geq 0$. Applying this under the normalized weight $t^{-2}dt/W_J$ yields

$$\|(I - P)M\|^2 \leq \frac{h^2}{4} \sum_J W_J = \frac{h^2}{8X} \leq \frac{1}{8}. \quad (50)$$

From sections 11 and 13,

$$M + \mathcal{T}_X = 2M(L) + Z_{\text{low}}, \quad (I - P)\mathcal{T}_X = -(I - P)M + (I - P)Z_{\text{low}}.$$

Projection removes the constant inherited term exactly. If $A_L = \sum_{k \leq L} |c_L(k)|$, the values of Z_{low} lie in an interval of length at most A_L : each fractional part lies in $[0, 1]$, with its coefficient's sign retained. The same variance estimate and (44) give

$$\|(I - P)Z_{\text{low}}\| \leq \frac{A_L}{\sqrt{8X}} \leq \frac{1 + \log L}{2}.$$

Consequently the calibrated reconstruction theorem is

$$\boxed{\mathcal{S}_X \leq \mathcal{Q}_X \leq \mathcal{S}_X + \left[\frac{1}{\sqrt{8}} + \frac{1 + \log L}{2} \right]^2.} \quad (51)$$

There are $X/h \asymp \sqrt{X}$ shadows, and the unobserved part costs only $O(\log^2 X)$. This is an unconditional upper reconstruction bound, not merely a lower bound on the energy supplied by projection. Together with (46), it proves that subpower $\mathcal{S}_X$ is equivalent to the existing historical-energy target. It does not assert the arithmetic size of $\mathcal{S}_X$.

15. Triangulation across scales with every product retained

The shadows in (48) can be calculated from the same authentic product coefficients. For a cell or larger interval $J = (a, b]$, define

$$q_J(k) = \left\lfloor \frac{a}{k} \right\rfloor + \frac{1}{W_J} \sum_{a < mk \leq b} \left(\frac{1}{mk} - \frac{1}{b} \right).$$

Finite integration of the floor jumps gives exactly

$$\tau_J = \frac{\beta_L \log(b/a)}{W_J} + \sum_{L < k \leq L^2} c_L(k) q_J(k). \quad (52)$$

A jump at b has zero weight. Inherited floor values at a , the smooth center, repeated denominators, and every coefficient are present.

Now recursively bisect $[X, 2X]$ until reaching the cells of width h . For an internal interval J with children J_-, J_+ , set

$$D_J = \sqrt{\frac{W_{J_-} W_{J_+}}{W_J}} (\tau_{J_-} - \tau_{J_+}), \quad D_0 = \frac{\tau_{[X, 2X]}}{\sqrt{2X}}.$$

The elementary weighted variance identity at each split telescopes to

$$\mathcal{S}_X = |D_0|^2 + \sum_{J \text{ internal}} |D_J|^2. \quad (53)$$

These are genuinely orthogonal observation channels. No orthogonality or independence of arithmetic colors is assumed. The whole-window mean D_0 must remain; zero-mean probes cannot see it.

Each contrast also has an exact compactly supported arithmetic kernel. Write

$$\psi_J = \sqrt{\frac{W_{J_+}}{W_{J_-} W_J}} \mathbf{1}_{J_-} - \sqrt{\frac{W_{J_-}}{W_{J_+} W_J}} \mathbf{1}_{J_+}, \quad K_J(t) = \int_a^t \psi_J(s) \frac{ds}{s^2}.$$

Then $K_J(a) = K_J(b) = 0$ and $D_J = \langle \mathcal{T}_X, \psi_J \rangle$. Integration by parts for the smooth term and floor jumps gives

$$D_J = -\beta_L \int_a^b K_J(t) dt - \sum_{L < k \leq L^2} c_L(k) \sum_{a < mk < b} K_J(mk). \quad (54)$$

The function K_J is a tent in the reciprocal coordinate. Substituting (40)-(41) retains every multiplicative dependency in each shadow.

What the triangulation establishes. Equations (51)-(54) determine the quadratic energy up to a proved polylogarithmic remainder. Its subpower estimate is equivalent to an upper bound on the complete squared shadow sum (53). One average, or bounds only from below, cannot replace that sum. This is the exact calibration required to infer size from indirect measurements.

A further possibility. The next two sections show that a different family of signed mean shadows can enter the analytic closing argument directly. That option does not require recovering all the quadratic energy first; it is an alternative sufficient input, not another condition appended to (53).

16. One sliding mean as a direct analytic observable

Keep the entire smooth center inside $\mathcal{T}_X$. Define its normalized signed mean

$$\mathfrak{a}(X) = \frac{1}{\sqrt{X}} \int_X^{2X} \frac{\mathcal{T}_X(t)}{t} dt. \quad (55)$$

For this section X ranges over **every integer** $X \geq 2$, not only powers of two. The finite convolution and bounds in sections 11 and 13 remain valid: $L = \lfloor \sqrt{2X} \rfloor \leq X$ and $(L+1)^2 > 2X$ are all those proofs require.

This shadow has a fully explicit arithmetic form. Put

$$\mathcal{J}_X(k) = \int_X^{2X} \left\lfloor \frac{t}{k} \right\rfloor \frac{dt}{t} = \sum_{1 \leq m \leq 2X/k} \log \frac{2X}{\max\{X, mk\}}. \quad (56)$$

Indeed $\lfloor t/k \rfloor = \sum_{m \geq 1} \mathbf{1}_{mk \leq t}$, so finite interchange proves the formula. Then

$$\boxed{\sqrt{X} \mathfrak{a}(X) = X\beta_L + \sum_{L < k \leq L^2} c_L(k) \mathcal{J}_X(k).} \quad (57)$$

The product coefficients, common denominators, and signed smooth center remain together. Thus a proposed subpower shadow bound has the precise shape

$$\boxed{\left| X\beta_L + \sum_{L < k \leq L^2} c_L(k) \mathcal{J}_X(k) \right| \ll_{\varepsilon} X^{1/2+\varepsilon} \text{ for every integer } X \geq 2, \varepsilon > 0.} \quad (58)$$

This is a smoothed signed first moment. No expansion of its square is required for the deduction below. Equation (58) is the arithmetic estimate proposed here; it has not been proved by the kernel identity.

To calibrate the shadow against the actual signed count, define

$$\mathcal{A}(x) = \int_1^x \frac{M(t)}{t} dt.$$

Cauchy-Schwarz on an interval of length X , together with $\|M + \mathcal{T}_X\|_{dt/t^2} \leq 3 + \log L$, proves

$$\boxed{\left| \mathfrak{a}(X) + \frac{\mathcal{A}(2X) - \mathcal{A}(X)}{\sqrt{X}} \right| \leq 3 + \log L.} \quad (59)$$

Consequently (58) gives, for every $\eta > 0$,

$$|\mathcal{A}(2X) - \mathcal{A}(X)| \ll_{\eta} X^{1/2+\eta}. \quad (60)$$

Use a smaller positive exponent in (58) if necessary to absorb the logarithm in (59). The bound (60) holds for all real $X \geq 2$ as well: $|\mathcal{A}'| = |M(x)|/x \leq 1$ almost everywhere, so replacing X by $\lfloor X \rfloor$ changes this difference by at most 3. This is why the family includes every integer starting position. Estimates only at isolated dyadic starting points would not justify that interpolation.

17. The shadow closes the same Mellin deduction, conditionally

The direct closing implication can now be proved in full. For real $x \geq 2$, write $x = 2^q r$ with $2 \leq r < 4$. Telescoping (60) gives

$$\mathcal{A}(x) = \mathcal{A}(r) + \sum_{j=0}^{q-1} [\mathcal{A}(2^{j+1}r) - \mathcal{A}(2^j r)].$$

The absolute values are bounded by a geometric series with exponent $1/2 + \eta$. Since $|\mathcal{A}(r)| \leq 3$,

$$\boxed{\mathcal{A}(x) = O_\eta(x^{1/2+\eta}) \text{ for every } \eta > 0.} \quad (61)$$

For $\Re s > 1/2$, define

$$\mathcal{F}(s) = s^2 \int_1^\infty \mathcal{A}(x) x^{-s-1} dx. \quad (62)$$

Given a compact subset of this half-plane, choose η smaller than its distance from the boundary. Equation (61) provides locally uniform absolute convergence, including the logarithmic factors needed for complex differentiation. Thus $\mathcal{F}$ is holomorphic there.

Initially for $\Re s > 1$, finite partial summation followed by absolute convergence and Fubini gives

$$\int_1^\infty \mathcal{A}(x) x^{-s-1} dx = \frac{1}{s} \int_1^\infty M(t) t^{-s-1} dt = \frac{1}{s^2 \zeta(s)}. \quad (63)$$

Hence $\mathcal{F} = 1/\zeta$ in that initial half-plane. The identity

$$(s-1)\zeta(s)\mathcal{F}(s) = s-1$$

extends throughout $\Re s > 1/2$, with the pole at 1 removed. At a zero ρ there, the left side would vanish and the right side would not. The functional equation then excludes nontrivial zeros to the left of the critical line as well. This proves the conditional implication from (58) to the critical-line conclusion.

Exact logical scope. A single signed mean on one window does not upper-bound its squared energy. Sections 16-17 use an entire sliding family of means as direct input to a Mellin transform of the integrated count. They do not infer (39) from one average. Conversely, the original all-scale historical-energy bound implies (58) by Cauchy-Schwarz, (46), and covering an arbitrary integer window by dyadic shells. Thus a proved original energy bound still suffices; the mean-shadow route supplies an alternative way to reach the same analytic conclusion.

What is established and what is still arithmetic. The reconstruction error in (51), the exact arithmetic kernels (52), (54), (56), the shadow comparison (59), and the implication (58) to the critical line are proved here. The signed first-moment estimate (58) itself remains unproved. The inspected square-packet source already proves fine-variance and diagonal allowances and retains the aggregate signed bound as its premise (active running proof, Part 0, equations C1-C13). Those results do not state (58).

The next direct arithmetic object is therefore the complete centered scalar sum in (58). Estimating its large-product part and $X\beta_L$ separately could destroy the cancellation under investigation. Its bound is an alternative to the full quadratic target; no additional energy, Fourier, lifetime-closing, or tail condition is required after it. The indirect-measurement proposal has been given a rigorous calibration and a complete conditional analytic finish, without promoting the unproved arithmetic rate to a theorem.

18. Direct scalar attack: a decaying calibration error

This continuation estimates actual parts of the signed shadow rather than assuming its desired size. Write

$$U(X) = \mathcal{A}(2X) - \mathcal{A}(X) = \int_X^{2X} M(t) \frac{dt}{t}, \quad C_L = \sum_{k \leq L} c_L(k), \quad \mathcal{H}_L = \sum_{j=1}^L \frac{1}{j}.$$

The logarithmic kernel already contains exact cancellation. If $\Phi(y) = \sum_{m \leq y} \log(y/m)$, then

$$\mathcal{J}_X(k) = \Phi(2X/k) - \Phi(X/k) = \sum_{\substack{m \leq 2X/k \\ m \text{ odd}}} \log \frac{2X}{mk}. \quad (64)$$

The even $m = 2r$ terms in the first sum are precisely the second sum. These are multiple indices; the product denominators k and their coefficients are unchanged.

There is a stronger calibration than the earlier $O(\log L)$ error for the normalized shadow. For every positive integer k , put

$$\rho_X(k) = \int_X^{2X} \{t/k\} \frac{dt}{t}.$$

The periodic continuous function

$$G(u) = \frac{\{u\}^2 - \{u\}}{2} + \frac{1}{16}$$

has $|G| \leq 1/16$ and derivative $\{u\} - 1/2$ almost everywhere. Integration by parts with $kG(t/k)$ proves

$$\rho_X(k) = \frac{\log 2}{2} + \epsilon_X(k), \quad |\epsilon_X(k)| \leq \frac{k}{8X}. \quad (65)$$

The two boundary terms contribute at most $3k/(32X)$ and the integral term at most $k/(32X)$.

Retain the signs when summing (65). Since

$$\sum_{k \leq L} k |c_L(k)| \leq \sum_{ab \leq L} ab \leq \frac{L^2}{2} (\mathcal{H}_L + 1),$$

and $L^2 \leq 2X$, we obtain

$$\int_X^{2X} Z_{\text{low}}(t) \frac{dt}{t} = \frac{\log 2}{2} C_L + r_X, \quad |r_X| \leq \frac{\mathcal{H}_L + 1}{8}. \quad (66)$$

For the middle inequality sum $a \sum_{b \leq L/a} b \leq L^2/(2a) + L/2$ over $a \leq L$.

Define the explicitly calibrated measurement

$$\tilde{\mathfrak{a}}(X) = \mathfrak{a}(X) - \frac{\log 2}{\sqrt{X}} \left(2M(L) + \frac{C_L}{2} \right).$$

Integrating $M + \mathcal{J}_X = 2M(L) + Z_{\text{low}}$ now gives

$$\tilde{\mathfrak{a}}(X) = -\frac{U(X)}{\sqrt{X}} + \frac{r_X}{\sqrt{X}}, \quad \frac{|r_X|}{\sqrt{X}} \leq \frac{\mathcal{H}_L + 1}{8\sqrt{X}}. \quad (67)$$

The subtracted calibration is $O(\log L)$ by $|M(L)| \leq L$ and $|C_L| \leq L(1 + \log L)$. Hence subpower bounds for $\mathfrak{a}$, $\tilde{\mathfrak{a}}$ and $U(X)/\sqrt{X}$ are equivalent. The error in (67) actually decays. This is a bound for measurement error, not yet for the surviving signed signal.

19. The modifier cancels the entire older prefix

The doubling identity underlying the Michaels Möbius modifier is essential here: it removes a full logarithmic background exactly. It is the same authentic identity already used in the active running proof, Part 0, section 4; this section applies it to the new scalar observable.

Let

$$\mathcal{A}_{\text{odd}}(x) = \sum_{\substack{n \leq x \\ n \text{ odd}}} \mu(n) \log(x/n).$$

Splitting by parity, using $\mu(2n) = -\mu(n)$ for odd n and zero contribution from multiples of four, proves

$$\mathcal{A}(x) = \mathcal{A}_{\text{odd}}(x) - \mathcal{A}_{\text{odd}}(x/2).$$

Consequently

$$U(X) = \mathcal{A}_{\text{odd}}(2X) - 2\mathcal{A}_{\text{odd}}(X) + \mathcal{A}_{\text{odd}}(X/2). \quad (68)$$

For every $n \leq X/2$, its coefficient is $\log(2X/n) - 2\log(X/n) + \log(X/(2n)) = 0$. All earlier-prefix terms cancel, with their authentic signs. Thus

$$U(X) = \sum_{\substack{X/2 < n \leq 2X \\ n \text{ odd}}} \mu(n) w_X(n), \quad (69)$$

where the continuous nonnegative logarithmic tent is

$$w_X(t) = \begin{cases} \log(2t/X), & X/2 < t \leq X, \\ \log(2X/t), & X < t \leq 2X, \\ 0, & \text{otherwise.} \end{cases} \quad (70)$$

Its peak is $\log 2$, its total variation is $2\log 2$, and direct integration gives $\int w_X(t) dt = X/2$.

Further prime dependencies remain available. For an odd prime p , splitting the odd indices into those coprime to p , those with exactly one factor p , and those divisible by p^2 gives

$$U(X) = \sum_{(n, 2p)=1} \mu(n) [w_X(n) - w_X(pn)]. \quad (71)$$

The weights vanish outside finite support. This is a signed difference of two related tents; its two parts are not independent populations.

An actual unconditional allowance. The elementary squarefree identity gives

$$\sum_{\substack{n \leq y \\ n \text{ odd}}} \mu(n)^2 = \sum_{\substack{d \leq \sqrt{y} \\ d \text{ odd}}} \mu(d) \#\{m \leq y/d^2 : m \text{ odd}\} = \frac{4}{\pi^2} y + O(\sqrt{y}).$$

Partial summation against (70) therefore yields

$$|U(X)| \leq \sum_{n \text{ odd}} \mu(n)^2 w_X(n) = \frac{2}{\pi^2} X + O(\sqrt{X}). \quad (72)$$

This uses the real squarefree support and arbitrary window endpoints. It is an upper allowance, not a prediction that the authentic signed sum is that large. The desired bound is $O_\varepsilon(X^{1/2+\varepsilon})$. The support estimate (72) does not give that saving; (71) identifies the signed dependency retained for further estimation.

20. Testing the remaining signed cancellation

The full multiplicative convolution gives more than a support restriction. Extend $\mathcal{A}(x) = 0$ for $x \leq 1$. Finite interchange and $\mu * \mathbf{1} = \delta_1$ give

$$\sum_{d \leq x} \mathcal{A}(x/d) = \sum_{r \leq x} \log(x/r) \sum_{n|r} \mu(n) = \log x.$$

Subtracting at $2X$ and X proves the complete signed identity

$$\boxed{\sum_{d \leq 2X} U(X/d) = \log 2, \quad X \geq 1.} \quad (73)$$

This is exact cancellation across the entire dilation family, including the inherited contributions. It is not an upper bound on each summand. In the attempted absolute-value induction, a proposed allowance $|U(y)| \leq Cy^\sigma$, $0 < \sigma < 1$, incurs the factor

$$\sum_{2 \leq d \leq X} d^{-\sigma} \asymp X^{1-\sigma}. \quad (74)$$

At square-root exponent this factor grows like $\sqrt{X}$, so this particular induction is not a contraction. The signed relation (73) must supply more information than its termwise absolute majorant.

A second exact check retains the reference square. Put $m_L = \sum_{a \leq L} \mu(a)/a$. Using $\mathcal{J}_X(k) = X/k - \rho_X(k)$ in (57),

$$\boxed{\sqrt{X} \mathfrak{a}(X) = X m_L^2 - \sum_{L < k \leq L^2} c_L(k) \rho_X(k).} \quad (75)$$

The required estimate asks for agreement of these two quantities to $O_\varepsilon(X^{1/2+\varepsilon})$. The elementary identity $L m_L = 1 + \sum_{a \leq L} \mu(a) \{L/a\}$ implies $|m_L| \leq 1$; and $\sum_k |c_L(k)| \leq L^2$. Taking absolute values in (75) consequently gives only an $O(X)$ allowance. The nonnegative square alone supplies no sign or closeness estimate for the second term.

Application of the stronger bound already in the archive. Part 0, equation C19, imports the classical unconditional estimate $M(t) = O_B(t/(\log t)^B)$ for every fixed $B > 0$. It is also stated in the primary source [5], introduction, equation (1). Integrating on $[X, 2X]$ gives

$$\boxed{U(X) = O_B\left(\frac{X}{(\log X)^B}\right), \quad \mathfrak{a}(X) = O_B\left(\frac{\sqrt{X}}{(\log X)^B}\right).} \quad (76)$$

For the second formula use (67) and absorb its logarithmic calibration into the displayed bound for sufficiently large X . This applies the archived theorem; it does not improve its exponent. For every fixed B and $0 < \varepsilon < 1/2$,

$$\frac{X^{1/2}/(\log X)^B}{X^\varepsilon} \rightarrow \infty.$$

Result of the direct attempt. The entire old prefix cancels in (69), the measurement error decays in (67), and the complete dilation cancellation is exact in (73). The bounds obtained are (72) and (76). They do not establish the subpower estimate (58). What remains is the rate of the signed compact tent sum, equivalently the centered comparison (75); these are alternative forms of the same scalar target. No further analytic condition follows a successful estimate.

5. E. S. Lee and N. Leong, New explicit bounds for Mertens function and the reciprocal of the Riemann zeta-function, [arXiv:2208.06141v4](https://arxiv.org/abs/2208.06141v4), introduction, equation (1). Only its classical asymptotic implication is used here; no numerical finite-range test supplies an all-scale premise.

21. Cancel the long boundary; keep the reciprocal correlations

The odd-index kernel in (64) admits a further exact resummation. It compresses the remaining signed term back to the square-root prefix and pays its boundary at the desired scale. Let

$$N = 2X, \quad L = \lfloor \sqrt{N} \rfloor, \quad Y = N/L,$$

and define the reciprocal-scale convolution

$$\mathcal{K}_L(y) = \int_{\max(1, y/L)}^{\min(L, y)} M(u) M(y/u) \frac{du}{u} \quad (1 \leq y \leq L^2), \quad (77)$$

with value zero outside this range. Both arguments of M are at most L . The exact identity is

$$U(X) = \mathcal{B}_X - \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \mathcal{K}_L(N/m), \quad (78)$$

where

$$\mathcal{B}_X = 2M(L) \int_L^Y M(u) \frac{du}{u} - M(L)^2 \log(Y/L). \quad (79)$$

Only finitely many terms of (78) are nonzero. The proof on the next page keeps every inherited term and shows how the long boundary cancels.

The remaining boundary is paid unconditionally. Put $\delta = Y - L$. Since N is an integer and $L = \lfloor \sqrt{N} \rfloor$, $N \leq (L+1)^2 - 1$, so $0 \leq \delta \leq 2$. Completing the square in (79) gives

$$\mathcal{B}_X = \int_L^Y \frac{M(u)^2 - [M(u) - M(L)]^2}{u} du.$$

The elementary inequalities $|M(u)| \leq u$ and $|M(u) - M(L)| \leq u - L$ imply

$$-\frac{\delta^3}{3L} \leq \mathcal{B}_X \leq L\delta + \frac{\delta^2}{2}, \quad -\frac{8}{3L} \leq \mathcal{B}_X \leq 2L + 2. \quad (80)$$

Thus $\mathcal{B}_X/\sqrt{X} = O(1)$ without any unproved cancellation bound.

Direct estimation of the remaining term. Multiplicative Cauchy-Schwarz, under $u \mapsto y/u$, proves

$$|\mathcal{K}_L(y)| \leq \sqrt{y} \int_1^L \frac{M(u)^2}{u^2} du = \sqrt{y} I_M(L). \quad (81)$$

Summing this absolute majorant in (78) costs $O(XI_M(L))$, because $\sum_{m \leq N} m^{-1/2} = O(\sqrt{N})$. It therefore does not prove the desired square-root bound. The common signed correlations across the odd m channels cannot be discarded.

The remaining estimate can be expressed as

$$\left| \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \mathcal{K}_L(2X/m) \right| \ll_{\varepsilon} X^{1/2+\varepsilon}. \quad (82)$$

By (78)-(80), this is equivalent to the scalar target for U . It is not an additional requirement. The boundary is now settled; the signed correlation estimate (82) is not established by (81).

22. Proof of the reciprocal resummation and current status

Use $\log_+ z = \max(0, \log z)$ and define the finite sums

$$\mathcal{A}_L(z) = \sum_{a \leq L} \mu(a) \log_+(z/a), \quad \mathcal{V}_L(y) = \sum_{a, b \leq L} \mu(a) \mu(b) \log_+(y/(ab)).$$

Expanding the two factors of M in (77), and integrating their indicator functions, gives

$$\boxed{\mathcal{K}_L(y) = \mathcal{V}_L(y) - 2M(L)\mathcal{A}_L(y/L) + M(L)^2 \log_+(y/L^2).} \quad (83)$$

For completeness, each pair initially supplies the interval $[a, y/b]$ in the measure du/u . Intersecting it with $[y/L, L]$ removes its two boundary portions, yielding the two $\mathcal{A}_L$ terms. When those removed portions overlap, their intersection contributes the last term back. This also shows that the right side cancels to zero outside $[1, L^2]$.

Integrating the exact finite convolution (35), using (64), proves

$$U(X) = 2M(L) \log 2 - \sum_{m \text{ odd}} \mathcal{V}_L(N/m). \quad (84)$$

The complete dilation identity for $\mathcal{A}$ yields

$$\sum_{m \text{ odd}} \mathcal{A}(Y/m) = \log Y - \log(Y/2) = \log 2, \quad Y \geq 2. \quad (85)$$

Since $Y/L = N/L^2 < 3$, only $m = 1$ can have $Y/m > L$. Therefore

$$\begin{aligned} \sum_{m \text{ odd}} \mathcal{A}_L(Y/m) &= \log 2 + \mathcal{A}_L(Y) - \mathcal{A}(Y), \\ \mathcal{A}(Y) - \mathcal{A}_L(Y) &= \int_L^Y \frac{M(u) - M(L)}{u} du. \end{aligned} \quad (86)$$

Also only $m = 1$ contributes to $\sum_{m \text{ odd}} \log_+(N/(mL^2))$, giving $\log(Y/L)$. Substitute (83) into (84), then use (85)-(86). The full $2M(L) \log 2$ term cancels, and the remaining boundary is exactly (79). Finally $\mathcal{K}_L(N) = 0$, because $N \geq L^2$, so the sum begins at $m = 3$. This proves (78).

What this direct attempt proves. It supplies the exact odd-index cancellation (64), decaying scalar calibration (67), complete removal of the older prefix (69), full dilation balance (73), reciprocal reconstruction (78), and square-root boundary bound (80). The classical bound in (76) is applied with its actual quantifiers. No numerical sweep or extrapolation is used.

What it does not prove. The all-scale signed estimate (82), equivalently the tent bound $U(X) = O_\varepsilon(X^{1/2+\varepsilon})$, remains unproved. The absolute-value attempts (72), (74), (75), and (81) identify exactly where their loss occurs. The next arithmetic task is the signed reciprocal-correlation sum, preserving the common prefix and the interaction among its odd channels. A successful bound activates sections 16-17 directly, with no extra energy premise afterward.

23. Take the combined signed sum first

Retain the definitions of sections 21-22: $N = 2X$, $L = \lfloor \sqrt{N} \rfloor$, $Y = N/L$, and $\mathcal{K}_L$ is the reciprocal convolution (77). Write

$$\mathcal{S}_X = \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \mathcal{K}_L(N/m). \quad (87)$$

Here $\mathcal{S}_X$ denotes this scalar correlation sum, not the cell-energy sum of section 14. The exact relation is $U(X) = \mathcal{B}_X - \mathcal{S}_X$, with the boundary already bounded in (80).

Absorb the endpoint before combining channels. Define

$$\nu_a = \mu(a) \quad (1 \leq a < L), \quad \nu_L = \mu(L) - M(L), \quad \sum_{a \leq L} \nu_a = 0. \quad (88)$$

The adjustment at L records the inherited endpoint; it is not a new arithmetic sequence. Expanding (83) gives, for every $y > 0$,

$$\mathcal{K}_L(y) = \sum_{a, b \leq L} \nu_a \nu_b \log_+ \frac{y}{ab}.$$

Introduce the complete odd kernel

$$\Theta(z) = \sum_{\substack{m \leq z \\ m \text{ odd}}} \log(z/m), \quad z \geq 1.$$

Because $\mathcal{K}_L(N) = 0$ and all sums are finite, we may include $m = 1$ and interchange the sums without an absolute-value estimate. Thus

$$\boxed{\mathcal{S}_X = \sum_{a, b \leq L} \nu_a \nu_b \Theta\left(\frac{N}{ab}\right)}. \quad (89)$$

Every product denominator, coefficient sign, and inherited boundary contribution is retained in (89).

Separate the continuous term exactly. Define

$$e(z) = \Theta(z) - \frac{z}{2} + \frac{\log 2}{2}.$$

The next section proves $|e(z)| \leq 1/(4z)$. Since $\sum \nu_a = 0$, the constant term cancels from (89). Consequently

$$\boxed{\mathcal{S}_X = \frac{N}{2} b'_L{}^2 + \mathcal{R}_X, \quad \mathcal{R}_X = \sum_{a, b \leq L} \nu_a \nu_b e\left(\frac{N}{ab}\right)}, \quad (90)$$

where partial summation gives

$$b'_L = \sum_{a \leq L} \frac{\mu(a)}{a} - \frac{M(L)}{L} = \int_1^L \frac{M(u)}{u^2} du. \quad (91)$$

The continuous contribution is a genuine nonnegative square. The lattice remainder is signed and remains attached to that square. Equation (90) is the result of combining the channels first; it does not assert that the remainder is small or positive.

24. The small-product remainder is bounded

Odd-kernel error lemma. For every real $z \geq 1$,

$$\left| \Theta(z) - \frac{z}{2} + \frac{\log 2}{2} \right| \leq \frac{1}{4z}. \quad (92)$$

Here is a proof including the centering constant. Put $r = \lfloor (z+1)/2 \rfloor$. The product of the first r odd positive integers is $(2r)!/(2^r r!)$, so

$$\Theta(z) = r \log z - \log \frac{(2r)!}{2^r r!}.$$

Stirling's formula and $|z - 2r| \leq 1$ show $\Theta(z) = z/2 - (\log 2)/2 + O(1/z)$; hence $e(z) \rightarrow 0$. Away from odd integers,

$$e'(z) = \frac{d(z)}{z}, \quad d(z) = \left\lfloor \frac{z+1}{2} \right\rfloor - \frac{z}{2}.$$

The period-two function d has mean zero. On $1 \leq z \leq 3$ it has the continuous periodic primitive

$$P(z) = \frac{z-1}{2} - \frac{(z-1)^2}{4} - \frac{1}{8}, \quad |P(z)| \leq \frac{1}{8}.$$

Integrating by parts to infinity gives

$$e(z) = \frac{P(z)}{z} - \int_z^\infty \frac{P(t)}{t^2} dt,$$

which proves (92), including at the continuous kernel's breakpoints.

Split $\mathcal{R}_X = \mathcal{R}_\leq(X) + \mathcal{R}_>(X)$ according to $ab \leq L$ and $ab > L$. Let $\mathcal{H}_L = \sum_{a \leq L} 1/a$. For the small products,

$$\begin{aligned} |\mathcal{R}_\leq(X)| &\leq \frac{1}{4N} \sum_{ab \leq L} ab |\nu_a \nu_b| \\ &\leq \frac{1}{4N} \left[\frac{L^2}{2} (\mathcal{H}_L + 1) + 2L |M(L)| \right] \\ &\leq \frac{\mathcal{H}_L + 1}{8} + \frac{1}{2}. \end{aligned} \quad (93)$$

Indeed, $\sum_{a \leq L} a \sum_{b \leq L/a} b \leq L^2 (\mathcal{H}_L + 1)/2$. The endpoint adjustment can affect only $(a, b) = (1, L), (L, 1)$ in this range, at an additional cost at most $2L |M(L)|$. Finally $N \geq L^2$ and $|M(L)| \leq L$.

Thus the complete signed sum has the proved form

$$\mathcal{S}_X = Xb'_L{}^2 + \mathcal{R}_>(X) + \mathcal{R}_\leq(X), \quad |\mathcal{R}_\leq(X)| = O(\log L). \quad (94)$$

The larger products and their smooth square remain together. Taking absolute values of them separately is not part of this reduction.

25. Only the combined negative deficit needs a bound

Define the complete larger-product comparison

$$\mathcal{W}_X = Xb'_L{}^2 + \mathcal{R}_>(X), \quad \mathcal{D}_X = (-\mathcal{W}_X)_+ = \max\{0, -\mathcal{W}_X\}. \quad (95)$$

This scalar deficit is not the earlier historical energy. It records only how far the signed remainder exceeds the available positive square in the negative direction.

One-sided closing criterion. The following arithmetic estimate suffices for the critical-line conclusion:

$$\boxed{\forall \varepsilon > 0 \exists C_\varepsilon < \infty \quad \mathcal{D}_X \leq C_\varepsilon X^{1/2+\varepsilon} \quad \text{for every integer } X \geq 2.} \quad (96)$$

Equivalently, with the same quantifiers, the allowed comparison is

$$\boxed{\mathcal{R}_>(X) \geq -Xb'_L{}^2 - C_\varepsilon X^{1/2+\varepsilon}.} \quad (97)$$

The entire square is retained as a positive contribution. A separate estimate for $|\mathcal{R}_>|$ would be stronger than what this argument needs.

By (93)-(94), (96) is equivalent, after changing constants, to the one-sided bound

$$\mathcal{S}_X \geq -C'_\varepsilon X^{1/2+\varepsilon}. \quad (98)$$

Combine it with $U = \mathcal{B} - \mathcal{S}$ and $\mathcal{B}_X \leq 2L + 2$ to obtain

$$U(X) \leq C''_\varepsilon X^{1/2+\varepsilon}. \quad (99)$$

This is only an upper bound on U ; we do not infer its absolute-value bound at this step.

To extend (99) to real starts, use $\mathcal{A}'(x) = M(x)/x$ almost everywhere, so $\mathcal{A}$ is 1-Lipschitz. Consequently $|U(x) - U(\lfloor x \rfloor)| \leq 3$ for $x \geq 2$. For $q = 1/2 + \varepsilon \in (1/2, 1)$, telescope from x down to $y = x/2^j \in [2, 4)$:

$$\mathcal{A}(x) = \mathcal{A}(y) + \sum_{k=1}^j U(x/2^k) \leq C_q x^q, \quad (100)$$

where the geometric series $\sum_{k \geq 1} 2^{-kq}$ converges. Enlarge C_q to cover $1 \leq x < 4$.

The next section proves that this one-sided estimate already completes the analytic deduction. The argument uses the standard positivity principle for Mellin transforms, rather than inserting an unproved absolute-value estimate.

Status. Equations (87)-(95) and the implication (96) $\Rightarrow$ (100) are proved here. The arithmetic assertion (96) itself is not proved. This is one sufficient input to the closing argument; it does not append a second energy requirement.

26. Proof of the one-sided analytic closure

Positivity lemma. Suppose $g \geq 0$ is locally integrable on $[1, \infty)$ and its Mellin integral

$$G(s) = \int_1^\infty g(x)x^{-s-1}dx$$

has a finite real abscissa of convergence c . Then G cannot extend holomorphically to a neighborhood of the real point c . This is the standard Landau positivity principle; see [6], section 2.4, Proposition 1.

For a self-contained proof, suppose such a neighborhood exists and take a real $a = c + \delta$ sufficiently close to c . The Taylor series of the extension at a converges at $a - 2\delta < c$. Differentiating the integral at a is justified by convergence at a slightly smaller real argument. Its signed Taylor coefficients are nonnegative moments:

$$(-1)^n G^{(n)}(a) = \int_1^\infty g(x)(\log x)^n x^{-a-1} dx \geq 0.$$

Monotone convergence in the Taylor series therefore gives

$$\begin{aligned} \sum_{n \geq 0} \frac{(2\delta)^n}{n!} (-1)^n G^{(n)}(a) &= \int_1^\infty g(x)x^{-a-1} \sum_{n \geq 0} \frac{(2\delta \log x)^n}{n!} dx \\ &= \int_1^\infty g(x)x^{-(a-2\delta)-1} dx < \infty, \end{aligned}$$

contradicting the definition of c .

Application to the signed scalar bound. Fix $q \in (1/2, 1)$ and assume (100). Put $g(x) = C_q x^q - \mathcal{A}(x) \geq 0$. Since $|\mathcal{A}(x)| \leq x$, the Mellin integral converges for $\Re s > 1$. The elementary integral of $\log(x/n)$ gives, initially there,

$$G(s) = \frac{C_q}{s - q} - \frac{1}{s^2 \zeta(s)}. \quad (101)$$

The expression on the right is regular at every real $s > q$: $1/\zeta$ is regular at 1; $\zeta(s) > 0$ for real $s > 1$; and for $0 < s < 1$ the alternating eta series is positive while $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$.

If the convergence abscissa of G exceeded q , equation (101), as a meromorphic identity on its original convergence half-plane, would extend G regularly through that real abscissa. The lemma excludes this. Thus the integral is holomorphic on $\Re s > q$. Hence

$$F_q(s) = s^2 \left[\frac{C_q}{s - q} - G(s) \right]$$

is holomorphic there and agrees with $1/\zeta(s)$ on $\Re s > 1$. The identity theorem applied to $(s - 1)\zeta(s)F_q(s) = s - 1$ rules out every zero with real part greater than q . Let $q \downarrow 1/2$ and use the functional equation to reflect zeros on the left. This proves the claimed conditional closure from (96).

6. M. Suzuki, On variants of Chebyshev's conjecture, Ramanujan Journal **68**, 95 (2025), section 2.4, Proposition 1. [doi:10.1007/s11139-025-01238-9](https://doi.org/10.1007/s11139-025-01238-9). The principle is classical; the application to (95) is derived above.

27. Direct estimate of the combined remainder: what it gives

The kernel bound (92), applied after the exact resummation, gives the unconditional estimate

$$|\mathcal{R}_X| \leq \frac{1}{4N} \left(\sum_{a \leq L} a |\nu_a| \right)^2 \leq \frac{(3L+1)^2}{16} = O(X). \quad (102)$$

Here $\sum a |\nu_a| \leq L(L+1)/2 + L|M(L)|$ and $N \geq L^2$. Since the square in (90) is nonnegative, this also bounds $(-\mathcal{S}_X)_+$. It is a valid aggregate allowance, but its power is too large for (96).

Applying the stronger classical estimate already recorded in (76), together with $U = \mathcal{B} - \mathcal{S}$ and the lower boundary bound (80), improves the actual conclusion to

$$\boxed{(-\mathcal{S}_X)_+, \mathcal{D}_X = O_B \left(\frac{X}{(\log X)^B} \right) \text{ for each fixed } B > 0.} \quad (103)$$

This does not imply (96): for $0 < \varepsilon < 1/2$ and every fixed B , $X^{1/2-\varepsilon}/(\log X)^B \rightarrow \infty$.

Audit of the full odd-dilation cancellation. There is a useful exact check on the combined sum. Multiplicativity at 2 and $\mu * \mathbf{1} = \delta_1$ give

$$\sum_{m \text{ odd}} M(z/m) = \mathbf{1}_{[1,2)}(z).$$

For $z = N/u \geq Y \geq 2$, it follows that

$$\boxed{\mathcal{S}_X = - \int_1^L \frac{M(u)}{u} \sum_{\substack{m \text{ odd} \\ m < Y/u}} M\left(\frac{N}{mu}\right) du.} \quad (104)$$

Endpoint equalities have measure zero. The cancellation is complete, but the remaining mirror arguments exceed L . Changing variables $v = mu$ returns

$$\mathcal{S}_X = -U(X) + \int_L^Y M(v)M(N/v) \frac{dv}{v} = -U(X) + \mathcal{B}_X. \quad (105)$$

For the last equality, $N < (L+1)^2$ ensures that v and N/v cannot both reach $L+1$. Therefore $[M(v) - M(L)][M(N/v) - M(L)] = 0$ almost everywhere, and expansion with $v \leftrightarrow N/v$ gives (79). This route verifies the boundary identity; it does not supply an additional growth estimate.

Present conclusion. Combining the signed channels yields the exact square plus remainder (90), the logarithmically bounded small-product part (93), and the one-sided closing criterion (96), with its full proof. The direct bounds obtained are (102)-(103). The unresolved arithmetic estimate is the negative deficit of the complete expression $Xb'_L{}^2 + \mathcal{R}_>(X)$, with the authentic endpoint adjustment and all interactions retained. Neither $\mathcal{R}_>$ alone nor each odd channel has to satisfy the requested bound separately.

28. A larger portion of the product remainder is bounded

The deficit rate (96) closes the selected argument. It is one sufficient input; sections 31-32 below prove other sufficient inputs that permit fewer observations or averaged control. We first improve the direct estimate without assuming any of them.

Retain $N = 2X$, $L = \lfloor \sqrt{N} \rfloor$, the endpoint coefficients ν_a in (88), and the exact error e in (92). For any real $L \leq Z < L^2$, set

$$\mathcal{R}_{\leq Z}(X) = \sum_{\substack{a, b \leq L \\ ab \leq Z}} \nu_a \nu_b e\left(\frac{N}{ab}\right).$$

Then

$$|\mathcal{R}_{\leq Z}(X)| \leq \frac{Z^2(\mathcal{H}_L + 5)}{8N}. \quad (106)$$

Proof. The interior coefficients satisfy $|\mu(a)| \leq 1$, and

$$\sum_{a \leq L} a \sum_{b \leq \min(L, Z/a)} b \leq \frac{Z^2}{2} \mathcal{H}_L + \frac{ZL}{2} \leq \frac{Z^2}{2} (\mathcal{H}_L + 1).$$

The endpoint adjustment adds at most

$$2L|M(L)| \sum_{b \leq Z/L} b \leq |M(L)| \left(\frac{Z^2}{L} + Z \right) \leq 2Z^2.$$

The double endpoint (L, L) is excluded because $Z < L^2$. Multiply the resulting bound by $1/(4N)$ from (92). This proves (106).

Choose $Z = L^{3/2}$. Since $L \geq 2$ and $N \geq L^2$,

$$|\mathcal{R}_{\leq L^{3/2}}(X)| \leq \frac{L}{8} (\mathcal{H}_L + 5) = O(\sqrt{X} \log X). \quad (107)$$

This is within $O_\varepsilon(X^{1/2+\varepsilon})$ for every $\varepsilon > 0$. The remaining exact expression is

$$\mathcal{S}_X = Xb'_L{}^2 + \sum_{\substack{a, b \leq L \\ ab > L^{3/2}}} \nu_a \nu_b e\left(\frac{N}{ab}\right) + O(\sqrt{X} \log X). \quad (108)$$

Every surviving index satisfies $a, b > \sqrt{L}$. Thus the part involving an index at most $\sqrt{L}$ is already bounded at the required scale. The positive square is retained in full; it has not been truncated along with the remainder.

This extends the earlier cutoff $ab \leq L$ to $ab \leq L^{3/2}$ by a proved estimate. It does not infer that the complementary larger-product sum has a favorable sign. For those products the odd kernel has at most $N/L^{3/2} = O(\sqrt{L})$ possible multipliers, but counting those multipliers alone is not a cancellation estimate.

29. Common-factor families can also be bounded

There is a second bounded portion, including terms above the cutoff in (107). Let $R \geq 1$ be an integer. Group ordered pairs by

$$a = gr, \quad b = gs, \quad (r, s) = 1, \quad \max(r, s) \leq R.$$

For $a, b < L$, a nonzero coefficient satisfies $\mu(a)\mu(b) = \mu(r)\mu(s)\mu(g)^2$. Thus the shared factor contributes a nonnegative weight; the reduced pair retains its authentic sign.

Put $q = \max(r, s)$ and $p = \min(r, s)$. The elementary error estimate gives

$$\begin{aligned} \sum_{g \leq (L-1)/q} |\mu(gr)\mu(gs)e(N/(g^2rs))| &\leq \frac{rs}{4N} \sum_{g \leq (L-1)/q} g^2 \\ &\leq \frac{L}{4} \frac{p}{q^2}. \end{aligned} \tag{109}$$

We used $\sum_{g \leq T} g^2 \leq T^3$ for $T \geq 1$ and $N \geq L^2$. The case $q = 1$ costs $L/4$. For each $q \geq 2$, the sum of p/q^2 over the two ordered reduced-pair orientations is at most $2 \sum_{p < q} p/q^2 \leq 1$. Hence all interior pairs in this family cost at most $LR/4$.

For a mixed endpoint pair, write $a = L$, $b = Ls/r < L$ with $r > s$ coprime and $r \mid L$. Since $|\nu_L| = |M(L-1)| \leq L$, the two ordered terms together cost at most

$$\frac{2L \cdot Lb}{4N} \leq \frac{Ls}{2r}.$$

Summing even over all $r \leq R$, rather than only divisors of L , costs at most $LR(R-1)/8$. Therefore

$$\boxed{|\text{contribution from these reduced pairs, excluding } (L, L)| \leq \frac{LR(R+1)}{8}.} \tag{110}$$

The estimate also bounds the absolute sum of any subfamily. It can therefore be applied only above $ab = L^{3/2}$, avoiding double counting with (107).

Taking $R = \lfloor (1 + \log L)^c \rfloor$ for any fixed $c > 0$ gives another $O(\sqrt{X} \text{ polylog } X)$ allowance. After both removals, the remaining object consists of the positive square, the endpoint diagonal, and pairs satisfying

$$a, b > \sqrt{L}, \quad ab > L^{3/2}, \quad \max\left(\frac{a}{(a, b)}, \frac{b}{(a, b)}\right) > R. \tag{111}$$

The endpoint diagonal $\nu_L^2 e(N/L^2)$ stays attached to the complete expression. Its separate size need not be small. Equations (107) and (110) are established allowances; a bound for the negative deficit of the remaining combination is still required along this route.

30. Packet pairs cancel except at product thresholds

This is an alternative representation of the **whole sum** (89), not an assertion about a sharply truncated remainder. Partition $(\nu_1, \dots, \nu_L)$ into successive nonzero first-return packets of total zero. The last packet is closed algebraically by $\nu_L = -M(L-1)$ if needed. Zero singleton packets may be omitted. Actual packet lengths are unrestricted.

For a packet i supported on integer indices from α_i to β_i , define its cumulative response

$$F_i(u) = \sum_{\substack{n \text{ in packet } i \\ n \leq u}} \nu_n = \sigma_i q_i(u), \quad q_i \geq 0, \quad \sigma_i \in \{-1, 1\}.$$

It is zero outside $[\alpha_i, \beta_i]$, and has one sign internally. This uses the first-return property; the forced terminal jump closes the last response at zero without overshooting.

Define $\bar{\Theta}(z) = \sum_{m \geq 3, m \text{ odd}} \log_+(z/m)$. Since $\sum \nu_a = 0$, the $m = 1$ logarithm has zero total contribution. Thus (89) holds with Θ replaced by $\bar{\Theta}$. For each pair of packets, two finite summations by parts yield

$$W_{ij} = \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \int_0^\infty q_i(u) q_j\left(\frac{N}{mu}\right) \frac{du}{u} \geq 0,$$

$$\boxed{\mathcal{S}_X = \sum_i W_{ii} + 2 \sum_{i < j} \sigma_i \sigma_j W_{ij}.} \quad (112)$$

For the underlying log hinge, the mixed derivative in u, v is the positive measure $\delta(c - uv)$, $c = N/m$; its integral is exactly the displayed reciprocal integral. This proves the formula with its signs and endpoints.

Exact zero contribution away from crossings. A packet rectangle contributes zero if its product range crosses no threshold $ab = N/m$ with odd $m \geq 3$. Within such a rectangle the active set of logarithms is constant, so the kernel is $C - r \log a - r \log b$; both packet totals are zero. Therefore only crossing rectangles survive. Same-sign pairs, including self-pairs, contribute nonnegatively. Opposite-sign pairs supply the negative terms.

There is an aggregate allowance without a separate multiplier-count factor. Put $H_i = \sup q_i$. Then

$$W_{ij} \leq H_i H_j \mathcal{L}_{ij}, \quad (113)$$

where the nonnegative crossing weight is exactly

$$\begin{aligned} \mathcal{L}_{ij} = & \bar{\Theta}\left(\frac{N}{\alpha_i \alpha_j}\right) - \bar{\Theta}\left(\frac{N}{\beta_i \alpha_j}\right) \\ & - \bar{\Theta}\left(\frac{N}{\alpha_i \beta_j}\right) + \bar{\Theta}\left(\frac{N}{\beta_i \beta_j}\right). \end{aligned} \quad (114)$$

All multipliers are already included in this four-corner weight. The target is a bound on opposite-sign crossing contributions **after** subtracting the same-sign credits in (112). The inequality (113) alone does not prove that bound.

31. Fourth-power starting positions suffice

The original one-sided criterion asked for every integer starting position. The one-step arithmetic rule lets us recover all intermediate positions from a much sparser family, with an error already at square-root scale.

Recall $U(x) = \mathcal{A}(2x) - \mathcal{A}(x)$. Almost everywhere,

$$U'(x) = \frac{M(2x) - M(x)}{x}. \quad (115)$$

Let $a \geq 1$ and $h > 0$. Away from integer and half-integer breakpoints, the numerator is constant and has absolute value at most $x + 1 \leq 2x$. The total smooth variation of U' on $[a, a + h]$ is at most $2h/a$. The jumps coming from $M(2x)$ and $M(x)$ have total absolute mass at most $(3h + 2)/a$. Hence

$$\boxed{\text{Var}_{[a, a+h]} U' \leq \frac{5h + 2}{a}.} \quad (116)$$

Take one-sided representatives at the finitely many breakpoints; their values do not affect U or the interpolation argument.

If $\ell(x)$ is the line joining the two endpoint values of U , the difference between U' and its average on the interval is bounded almost everywhere by $\text{Var } U'$. Integrating gives the sufficient estimate

$$\boxed{\sup_{a \leq x \leq a+h} |U(x) - \ell(x)| \leq \frac{5h^2 + 2h}{a}.} \quad (117)$$

Now set $a_j = j^4$ and $h_j = (j + 1)^4 - j^4 \leq 15j^3$. The right side of (117) is $O(j^2)$, with an absolute constant. Also (80), (93) and (95) give

$$U(n) \leq \mathcal{D}_n + C\sqrt{n} \quad (n \geq 2). \quad (118)$$

Consequently the following arithmetic estimate is sufficient:

$$\boxed{\mathcal{D}_{j^4} \ll_{\varepsilon} j^{2+\varepsilon} \quad \text{for every } j \geq 2 \text{ and every } \varepsilon > 0.} \quad (119)$$

Indeed, interpolate the endpoint upper bounds across $[j^4, (j+1)^4]$ and add $O(j^2)$. This gives $U(x) \ll_{\varepsilon} x^{1/2+\varepsilon/4}$; relabel the arbitrary exponent and apply sections 25-26.

Only $O(N^{1/4})$ starting positions through N enter (119). The intervening scales are covered analytically. This is not a finite verification of an infinite assertion: (119) must still be proved for the complete sequence of fourth powers. Its rate has not been derived here. The result is a proved reduction in the observations required, not a new estimate of their arithmetic values.

32. Averaged deficits offer another sufficient route

Large deficits can also be allowed if their total contribution is controlled. A sufficient arithmetic input is

$$\boxed{\sum_{2 \leq n \leq T} \mathcal{D}_n \ll_{\varepsilon} T^{3/2+\varepsilon} \text{ for every } \varepsilon > 0.} \quad (120)$$

For nonnegative $\mathcal{D}_n$, this is equivalent, with the same all-positive-exponent convention, to $\sum_{n \geq 2} \mathcal{D}_n n^{-3/2-\varepsilon} < \infty$ for every $\varepsilon > 0$. Partial summation proves one direction; bounding the partial sum by $T^{3/2+\varepsilon}$ times the convergent series proves the other.

Here is the full conditional transfer. Equation (118) and the 3-Lipschitz property of U give

$$\int_1^T U(t)_+ dt \ll_{\varepsilon} T^{3/2+\varepsilon}. \quad (121)$$

Extend $\mathcal{A}(x) = 0$ for $x \leq 1$ and use dyadic telescoping. Integrating the upper bound for the positive window terms, including the bounded initial interval, gives

$$\begin{aligned} \mathcal{C}(x) &:= \frac{1}{x} \int_1^x \mathcal{A}(t) dt \leq O(1) + \frac{1}{x} \sum_{k \leq \log_2 x} 2^k \int_1^{x/2^k} U(t)_+ dt \\ &\ll_{\varepsilon} x^{1/2+\varepsilon} \sum_{k \geq 1} 2^{-k(1/2+\varepsilon)} + O(1). \end{aligned} \quad (122)$$

The final series converges. Initially for $\Re s > 1$, Fubini gives

$$\int_1^{\infty} \mathcal{C}(x) x^{-s-1} dx = \frac{1}{s^2(s+1)\zeta(s)}. \quad (123)$$

Apply the positivity lemma of section 26 to $C_q x^q - \mathcal{C}(x) \geq 0$, for $q > 1/2$. The extra factor $s+1$ has no zero in that half-plane, so exactly the same argument excludes zeta zeros there and gives the critical-line conclusion. No energy bound is required after (120).

Averaging only fourth-power observations also suffices. Combining (117)-(118), the positive part of U between successive fourth powers is bounded by the linear interpolation of the two endpoint deficits plus $C\sqrt{x}$. The interval lengths are $O(j^3)$. Therefore the still weaker sampling input

$$\boxed{\sum_{2 \leq j \leq J} j^3 \mathcal{D}_{j^4} \ll_{\varepsilon} J^{6+\varepsilon} \text{ for every } \varepsilon > 0} \quad (124)$$

implies (121), after rescaling the arbitrary exponent, and hence closes by (122)-(123). A possible last incomplete interval is bounded by the entire next interval; $J+1$ and J have comparable powers.

Equations (120) and (124) allow occasional larger deficits. They are alternative sufficient arithmetic inputs, not proved rates or extra requirements after (96). The interpolation and Mellin transfers themselves are proved.

33. Import the existing large-block and record work

The current deficit connects directly to the complete historical energy already studied. From $\mathcal{W}_X = \mathcal{S}_X - \mathcal{R}_{\leq}(X)$, $\mathcal{S}_X = \mathcal{B}_X - U(X)$, and the lower boundary estimate in (80),

$$\mathcal{D}_X \leq U(X)_+ + \frac{8}{3L} + |\mathcal{R}_{\leq}(X)|.$$

Cauchy-Schwarz on $[X, 2X]$ therefore proves

$$\mathcal{D}_X \leq \sqrt{X \int_X^{2X} \frac{M(t)^2}{t^2} dt} + O(\log X). \quad (125)$$

This transfers an actual complete energy estimate to the current signed deficit without bounding the larger-product remainder separately.

Existing long/open allowances remain available. The archived note Michaels persistent-level absorption and transfer to short records (8 September 2026), equations (1)-(3) and (8), proves

$$\sqrt{\max_{n \leq Z} R(n)} \leq \sqrt{60(1 + \log Z)^2 \mathcal{H}_Z + 4\mathfrak{C}_{\text{short}}(Z)} + \log(1 + \log Z). \quad (126)$$

Here $R(n) = \int_1^n M(t)^2 t^{-2} dt + M(n)^2/n$; $\mathcal{H}_Z$ is the harmonic sum. The retained cost $\mathfrak{C}_{\text{short}}$ is the source's maximum over short strict energy records, with its genuine opening positions, heights, complete histories, and unfinished endpoints. Its large and long families have already been treated in that source.

Combining (125)-(126) yields

$$\mathcal{D}_X \leq \sqrt{X [\sqrt{60(1 + \log(2X))^2 \mathcal{H}_{2X} + 4\mathfrak{C}_{\text{short}}(2X)} + \log(1 + \log(2X))]} + O(\log X). \quad (127)$$

Thus a subpower estimate for that retained cost would supply the present deficit rate. This is an alternative sufficient route, not a new demand after the direct one-sided criterion.

The same source, equations (9)-(11), explains its limitation: a surviving lower stack contributes identically to both record endpoints, so a small change in record energy does not bound the absolute size of that stack. Its statement explicitly leaves the retained arithmetic rate unproved. This continuation preserves that scope.

The full note Authentic cross-term cancellation on square windows (3 October 2026), equations (9)-(11), also supplies genuine aggregate cancellation with remainder $O_A(X/(\log X)^A)$. Its transfer through (125) gives the same arbitrary fixed logarithmic saving already recorded in (103), after relabeling A . It does not establish the desired power. Both source documents were read in full for this integration; this scoped audit does not claim to exhaust the archive.

The source archive accompanying this note includes the exact retrieved text of these two records and their identifiers, so the imported definitions and hypotheses can be checked without reconstructing them from a summary.

34. What the parallel attempts establish

These are alternative ways into the same closing deduction. Success along one is sufficient; they are not a list of simultaneous obligations.

Approach	Proved result	Arithmetic estimate still needed
Larger-product localization	All products $ab \leq L^{3/2}$ cost $O(\sqrt{X} \log X)$	Negative deficit of the complementary sum with its square and endpoint retained
Common-factor grouping	Reduced-ratio families through R cost at most $LR(R+1)/8$, apart from the retained endpoint diagonal	Remaining authentic large reduced-ratio interactions
Packet crossings	Noncrossing packet rectangles cancel exactly; same-sign pairs give positive credits	Opposite-sign crossing weight minus the same-sign credits
Sparse observations	Fourth-power interpolation costs only $O(\sqrt{X})$	Rate (119) at those sample starts
Averaged observations	Either (120) or (124) implies the Mellin closure	A bound for the cumulative deficit
Existing records	The archived complete-energy allowances transfer by (127)	The source's retained short-record arithmetic cost

The most concrete next arithmetic object. With the packet weights of section 30, put

$$\mathcal{P}_X = \sum_i W_{ii} + 2 \sum_{\substack{i < j \\ \sigma_i = \sigma_j}} W_{ij}, \quad \mathcal{N}_X = 2 \sum_{\substack{i < j \\ \sigma_i \neq \sigma_j}} W_{ij}.$$

Then

$$\boxed{\mathcal{S}_X = \mathcal{P}_X - \mathcal{N}_X, \quad (-\mathcal{S}_X)_+ = (\mathcal{N}_X - \mathcal{P}_X)_+} \quad (128)$$

The required estimate concerns this **net** quantity. Every packet length and the algebraically closed terminal packet remain included. Equations (113)-(114) expose its actual heights and product-threshold overlap; no assumption of equal packet lengths or orthogonality is made. The analytic targets may be imposed on all integer starts, on fourth-power starts, or through the corresponding averaged allowance.

Why a generic positive-kernel shortcut was rejected. For the model cumulative input $F(u) = \mathbf{1}_{[29,31)}(u) - \mathbf{1}_{[39,41)}(u)$, take $N = 3600$, $L = 60$. Each bump's self-product interval misses every odd threshold $3600/m$. The mixed product interval crosses only $m = 3$, giving the exact signed total $-2 \log(41/39) < 0$. Thus its two self-contributions vanish while the mixed contribution is negative. A Gram-type inequality bounding that cross term by the self-terms is false for general inputs. This model is not the Möbius sequence; it identifies where authentic arithmetic information must enter.

Result of this continuation. Several substantial portions are now bounded or cancel identically, and sparse or averaged observations suffice for the analytic finish. No argument above establishes the remaining all-scale arithmetic rate. In particular, the formal completeness of the multiplicative identities, the sign of individual credits, and the already proved treatment of large blocks do not by themselves estimate (128). The next attempt can focus on the actual opposite-sign threshold crossings and their available credits, using the existing record work with its original weights.

35. The crossing allowance uses the existing block quantities

The packet weights in (112) can be estimated with the archived block heights, reciprocal charges, and exact savings. This supplies an explicit transfer, including the terminal packet.

For packet i , define

$$b_i = \int \frac{q_i(u)}{u^2} du = \left| \sum_{n \text{ in packet } i} \frac{\nu_n}{n} \right|, \quad E_i = \int \frac{q_i(u)^2}{u^2} du.$$

Partial summation proves the charge identity because the packet total is zero. With $H_i = \sup q_i$,

$$E_i = H_i b_i - V_i, \quad V_i = \int \frac{q_i(u)(H_i - q_i(u))}{u^2} du \geq 0. \quad (129)$$

This is the same height-times-charge minus saving structure used in the earlier block estimates.

The exact four-corner crossing measure in (114) also has a width allowance. Let $\Delta_i = 1/\alpha_i - 1/\beta_i$ and $\lambda_i = \log(\beta_i/\alpha_i)$. Counting odd integers in $N/(u\beta_j) < m < N/(u\alpha_j)$, and then integrating du/u , gives

$$\mathcal{L}_{ij} \leq \frac{N}{2} \Delta_i \Delta_j + \min(\lambda_i, \lambda_j). \quad (130)$$

The count is at most its interval length divided by two, plus one; symmetry gives the minimum. No upper bound on either packet's length is assumed.

There is an energy allowance on the full joint incidence set. Put

$$\mathcal{Q}_{ij} = \sup_{u \in [\alpha_i, \beta_i]} \sum_{\substack{m \geq 3 \text{ odd} \\ N/(mu) \in [\alpha_j, \beta_j]}} m^{-1/2}.$$

Using $v = N/(mu)$, one Cauchy-Schwarz inequality on all (u, m) incidences gives

$$W_{ij} \leq \sqrt{N} \sqrt{\mathcal{Q}_{ij} \mathcal{Q}_{ji} E_i E_j}. \quad (131)$$

Indeed, write each integrand as $\sqrt{N/m} [q_i(u)/\sqrt{u}] [q_j(v)/\sqrt{v}] du/u$; Cauchy-Schwarz leaves the two weighted sums of squares, bounded by $\mathcal{Q}_{ij} E_i$ and $\mathcal{Q}_{ji} E_j$ after the reciprocal substitution.

Thus the negative pairs admit the combined allowance

$$W_{ij} \leq \min \left\{ H_i H_j \mathcal{L}_{ij}, \sqrt{N} \sqrt{\mathcal{Q}_{ij} \mathcal{Q}_{ji} (H_i b_i - V_i)(H_j b_j - V_j)} \right\}. \quad (132)$$

Keep the actual positive terms $\mathcal{P}_X$ in (128) when applying this bound to opposite-sign pairs. No extra multiplier-count factor is appended: the incidence weights already account for all multipliers.

The terminal closure is algebraic. Before L its response is the actual unfinished walk, but its charge need not equal the physical unfinished charge. If h_{open} is the actual block's sum $\sum_{n \leq L} \mu(n)/n$, then

$$\sigma_i b_i = h_{\text{open}} - \frac{M(L)}{L}, \quad E_{\text{open}} = E_i + \frac{M(L)^2}{L}.$$

Thus physical unfinished-energy bounds remain upper bounds for E_i . Evaluate them with their original physical height, charge and terminal saving; do not identify those data with the artificially closed packet's data. The remaining issue is to sum the negative incidences against the positive credits sharply enough. Equation (132) does not itself settle that comparison.

36. Remove the artificial endpoint coefficients

This continuation records the six parallel analytical attempts following section 35. The allowances below are proved reductions of the arithmetic expression. The final deficit rate remains a conditional input to sections 25-26 or 32.

Retain $N = 2X \in \mathbb{N}$, $L = \lfloor \sqrt{N} \rfloor \geq 2$, $Y = N/L$, and define the ordinary-coefficient form

$$Q_L(N) = \sum_{a,b \leq L} \mu(a)\mu(b)\Theta\left(\frac{N}{ab}\right), \quad m_L = \sum_{a \leq L} \frac{\mu(a)}{a}. \quad (\text{R1})$$

Unlike ν_L , every coefficient in this double sum has absolute value at most one. The inherited endpoint must first be accounted for exactly.

Single-convolution identity. Dirichlet convolution gives $\mu * \mathbf{1}_{\text{odd}} = \delta_1 - \delta_2$. Consequently, for every $z \geq 1$,

$$\sum_{a \leq z} \mu(a)\Theta(z/a) = \log z - \log_+(z/2). \quad (\text{R2})$$

Here $\log_+ t = \max(0, \log t)$. The equality follows by grouping the finite sum by am ; it does not require an estimate of a Möbius sum.

Since $L \leq Y \leq L + 2$ and $Y \geq 2$, define

$$T_L = \int_L^Y \frac{M(u) - M(L)}{u} du.$$

The omitted terms $a > L$ in (R2), with $z = Y$, equal T_L : throughout this omitted range $Y/a < 2$, so only the odd multiplier 1 occurs. Also $|T_L| \leq 2/L$. Thus

$$\sum_{a \leq L} \mu(a)\Theta(Y/a) = \log 2 - T_L. \quad (\text{R3})$$

Expanding $\nu = \mu - M(L)\delta_L$ in (89), and using $\Theta(N/L^2) = \log(Y/L)$, proves

$$\boxed{\mathcal{S}_X = Q_L(N) - 2(\log 2)M(L) + 2M(L)T_L + M(L)^2 \log(Y/L)}. \quad (\text{R4})$$

For the last kernel evaluation, $N/L^2 < 3$ for every $L \geq 2$. Since $\log(Y/L) \leq 2/L$, this implies

$$|\mathcal{S}_X - Q_L(N)| \leq (2 + 2\log 2)L + 4, \quad \mathcal{S}_X \geq Q_L(N) - 2(\log 2)L - 4. \quad (\text{R5})$$

The terms involving T_L and $\log(Y/L)$ are precisely the original boundary $\mathcal{B}_X$. This also checks compatibility with $U = \mathcal{B} - \mathcal{S}$.

The complete new centering. Substituting the definition of e into (R1) yields

$$\boxed{Q_L(N) = \frac{N}{2}m_L^2 - \frac{\log 2}{2}M(L)^2 + \sum_{a,b \leq L} \mu(a)\mu(b)e\left(\frac{N}{ab}\right)}. \quad (\text{R6})$$

Both centering terms remain. Removing the exceptional endpoint coefficient converts a nonnegative square in the ν representation into the displayed signed pair of centering terms. The negative $M(L)^2$ term cannot be discarded. The benefit is that bounded ordinary Möbius coefficients are now available throughout the remainder, including the last row and column.

37. Complete product rectangles through the four-fifths cutoff

We prove an oscillatory allowance for complete dyadic rectangles. It is applied before any irregular masking of a rectangle.

Let $B_2(t) = t^2 - t + 1/6$ and set

$$P_0(z) = -B_2\left(\left\{\frac{z+1}{2}\right\}\right) = -\frac{1}{\pi^2} \sum_{k \geq 1} \frac{(-1)^k \cos(\pi k z)}{k^2}. \quad (\text{R7})$$

This continuous periodic primitive has derivative $\lfloor (z+1)/2 \rfloor - z/2$ away from breakpoints. Integration by parts, using $e(z) \rightarrow 0$, gives

$$e(z) = \frac{1}{z} \left[P_0(z) - \int_1^\infty P_0(vz) \frac{dv}{v^2} \right], \quad z \geq 1. \quad (\text{R8})$$

The Fourier series is absolutely convergent. Hence all forthcoming interchanges with the finite arithmetic sums and the v^{-2} integral are justified.

Reciprocal phase estimate. Let $a \in [A, 2A)$, $b \in [B, 2B)$ be integer indices, with $A \geq B \geq 1$, coefficients $|\alpha_a|, |\beta_b| \leq 1$, and write $P = AB$, $T = N/P$. For $q = wT$ with $1 \leq q \leq A/4$,

$$\left| \sum_{a,b} \alpha_a \beta_b \exp\left(\frac{i\pi w N}{ab}\right) \right| \ll AB \left[B^{-1/2} + \sqrt{\frac{\log(2q)}{q}} \right]. \quad (\text{R9})$$

To prove this, apply Cauchy-Schwarz in a and expand the square over b, b' . The diagonal contributes $O(A^2 B)$. For $h = |b - b'| > 0$, the derivative in a of the normalized phase is monotone, of magnitude comparable to $qh/(AB)$, and less than $1/8$. The elementary first-derivative estimate therefore bounds its a sum by $O(\min\{A, AB/(qh)\})$.

For completeness, that first-derivative estimate follows by writing consecutive terms as a telescoping difference divided by $\exp(2\pi i \Delta f) - 1$, and using summation by parts. Monotonicity of the derivative makes the increments monotone; if their distance from an integer is at least λ , the reciprocal denominator and its total variation are $O(1/\lambda)$. This gives $O(1/\lambda)$, together with the trivial length bound. No assertion about cancellation in the coefficients is inserted at this step: Cauchy-Schwarz has removed the α_a coefficients from these inner sums.

There are $O(B)$ ordered pairs for each h . Split the h sum at B/q when that number exceeds one. The result is

$$\sum_{1 \leq h \leq B} \min\{A, AB/(qh)\} \ll \frac{AB}{q} \log(2q).$$

When $q > B$, use $\log(2B) \leq \log(2q)$. Taking square roots proves (R9). For $q < 1$ or $q > A/4$ we use the trivial $O(AB)$ bound instead.

Remainder on one rectangle. Put

$$R_{A,B} = \sum_{A \leq a < 2A, B \leq b < 2B, a, b \leq L} \mu(a) \mu(b) e(N/(ab)).$$

The last dyadic intervals are completed by zero coefficients; this keeps the domain a product of two intervals. The factor ab/N in (R8) can be absorbed into bounded separate coefficients a/A and b/B , at a fixed absolute cost. If $T \geq 1$ and $A/T \geq 4$, (R7)-(R9) give

$$|R_{A,B}| \ll \frac{P^2}{N} \left[B^{-1/2} + \sqrt{\frac{\log(2T)}{T}} + \frac{T}{A} \left(1 + \log \frac{A}{T} \right) \right]. \quad (\text{R10})$$

Indeed, the portion $kTv \leq A/4$ is covered by (R9); summing its $k^{-2}v^{-2}$ weights preserves the first two displayed terms. In the complementary portion, trivial estimation gives

$$\sum_{k \geq 1} \frac{1}{k^2} \min\left(1, \frac{4kT}{A}\right) \ll \frac{T}{A} \left(1 + \log \frac{A}{T}\right).$$

This also covers the nonintegrated Fourier tail $kT > A/4$. These tails are part of the proof, not omitted high frequencies.

For $P \leq N^{3/4}$, the elementary estimate $|e(N/(ab))| \leq ab/(4N)$ already gives $|R_{A,B}| \ll P^2/N \leq \sqrt{N}$. For $N^{3/4} < P \leq N^{4/5}$, orient the rectangle so $A \geq B$. Since $A \leq L \leq \sqrt{N}$, $B \geq P/\sqrt{N}$, while $A \geq \sqrt{P}$. Therefore the three terms in (R10) are respectively

$$O(N^{9/20}), \quad O(\sqrt{N}\sqrt{\log(2N)}), \quad O(N^{2/5} \log(2N)).$$

Here $A/T \geq N^{1/8}$; the finitely many smaller N for which it is below four are absorbed by an absolute constant. Summing the $O((1 + \log N)^2)$ complete rectangles proves

$$\boxed{\sum_{\substack{\text{complete dyadic rectangles} \\ AB \leq N^{4/5}}} |R_{A,B}| \ll \sqrt{N}(1 + \log N)^3.} \tag{R11}$$

The selected union contains every pair $ab \leq N^{4/5}$ and may contain additional pairs in the boundary rectangles. Formula (R11) bounds the signed sum on each whole rectangle. It does not bound the sum of the absolute values of its individual terms, or an arbitrary filtered subfamily. Subsequent removals must respect this distinction.

38. Common-factor wedge and the complete surviving expression

After section 36, all coefficients in the remainder have magnitude at most one. This permits an absolute estimate which can safely be applied outside the rectangles already treated.

Write $a = gr$, $b = gs$, $(r, s) = 1$, $r \geq s$. Then

$$\begin{aligned} \sum_{g \leq L/r} |\mu(gr)\mu(gs)e(N/(g^2rs))| &\leq \frac{rs}{4N} \sum_{g \leq L/r} g^2 \\ &\leq \frac{L^3}{4N} \frac{s}{r^2}. \end{aligned} \tag{R12}$$

The last inequality uses $\sum_{1 \leq g \leq t} g^2 \leq t^3$ for real $t \geq 1$ (first prove it at integers, then use monotonicity). No squarefree density or independence approximation is involved.

Let $B \geq 1$ and select all pairs with

$$\min(r, s)^2 \leq B \max(r, s). \tag{R13}$$

For a fixed larger factor r , sum $s \leq \min(r, \sqrt{Br})$, with both orientations. Dropping the coprimality condition only enlarges the bound. If $t = \lfloor \min(r, \sqrt{Br}) \rfloor$, then $t(t+1) \leq Br + \sqrt{Br}$. Counting the diagonal twice is harmless. Thus the absolute contribution of the entire selected family is at most

$$\boxed{\frac{L^3}{4N} [B\mathcal{H}_L + \sqrt{B}\zeta(3/2)]}. \tag{R14}$$

Taking $B = (1 + \log N)^2$ gives $O(\sqrt{N}(1 + \log N)^3)$. This controls a wedge of reduced-factor ratios, not only finitely many small reduced factors. It also includes the diagonal. Because (R14) is an absolute sum, it remains valid when restricted to pairs outside the already treated rectangles.

Let $\mathcal{C}_N$ be the complement, in $1 \leq a, b \leq L$, of the complete rectangles in (R11) and the wedge (R13). Combining (R5), (R6), (R11), and (R14) proves

$$\begin{aligned} \mathcal{S}_X &= \frac{N}{2} m_L^2 - \frac{\log 2}{2} M(L)^2 \\ &\quad + \sum_{(a,b) \in \mathcal{C}_N} \mu(a)\mu(b)e(N/(ab)) + O(\sqrt{N}(1 + \log N)^3). \end{aligned} \tag{R15}$$

Every surviving pair satisfies

$$ab > N^{4/5}, \quad a, b > N^{3/10}, \quad \min(r, s)^2 > (1 + \log N)^2 \max(r, s). \tag{R16}$$

These are necessary conditions for membership in $\mathcal{C}_N$, not a replacement definition of that set: the complete rectangle removals can remove some additional pairs.

The estimate still needed along this route is a lower bound, for every $\varepsilon > 0$, of the form

$$\frac{N}{2} m_L^2 - \frac{\log 2}{2} M(L)^2 + \sum_{(a,b) \in \mathcal{C}_N} \mu(a)\mu(b)e(N/(ab)) \geq -C_\varepsilon N^{1/2+\varepsilon}. \tag{R17}$$

The constants may depend on ε but not on N . The proved error in (R15) fits this allowance. Both smooth centering terms are retained inside (R17); neither is a separately established bound for the surviving arithmetic interaction. Once (R17) is established, the conditional analytic deduction already given applies.

39. Reflection and triangle credits in the packet form

The packet representation (112) keeps all signs before taking a deficit. Two further identities exhibit credits within that complete form. They do not supply their own aggregate growth estimate.

Reflection credit. In logarithmic coordinates let

$$f(s) = e^{-s/2} M(e^s) \mathbf{1}_{[0, \log L]}(s), \quad E = \|f\|_{L^2(\mathbb{R})}^2,$$

and define the unitary reflection $(R_m f)(s) = f(\log(N/m) - s)$. Put $w_m = m^{-1/2}$. For any two multipliers m, n , their reflections compose to a translation by $\delta = \log(n/m)$, up to the harmless convention for the sign of a translation. Weighted completion of a square gives

$$\begin{aligned} w_m \langle f, R_m f \rangle + w_n \langle f, R_n f \rangle &\geq - (w_m + w_n) E \\ &\quad + \frac{w_m w_n}{2(w_m + w_n)} \|f - T_\delta f\|^2 \\ &\geq - (w_m + w_n) E + \frac{\min(w_m, w_n)}{4} \|f - T_\delta f\|^2. \end{aligned} \tag{R18}$$

For a direct proof set $A = (I + R_m)f$ and $B = R_m(I + R_n)f$. Then $A - B = f - R_m R_n f$ and $w_m \|A\|^2 + w_n \|B\|^2 \geq w_m w_n \|A - B\|^2 / (w_m + w_n)$. Expanding both squared norms proves the first inequality, and the second follows from elementary algebra.

If $\delta \neq 0$, finite support supplies a strictly positive displacement cost. With $k = \lceil \log L / |\delta| \rceil + 1$, the supports of f and $T_{k\delta} f$ are disjoint. Telescoping and Cauchy-Schwarz give

$$2E = \|f - T_{k\delta} f\|^2 \leq k^2 \|f - T_\delta f\|^2. \tag{R19}$$

These are valid whole-space credits. Restricting each channel to its active interval changes the relevant energies from one common E to channel-dependent quantities. Differences of those active energies are boundary terms that must be retained. Equations (R18)-(R19) do not justify spending a whole-space credit while dropping that boundary cost, nor do they overcome the sum of the negative baseline terms automatically.

Triangle credit. For three packet signs and nonnegative edge weights a, b, c ,

$$a\sigma_1\sigma_2 + b\sigma_2\sigma_3 + c\sigma_1\sigma_3 \geq -(a + b + c) + 2 \min(a, b, c). \tag{R20}$$

Indeed, the product of the three sign products is 1. Either all three are positive, or exactly two are negative, leaving at least the smallest weight positive. Thus three pairwise adverse signs cannot occur in one triangle.

More generally choose a nonnegative allocation τ_T to each selected packet triangle, such that the sum of allocations on each edge is at most its actual W_{ij} . Apply (R20) to equal weights τ_T within each triangle and retain the residual edges. This proves

$$\boxed{\mathcal{S}_X \geq \sum_i W_{ii} - 2 \sum_{i < j} W_{ij} + 4 \sum_T \tau_T.} \tag{R21}$$

One can preserve the actual signs of unallocated edges for a sharper bound. The edge-capacity condition prevents using the same crossing credit more than once. Neither arbitrary packet geometry nor (R21) yet bounds the unmatched adverse crossing weight at the rate (96). The geometric counterexample in section 34 remains relevant: authentic multiplicative restrictions must enter the estimate where a general input can concentrate on opposite reciprocal pulses.

40. Short positive excursions and an adapted fourth-power grid

An averaged closing estimate can allow large individual values when their total positive area is controlled. The regularity already proved in section 31 pays an entire class of such areas.

Let $I = (a, b)$ be a complete connected component of $\{x : U(x) > 0\}$ contained in $[X, 2X]$, and write $h = b - a$. Continuity gives $U(a) = U(b) = 0$. Apply (117) with these endpoints to obtain

$$0 \leq \int_I U(x) dx \leq \frac{5h^3 + 2h^2}{X}. \quad (\text{R22})$$

For any disjoint family of these excursions with $h \leq W$, their widths sum to at most X . Hence

$$\sum_{I: |I| \leq W} \int_I U(x) dx \leq 5W^2 + 2W. \quad (\text{R23})$$

Choosing $W = X^{3/4}$ gives a collective $O(X^{3/2})$ allowance. This is a sum over the entire family, so there is no further block-count multiplier.

Let $\mathcal{L}_X$ be the union of the portions in $[X, 2X]$ of the remaining positive components: complete components of width greater than $X^{3/4}$ and those crossing a shell boundary. There are at most $X^{1/4}$ complete long components and at most two boundary components. From (125)'s preceding comparison $\mathcal{D}_n \leq U(n)_+ + O(\log n)$ and the 3-Lipschitz property of U , integral-to-integer comparison yields

$$\sum_{X \leq n < 2X} \mathcal{D}_n \leq CX^{3/2} + \int_{\mathcal{L}_X} U(x)_+ dx. \quad (\text{R24})$$

The errors $O(X \log X)$ and $O(X)$ are included in the first term. Thus a sufficient remaining input in this version is $\int_{\mathcal{L}_X} U_+ \ll_\varepsilon X^{3/2+\varepsilon}$ for every $\varepsilon > 0$. Sum dyadic shells and use section 32. These excursions of U are a different indexed family from first-return blocks of M or prime blocks; their widths must not be identified.

Adapted sparse observations. The interpolation proof of section 31 works equally for $X_j = 2j^4$. Its gaps are $O(j^3)$ and its interpolation error is $O(j^2)$. At these starts,

$$N = 4j^4 = L^2, \quad L = 2j^2, \quad Y = L, \quad \mathcal{B}_{X_j} = 0. \quad (\text{R25})$$

Then $T_L = 0$ and $\Theta(1) = 0$ in (R4). We have the exact identities

$$\mathcal{S}_{X_j} = Q_L(L^2) - 2(\log 2)M(L), \quad U(X_j) = 2(\log 2)M(L) - Q_L(L^2). \quad (\text{R26})$$

The terminal carry has absolute size at most $2(\log 2)L$, already at square-root scale. Therefore a lower bound $Q_{2j^2}(4j^4) \geq -C_\varepsilon j^{2+\varepsilon}$ for every j and every positive ε suffices after interpolation. It is an alternative to the averaged long-excursion input. Vanishing of the reconstruction boundary at these starts does not assert positivity of Q_L .

The known fixed-logarithmic estimates give dense cumulative allowances of size $O_B(T^2/(\log T)^B)$, and the corresponding weighted fourth-power allowance of size $O_B(J^8/(\log J)^B)$. They remain larger in power than the sufficient targets $T^{3/2+\varepsilon}$ and $J^{6+\varepsilon}$. The short-excursion payment above does not supply the missing estimate on the remaining areas.

41. Exact operator transfer of the archived block quantities

The packet estimates in section 35 have an operator formulation that shows precisely how the existing energy enters the present signed expression.

Work in $\mathcal{H} = L^2((0, \infty), du/u)$. Let P_L restrict to $[1, L]$, and let $(J_c f)(u) = f(c/u)$; J_c is a unitary self-adjoint reflection. Define

$$\mathsf{T}_{N,L} = \sqrt{N} \sum_{3 \leq m \leq N, m \text{ odd}} \frac{1}{\sqrt{m}} P_L J_{N/m} P_L. \quad (\text{R27})$$

This is a finite self-adjoint operator. For the cumulative packets q_i of section 30 put $f_i(u) = q_i(u)/\sqrt{u}$ and $f = \sum_i \sigma_i f_i$. Then direct substitution gives

$$\boxed{W_{ij} = \langle f_i, \mathsf{T}_{N,L} f_j \rangle, \quad \mathcal{S}_X = \langle f, \mathsf{T}_{N,L} f \rangle, \quad \|f_i\|^2 = E_i = H_i b_i - V_i.} \quad (\text{R28})$$

The disjoint packet supports imply $\|f\|^2 = \sum_i E_i = \int_1^L M(u)^2 u^{-2} du =: I_M(L)$. Orthogonality here refers to the ordinary Hilbert norm. It does not diagonalize $\mathsf{T}_{N,L}$.

The elementary norm bounds are

$$\|\mathsf{T}_{N,L}\| \leq 2N, \quad \|P_{[a,L]} \mathsf{T}_{N,L} P_{[a,L]}\| \leq \frac{2N}{a} \quad (1 \leq a \leq L). \quad (\text{R29})$$

Indeed $\sum_{m \leq N} m^{-1/2} \leq 2\sqrt{N}$. In the second restriction, a nonzero reciprocal incidence has both $u, v \geq a$, hence $m = N/(uv) \leq N/a^2$; apply the same sum bound. These estimates are valid for arbitrary lengths but do not themselves have the closing scale.

For a finite explicit comparison, omit any zero-energy packets, and form the symmetric matrix with zero diagonal

$$A_{ij} = \begin{cases} \min \left\{ \frac{H_i H_j \mathcal{L}_{ij}}{\sqrt{E_i E_j}}, \sqrt{N} \sqrt{\mathcal{Q}_{ij} \mathcal{Q}_{ji}} \right\}, & \sigma_i \neq \sigma_j, \\ 0, & \sigma_i = \sigma_j. \end{cases} \quad (\text{R30})$$

Its entries are nonnegative. With $z_i = \sqrt{E_i}$, (132) gives $\mathcal{N}_X \leq z^\top A z \leq \rho(A) I_M(L)$, where $\rho(A)$ is the spectral radius. Retaining the exact same-sign credit $\mathcal{P}_X$ from (128) yields

$$\boxed{(-\mathcal{S}_X)_+ \leq [\rho(A) I_M(L) - \mathcal{P}_X]_+}. \quad (\text{R31})$$

This imports the actual height, charge, and saving data into a full crossing comparison. It neither proves a suitably small $\rho(A)$ nor permits discarding the credit $\mathcal{P}_X$ without loss.

If P is an orthogonal projection motivated by an archived decomposition, its use in this operator requires the full identity

$$\begin{aligned} \langle f, \mathsf{T} f \rangle &= \langle P f, \mathsf{T} P f \rangle + 2 \operatorname{Re} \langle P f, \mathsf{T} (I - P) f \rangle \\ &\quad + \langle (I - P) f, \mathsf{T} (I - P) f \rangle. \end{aligned} \quad (\text{R32})$$

The ordinary orthogonality $P(I - P) = 0$ does not imply $P \mathsf{T} (I - P) = 0$. Thus a projection imported from a historical-energy or cofactor estimate must either preserve this operator or explicitly pay its mixed term. This is the exact compatibility requirement, not an additional demand after a sufficient direct deficit estimate.

Finally, preserve the terminal dictionary. The artificially closed final packet has charge $\sigma_i b_i = h_{\text{open}} - M(L)/L$ and energy E_i , while the physical unfinished block includes $M(L)^2/L$ in E_{open} . The existing estimates for long intervals, retained histories, and unfinished terminals remain available with their

original definitions. One may use $E_i \leq E_{\text{open}}$, but one may not equate the two sets of height and charge data.

Scope of the first six-route continuation. Sections 36-41 establish the raw-coefficient conversion, the four-fifths rectangle allowance, the additional absolute common-factor removal, exact sign credits, a collective short-excursion allowance, an adapted sparse grid, and the operator transfer. Each offers information about the same sufficient arithmetic input. The remaining all-scale rate in (R17), or one of the established sparse or averaged alternatives, has not been proved by these reductions.

42. Exact raw identity and a complete terminal-strip allowance

This continuation records the second round of parallel analytical work. Its estimates concern the same arithmetic observable and the same closing implication as sections 25-26. They do not add another condition after that implication.

Throughout sections 42-51, $N = 2X$ is an integer, $L = \lfloor \sqrt{N} \rfloor$, and

$$\Theta(z) = \sum_{\substack{m \leq z \\ m \text{ odd}}} \log(z/m), \quad Q_J(N) = \sum_{a, b \leq J} \mu(a)\mu(b)\Theta(N/(ab)).$$

Set $\Theta(z) = 0$ for $z < 1$. The symbol e continues to denote the odd-counting remainder, not the prime discrepancy $\psi(x) - x$:

$$e(z) = \Theta(z) - z/2 + \frac{\log 2}{2}.$$

For $z \geq 1$, $|e(z)| \leq 1/(4z)$.

Exact hyperbola identity. For $L \geq 2$,

$$\boxed{U(X) = 2(\log 2)M(L) - Q_L(N)}. \quad (\text{S1})$$

Indeed, the Dirichlet convolution identity

$$\mu * \mathbf{1}_{\text{odd}} = \delta_1 - \delta_2$$

gives, for $z \geq 1$,

$$\sum_{b \leq z} \mu(b)\Theta(z/b) = \log z - \log_+(z/2).$$

The full double sum over $ab \leq N$ is $\mathcal{A}(N) - \mathcal{A}(N/2) = U(N/2)$, where $\mathcal{A}(x) = \int_1^x M(t) dt/t$ and $\mathcal{A} = 0$ below 1. No pair with both $a, b > L$ has product at most N . Splitting at L therefore gives the full sum as $2 \sum_{a \leq L} \mu(a) \log 2 - Q_L(N)$, since $N/a \geq 2$. This proves (S1) for nonsquare as well as square N .

Terminal-strip theorem. Suppose $L \geq 8$ and H is an integer with $1 \leq H \leq L/4$. Put $K = L - H$. Then

$$\boxed{|U(X) - \{2(\log 2)M(K) - Q_K(N)\}| \leq \frac{6H^3}{L}}. \quad (\text{S2})$$

The corresponding raw difference satisfies

$$|Q_L(N) - Q_K(N)| \leq 2H \log 2 + \frac{6H^3}{L}. \quad (\text{S3})$$

Here is a proof retaining the intersection of the removed rows and columns. Let $I = \{K + 1, \dots, L\}$ and

$$F_a = \sum_{b \leq L} \mu(b)\Theta(N/(ab)).$$

For $a \in I$, all omitted terms with $b > L$ have only the $m = 1$ logarithm. Hence

$$F_a = \log 2 - T_a, \quad T_a = \int_L^{N/a} \frac{M(t) - M(L)}{t} dt.$$

Write $k = L - a$ and $r_a = N/a - L$. Since $N - L^2 \leq 2L$ and $L \geq 4H$,

$$r_a \leq \frac{L(k+2)}{L-k} \leq \frac{4H}{3H+1}(k+2).$$

The exact finite sum $\sum_{k=0}^{H-1} (k+2)^2 = H(2H^2 + 9H + 13)/6$ implies $\sum_{a \in I} r_a^2 \leq 4H^3$. The unit-step inequality $|M(t) - M(L)| \leq t - L$ gives

$$|T_a| \leq r_a^2/(2L), \quad 2 \sum_{a \in I} |T_a| \leq 4H^3/L.$$

Every pair in I^2 satisfies $1 \leq N/(ab) < 3$, so only $m = 1$ is present. Put $C = \sum_{a \in I} \mu(a)$ and $D = \sum_{a \in I} \mu(a) \log(L/a)$. The intersection contribution is

$$Q_{I,I} = C^2 \log(N/L^2) + 2CD.$$

Using $|C| \leq H$, $\log(N/L^2) \leq 2/L$, and $|D| \leq 2H(H-1)/(3L)$, one obtains

$$|Q_{I,I}| \leq 2H^2/L + 4H^2(H-1)/(3L) \leq 2H^3/L.$$

Finally,

$$Q_L - Q_K = 2(\log 2)C - 2 \sum_{a \in I} \mu(a)T_a - Q_{I,I}.$$

This proves (S3). Substitution into (S1) cancels the linear term $2(\log 2)C$ and proves (S2). The constant 6 is the independently checked refinement of the earlier constant 12.

The exact remainder in (S2) is

$$\boxed{G_K(N) = \sum_{\substack{a,b > K \\ ab \leq N}} \mu(a)\mu(b) \log \frac{N}{ab}, \quad U(X) = 2(\log 2)M(K) - Q_K(N) + G_K(N).} \quad (\text{S4})$$

The reason this remainder contains only $m = 1$ is $N/(K+1)^2 < 3$.

Taking

$$H = \min\{\lfloor L^{2/3} \rfloor, \lfloor L/4 \rfloor\}$$

makes the complete strip cost $O(L)$. It includes all interactions of the removed indices with the retained indices. Repeated stripping does not make the cost disappear: the estimate depends on the total width removed.

43. Keep the center attached after every removal

With the reduced cutoff $K = L - H$, the raw form has the exact decomposition

$$Q_K(N) = \frac{N}{2}m_K^2 - \frac{\log 2}{2}M(K)^2 + \sum_{a,b \leq K} \mu(a)\mu(b)e(N/(ab)), \quad m_K = \sum_{a \leq K} \frac{\mu(a)}{a}. \quad (\text{S5})$$

Both terms of the center are compulsory. In particular, the negative $-(\log 2)M(K)^2/2$ term is not an error that has already been paid.

The correct order of the combined reduction is:

1. Remove the complete raw terminal strip using (S2).
2. Apply the oscillatory estimate of section 37 to complete dyadic rectangles in the remaining e -sum.
3. Apply the absolute common-factor allowance of section 38 only to pairs outside those rectangles.

Let $\mathcal{C}_{N,K}$ be the remaining pairs. Using the same dyadic base convention as section 37, they lie outside every removed rectangle with $AB \leq N^{4/5}$. They also lie outside

$$\min(r, s)^2 \leq (1 + \log N)^2 \max(r, s), \quad a = gr, \quad b = gs, \quad (r, s) = 1.$$

Consequently,

$$Q_K(N) = \frac{N}{2}m_K^2 - \frac{\log 2}{2}M(K)^2 + \sum_{(a,b) \in \mathcal{C}_{N,K}} \mu(a)\mu(b)e(N/(ab)) + O(\sqrt{N}(1 + \log N)^3). \quad (\text{S6})$$

All remaining indices satisfy $a, b > N^{3/10}$ and $a, b \leq K$. The surviving product set is defined by complete dyadic rectangles; one must not replace their proved estimate by an estimate for arbitrary masked subsets.

The direct sufficient target is a lower bound of order $-N^{1/2+\varepsilon}$ for the complete center plus the remaining signed sum in (S6), for every $\varepsilon > 0$. The known prefix term $2(\log 2)M(K)$ costs $O(\sqrt{N})$. Thus (S2) and (S6) feed directly into the existing one-sided closing argument.

Exact local-energy insertion. The terminal-strip proof can retain the actual two-window interaction rather than the unit-step majorant. Define

$$\begin{aligned} V_-(u) &= M(u) - M(K), \quad V_+(t) = M(t) - M(L), \\ Y &= N/L, \quad Z = N/K, \\ E &= \sum_{K < a \leq L} \mu(a) \int_L^{N/a} V_+(t) \frac{dt}{t}. \end{aligned}$$

Abel summation gives

$$E = V_-(L) \int_L^Y V_+(t) \frac{dt}{t} + \int_K^L V_-(u) V_+(N/u) \frac{du}{u}. \quad (\text{S7})$$

Therefore

$$|E| \leq \frac{2H}{L} + \sqrt{N} \left(\int_K^L \frac{V_-(u)^2}{u^2} du \right)^{1/2} \left(\int_Y^Z \frac{V_+(v)^2}{v^2} dv \right)^{1/2}. \quad (\text{S8})$$

The opening levels, reciprocal denominators, and reflected cutoffs are exact. The unit-step bounds recover the cubic-over- L allowance. An improvement using older block results must apply to these actual windows and inherited levels, or directly to their signed interaction.

44. A concrete historical projection pays its mixed remainder

The previous operator formulation identified a transport issue: orthogonality in the historical-energy norm does not eliminate a mixed contribution to the reciprocal kernel. Here the relevant short-prefix projection is instantiated and its actual mixed remainder is bounded.

Let $a = \lfloor \sqrt{L} \rfloor$, $L \geq 4$, and

$$f(u) = M(u)u^{-1/2}\mathbf{1}_{[1,L)}(u), \quad Pf = \mathbf{1}_{[1,a)}f$$

in $L^2(du/u)$. Its endpoint-adjusted coefficient vector is

$$\nu_n^{\text{lo}} = \begin{cases} \mu(n), & n < a, \\ -M(a-1), & n = a, \\ 0, & n > a. \end{cases}$$

With ν as in section 23, the complementary vector is

$$\nu_n^{\text{hi}} = \begin{cases} 0, & n < a, \\ M(a), & n = a, \\ \mu(n), & a < n < L, \\ -M(L-1), & n = L. \end{cases}$$

Both vectors have total zero, and $\nu = \nu^{\text{lo}} + \nu^{\text{hi}}$. The high part includes the inherited atom $M(a)$.

For

$$\mathcal{R}(v, w) = \sum_{r,s \leq L} v_r w_s e(N/(rs)),$$

the pointwise remainder bound implies

$$|\mathcal{R}(v, w)| \leq \frac{1}{4N} \left(\sum_r r |v_r| \right) \left(\sum_s s |w_s| \right).$$

The authentic projected coefficients satisfy

$$A := \sum_r r |\nu_r^{\text{lo}}| \leq \frac{3}{2}a^2, \quad B := \sum_s s |\nu_s^{\text{hi}}| \leq \frac{3L^2 + a^2}{2}.$$

It follows that

$$\begin{aligned} & |\mathcal{R}(\nu^{\text{lo}}, \nu^{\text{lo}}) + 2\mathcal{R}(\nu^{\text{lo}}, \nu^{\text{hi}})| \\ & \leq \frac{A^2 + 2AB}{4N} \leq \frac{9a^2L^2}{8N} + \frac{15a^4}{16N} \leq \frac{9}{8}L + \frac{15}{16}. \end{aligned} \tag{S9}$$

Here $a^2 \leq L$ and $N \geq L^2$.

Keeping the entire continuous square gives

$$\mathcal{S}_X = Xb_L'^2 + \mathcal{R}(\nu^{\text{hi}}, \nu^{\text{hi}}) + \mathcal{E}_X, \quad |\mathcal{E}_X| \leq 9L/8 + 15/16. \tag{S10}$$

Its complete mixed term is retained inside $Xb_L'^2$; only the remainder is split. If

$$\mathcal{D}_X^{\text{hi}} = [-Xb_L'^2 - \mathcal{R}(\nu^{\text{hi}}, \nu^{\text{hi}})]_+,$$

the original deficit and this deficit differ by $O(\sqrt{X})$, including the prior $O(\log X)$ small-product correction.

This is a proved map from historical coefficients into the current form. It is not a claim that $PT(I - P) = 0$. It also does not remove a larger region than the product cutoff already recorded; its value is the explicit accounting of the inherited atom and cross term.

Within the packet representation, a retained one-sign lower stack has nonnegative self-contribution W_{ii} . Bounding the entire stack separately can therefore be stronger than necessary for this direct route. Its adverse interactions with opposite-sign packets are the contributions that need comparison.

45. Squarefree lifts and their complete inherited correction

For $R \geq 1$, define

$$K_R(y) = \begin{cases} \int_{\max(1, y/R)}^{\min(R, y)} M(u)M(y/u) \frac{du}{u}, & 1 \leq y \leq R^2, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{S11})$$

An empty interval contributes zero; reversed integration limits are never used. Set $M(x) = 0$ for $x < 1$.

Let $r = \sqrt{N}$. The current quantity is

$$\mathcal{S}_X = \sum_{\substack{m \geq 3 \\ m \text{ odd}}} K_L(N/m).$$

Its $m = 1$ term is zero since $N \geq L^2$. Define a balanced squarefree child by

$$Q_{\text{sf}}(Y) = \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \mu(d)^2 K_{\sqrt{Y}}(Y/d). \quad (\text{S12})$$

The $d = 1$ term of this balanced child is also zero.

Put

$$A_2(v) = \sum_{\substack{h \leq v \\ h \text{ odd}}} \mu(h), \quad G(z) = A_2(\sqrt{z}) - A_2(\sqrt{z/2}).$$

The exact squarefree convolution is

$$\boxed{\sum_{\substack{d \leq z \\ d \text{ odd}}} \mu(d)^2 M(z/d) = G(z).} \quad (\text{S13})$$

It follows by checking prime powers, or from the Dirichlet series

$$\left(\sum_{d \text{ odd}} \frac{\mu(d)^2}{d^s} \right) \frac{1}{\zeta(s)} = (1 - 2^{-s}) \sum_{h \text{ odd}} \frac{\mu(h)}{h^{2s}}, \quad \Re s > 1.$$

The surviving coefficients occur at odd squares and twice odd squares.

Every odd integer has a unique representation $m = k^2 d$, where k is odd and d is odd squarefree. There is no coprimality condition between k and d . With these conventions,

$$\boxed{\mathcal{S}_X = \sum_{\substack{k \leq r \\ k \text{ odd}}} Q_{\text{sf}}(N/k^2) + B_N + D_N,} \quad (\text{S14})$$

where

$$\begin{aligned} B_N &= 2 \sum_{\substack{k \leq r \\ k \text{ odd}}} \int_{r/k}^r \frac{M(u)}{u} G\left(\frac{N}{k^2 u}\right) du, \\ D_N &= 2 \int_L^r \frac{M(u)M(N/u)}{u} du, \quad |D_N| < 2r. \end{aligned} \quad (\text{S15})$$

To verify (S14), first use cutoff r . For $Y = N/k^2$ and $y = Y/d$, the difference between the cutoff r and the balanced cutoff $\sqrt{Y} = r/k$ consists of two reciprocal arms. Both coordinates cannot exceed

$\sqrt{Y}$ because $y \leq Y$. Thus

$$K_r(y) - K_{\sqrt{Y}}(y) = 2 \int_{\sqrt{Y}}^r M(u)M(y/u) \frac{du}{u}.$$

Summing the squarefree d channels and using (S13) gives B_N .

For the change from r to L , the full odd-ancestor identity gives $\sum_{m \geq 3, m \text{ odd}} M(z/m) = -M(z)$ for $z \geq 2$. This produces the plus sign of D_N in (S14). On $L \leq u < r < L + 1$, $M(u) = M(L)$ and $|M(L)| \leq L$, so

$$|D_N| \leq 2LN \int_L^r u^{-2} du = 2r(r - L) < 2r.$$

In an unbalanced child, $d = 1$ can contribute: it accounts for original square multipliers $m = k^2$. Those terms belong to B_N and have not been discarded.

46. Sum the square lifts before applying an energy bound

The complete correction in (S15) satisfies

$$|B_N| \leq 10\sqrt{N} \sqrt{I_M(\sqrt{N}) \log \sqrt{N}}. \quad (\text{S16})$$

This estimate uses the signed odd Möbius harmonic sum before Cauchy-Schwarz.

Fix $1 \leq u \leq r$ and write

$$t = r/\sqrt{u}, \quad a = r/u, \quad b = \sqrt{u}.$$

Reversing the order of the square-lift sum and integral leaves

$$H(u) = \sum_{\substack{k > a \\ k \text{ odd}}} [A_2(t/k) - A_2(t/(\sqrt{2}k))].$$

The upper restriction $k \leq r$ is redundant: a nonzero summand requires $k \leq t \leq r$. Expanding the odd prefix and counting odd k gives

$$\begin{aligned} H(u) &= \frac{t}{2} [m_o(b) - 2^{-1/2} m_o(b/\sqrt{2})] \\ &\quad - \frac{a}{2} [A_2(b) - A_2(b/\sqrt{2})] + E(u), \quad |E(u)| \leq b, \end{aligned} \quad (\text{S17})$$

where $m_o(v) = \sum_{h \leq v, h \text{ odd}} \mu(h)/h$, and $m_o = 0$ below 1.

For completeness, the elementary identity

$$\sum_{\substack{h \leq v \\ h \text{ odd}}} \mu(h) \left\lfloor \frac{v/h + 1}{2} \right\rfloor = 1 \quad (v \geq 1)$$

and the error bound $|\lfloor (y+1)/2 \rfloor - y/2| \leq 1/2$ imply

$$|m_o(v)| \leq \frac{2}{v} + \frac{\#\{h \leq v : h \text{ odd}\}}{v} \leq 3. \quad (\text{S18})$$

In (S17), $ab = t$, $b \leq t$, and $|A_2(b) - A_2(b/\sqrt{2})| \leq b$. Hence

$$|H(u)| \leq \left(3 + \frac{3}{2\sqrt{2}}\right) t < 5t.$$

The complete inherited correction consequently satisfies

$$|B_N| \leq 10r \int_1^r \frac{|M(u)|}{u^{3/2}} du \leq 10r \left(\int_1^r \frac{M(u)^2}{u^2} du \right)^{1/2} \sqrt{\log r},$$

which proves (S16).

Since M is constant between L and r ,

$$I_M(r) \leq I_M(L) + 1.$$

Thus (S14) gives the explicit lower estimate

$$\begin{aligned} \mathcal{S}_X &\geq \sum_{\substack{k \leq \sqrt{N} \\ k \text{ odd}}} Q_{\text{sf}}(N/k^2) - 2\sqrt{N} \\ &\quad - 10\sqrt{N} \sqrt{(I_M(L) + 1) \log \sqrt{N}}. \end{aligned} \quad (\text{S19})$$

The gain is square-root dependence on the complete historical energy, with all square lifts combined. With only the elementary bound $I_M(L) \leq L$, the correction is $O(N^{3/4}\sqrt{\log N})$. The child sum is still signed. No positivity or contracting recurrence for that child sum has been established by this decomposition.

47. Long positive intervals have a finite-capacity transfer

The averaged route in section 40 pays the complete short positive intervals of U . The remaining long intervals can be combined without repeatedly assigning the same ancestral contribution.

Let $\mathcal{L}$ be disjoint intervals $I = (a, b)$ inside $[X, 2X]$ on which U is positive, and let their total width be H . Boundary-crossing pieces can be included after clipping to this shell. Define

$$w_I(t) = |I \cap [t/2, t]|, \quad k(t) = \frac{1}{t} \sum_{I \in \mathcal{L}} w_I(t).$$

Fubini's theorem gives the exact signed transfer

$$\sum_{I \in \mathcal{L}} \int_I U(x) dx = \int_X^{4X} M(t) k(t) dt. \quad (\text{S20})$$

Because the intervals are disjoint,

$$0 \leq k(t) \leq \min(1/2, H/X), \quad \int_X^{4X} k(t) dt = H \log 2. \quad (\text{S21})$$

Thus the same bounded coefficient multiplies both signs:

$$\int M(t) k(t) dt = \int M_+(t) k(t) dt - \int M_-(t) k(t) dt.$$

There is no separate dilation multiplicity to append.

For one interval, put $h = b - a$. Since $b \leq 2a$,

$$w_I(t) = \begin{cases} t - a, & a \leq t \leq b, \\ h, & b \leq t \leq 2a, \\ b - t/2, & 2a \leq t \leq 2b, \\ 0, & \text{otherwise.} \end{cases}$$

Direct integration yields the exact calibration

$$\int w_I(t)^2 dt = ah^2.$$

Therefore

$$\left| \int_I U(x) dx \right| \leq h\sqrt{a} \left(\int_a^{2b} \frac{M(t)^2}{t^2} dt \right)^{1/2}.$$

Minkowski's inequality for the combined w_I gives

$$\sum_{I \in \mathcal{L}} \int_I U(x) dx \leq H\sqrt{2X} \sqrt{I_M(4X) - I_M(X)}. \quad (\text{S22})$$

The full common energy is retained before the final estimate.

An archived positive-level family with total occupation P contributes at most $\min(1/2, H/X)P$. For the height/lifetime range previously retained in the occupation analysis, its recorded estimate

$$P \ll X^{3/2+\delta} / (\log X)^{1/200}$$

therefore suffices at scale $X^{3/2}$ whenever the total remaining width is $H \ll X^{1-\delta}$. This is an application of the earlier occupation allowance, not a new proof of that width bound.

Counting the number of long intervals does not bound their total width strongly enough: even a small number can occupy much of the shell. The count, width, and signed area are distinct quantities.

There is also a complete signed reserve at a weaker weight. The imported unconditional logarithmic estimates justify

$$\int_{1/2}^{\infty} U(x)x^{-2} dx = 0.$$

Since $U(x) = \log(2x)$ for $1/2 \leq x \leq 1$,

$$\boxed{\int_1^{\infty} U(x)x^{-2} dx = \log 2 - 1.} \tag{S23}$$

This identity does not justify replacing x^{-2} by the stronger closing weight $x^{-3/2-\varepsilon}$.

48. The complete signed average and its positive first channel

For real $L \geq 1$, write $Q(L) = Q_L(L^2)$. Its exact Mellin transform is

$$\boxed{\int_1^\infty Q(L)L^{-s-1} dL = \frac{2 \log 2}{s} \left(\frac{1}{\zeta(s)} - 1 \right) + \frac{2(1 - 2^{-s/2})}{s^2} \left(2 - \frac{1}{\zeta(s/2)} \right), \quad \Re s > 2.} \quad (\text{S24})$$

No cutoff terms are omitted.

For $L \geq \sqrt{2}$, (S1) becomes

$$Q(L) = 2(\log 2)M(L) - U(L^2/2).$$

For $1 \leq L < \sqrt{2}$, the correction to the right side is $4 \log L - 2 \log 2$. Its Mellin integral is

$$\frac{4(1 - 2^{-s/2})}{s^2} - \frac{2 \log 2}{s}.$$

Combining this with $\int_1^\infty M(L)L^{-s-1} dL = 1/(s\zeta(s))$ and

$$\int_0^\infty U(x)x^{-z-1} dx = \frac{2^z - 1}{z^2 \zeta(z)}, \quad \Re z > 1,$$

proves (S24).

The $m = 1$ theta channel has a definite signed average:

$$Q^{(1)}(L) = \sum_{a,b \leq L} \mu(a)\mu(b) \log \frac{L^2}{ab} = 2M(L)\mathcal{A}(L) = L \frac{d}{dL} \mathcal{A}(L)^2$$

almost everywhere. Integration by parts, with vanishing boundary terms for real $\sigma > 2$, gives

$$\boxed{\int_1^\infty Q^{(1)}(L)L^{-\sigma-1} dL = \sigma \int_1^\infty \mathcal{A}(L)^2 L^{-\sigma-1} dL \geq 0.} \quad (\text{S25})$$

This accounts for the whole averaged channel rather than discarding its individual signed terms.

The full signed average has the positive Abel limit

$$\boxed{\lim_{\sigma \downarrow 2} \int_1^\infty Q(L)L^{-\sigma-1} dL = \frac{1}{2} + \log 2 \left(\frac{1}{\zeta(2)} - 1 \right) > 0.} \quad (\text{S26})$$

This is a limit of the integrals already established for $\sigma > 2$. It does not assert that an undamped integral can be obtained by exchanging the limit and integration.

The pole structure explains the remaining limitation. If $\zeta(\rho) = 0$ with $\Re \rho > 1/2$, the meromorphic expression in (S24) has an uncanceled pole at $s = 2\rho$. The coefficient $1 - 2^{-\rho}$ is nonzero, and $1/\zeta(s)$ is analytic at 2ρ because $\Re(2\rho) > 1$. An analytic continuation of the expression alone does not prove convergence or a favorable sign of its defining integral in that larger region. A one-sided bound or a negative-part budget would supply the nonnegative function used in section 26.

49. Complete multiplicative dependencies and cutoff defects

The common-factor analysis can retain the full arithmetic dependencies before any absolute-value estimate. Let $a = gr$, $b = gs$, $(r, s) = 1$. Squarefreeness implies that a prime is assigned to at most one of g, r, s . The complete local choices give

$$\mathcal{F}(w; s, t) = \prod_p (1 + p^{-w} - p^{-s} - p^{-t})$$

in its domain of absolute convergence. On the coupled plane $w = s + t$,

$$\boxed{\mathcal{F}(s + t; s, t) = \frac{1}{\zeta(s)\zeta(t)}}. \quad (\text{S27})$$

This records a real cancellation lost by independently bounding each common-factor contribution. Finite numerical cutoffs still require their boundary terms.

For the squarefree density coefficients, put $\kappa_p = p/(p+1)$ and $\kappa(r) = \prod_{p|r} \kappa_p$. The corresponding product is

$$\begin{aligned} D(s, t) &= \frac{6}{\pi^2} \prod_p [1 - \kappa_p(p^{-s} + p^{-t})] \\ &= \frac{H(s, t)}{\zeta(s)\zeta(t)\zeta(s+t)}. \end{aligned} \quad (\text{S28})$$

The product for H converges absolutely in a neighborhood of $\Re s = \Re t = 1/2$: after division by $(1 - p^{-s})(1 - p^{-t})(1 - p^{-s-t})$, the local difference from 1 is bounded by a constant times

$$p^{-\Re s - 1} + p^{-\Re t - 1} + p^{-2\Re s - \Re t} + p^{-\Re s - 2\Re t}.$$

The identity first holds where the original products converge absolutely, then gives their meromorphic continuation. It is not a finite-cutoff growth estimate.

Why complete color positivity is insufficient by itself. For a complete finite set of prime choices, the conjugate spectral product is

$$\prod_{p \in P} \left(1 - \frac{2\sqrt{p}}{p+1} \cos(\tau \log p) \right) = \prod_{p \in P} \frac{|1 - p^{-1/2-i\tau}|^2}{1 + p^{-1}} > 0.$$

A numerical cutoff need not preserve those complete choices. At $L = 3$ and $\tau = 0$, the authentic truncated polynomial equals

$$\sum_{\substack{r, s \leq 3 \\ (r, s) = 1}} \frac{\mu(r)\mu(s)\kappa(r)\kappa(s)}{\sqrt{rs}} = 1 - \frac{4}{3\sqrt{2}} - \frac{3}{2\sqrt{3}} + \frac{1}{\sqrt{6}} < 0. \quad (\text{S29})$$

Completing the prime cube by adding 6 adds $1/\sqrt{6}$ and restores the positive product. Thus the missing completion terms need a bound; they cannot simply be assigned zero cost.

The ratio kernel also has a signed spectral weight. Define

$$J(q) = \int_q^\infty e(z) z^{-3/2} dz, \quad h(u) = \frac{1}{2} J(e^{|u|}).$$

For

$$E(s) = \int_1^\infty e(z) z^{-s-1} dz = \frac{(1 - 2^{-s})\zeta(s)}{s^2} - \frac{1}{2(s-1)} + \frac{\log 2}{2s},$$

with removable singularities interpreted by continuation, integration in logarithmic coordinates gives

$$\hat{h}(\tau) = -\frac{\Im E(1/2 + i\tau)}{\tau}, \quad (\text{S30})$$

with the value at zero defined by the limit. In particular,

$$J(1) = 4(1 - 2^{-1/2})\zeta(1/2) + 1 + \log 2 < 0. \quad (\text{S31})$$

One elementary verification writes $\zeta(1/2) = \eta(1/2)/(1 - \sqrt{2})$. The first six paired terms of the alternating eta series, plus the lower tail integral $\sqrt{14} - \sqrt{13}$, exceed 0.5996, whereas $\sqrt{2}(1 + \log 2)/4 < 0.5993$. The tail bound follows by integrating the positive decreasing function $(2x - 1)^{-1/2} - (2x)^{-1/2}$ from 7 to infinity. These strict rational-decimal comparisons can also be certified by squaring rational bounds for the finitely many radicals.

Because $h(0) < 0$, this integrable ratio kernel cannot have an everywhere nonnegative Fourier weight. The density main term therefore does not become positive merely by writing a positive complete prime-color product. Both the numerical cutoff and the signed kernel weight remain part of the original combined form.

Finally, in the centered ν coordinates the $m = 1$ term cancels exactly: $\sum \nu_a = 0$ makes its logarithmic quadratic form zero. The resulting kernel is supported on $ab \leq N/3$. Returning to raw μ coordinates restores

$$Q_L^{(1)}(N) = \log(N/L^2)M(L)^2 + 2M(L)\mathcal{A}(L).$$

That rank term must remain coupled. Removing $m = 1$ in centered coordinates does not separately pay the raw upper corner.

50. Further local tests: prime extraction and narrow self-blocks

The prime extraction test applies multiplicativity to the actual finite raw form. Fix a prime p . Writing every nonzero Möbius coefficient either at a p -free index or at p times such an index gives

$$Q_L(N) = \sum_{\substack{r,s \leq L \\ p \nmid rs}} \mu(r)\mu(s) \left[\Theta(N/(rs)) - \mathbf{1}_{pr \leq L} \Theta(N/(prs)) \right. \\ \left. - \mathbf{1}_{ps \leq L} \Theta(N/(prs)) + \mathbf{1}_{pr,ps \leq L} \Theta(N/(p^2rs)) \right]. \quad (\text{S32})$$

On the complete interior $r, s \leq L/p$, the bracket is a second finite difference of the convex function $t \mapsto \Theta(e^t)$. It is nonnegative and vanishes on its affine sectors. The multiplier $\mu(r)\mu(s)$ is still signed. On the outer corner $r, s > L/p$, the bracket retains its original single term. Therefore this extraction supplies an exact cancellation pattern, but no automatic power saving for the full numerical cutoff.

A separate local calculation bounds complete raw self-blocks. If $I = (a, a + h]$ is an integer interval, $1 \leq h \leq a$, $a + h \leq L$, and $c_n \in \mathbb{R}$ with $|c_n| \leq 1$, then

$$\sum_{r,s \in I} c_r c_s \Theta(N/(rs)) \geq -CN(h/a)^3 \quad (\text{S33})$$

for an absolute constant C . To see the scale, separate odd channels with threshold N/m above $(a + h)^2$ from those crossing the interval of products $[a^2, (a + h)^2]$.

For a fully active channel,

$$\log \frac{N/m}{rs} = \log \frac{N/m}{(a + h)^2} + \log \frac{a + h}{r} + \log \frac{a + h}{s}.$$

The first term gives a nonnegative square. The remaining cross term has magnitude $O(h^3/a)$, and there are $O(N/a^2)$ such channels. Crossing channels number $O(Nh/a^3 + 1)$; each costs at most $O(h^3/a)$. Since $h \leq a$ and $N \geq a^2$, these contributions are bounded by the scale in (S33).

A geometric partition with relative width of order $L^{-1/2}$, restricted to indices sufficiently above $\sqrt{L}$ so that integer widths are available, gives total negative self-block cost $O(L \log L)$. This estimate concerns complete raw self-blocks. It does not pay cross-cell interactions and cannot be added to an e -only decomposition without allocating the corresponding center exactly.

These two tests are preserved because they locate where the multiplicative structure and local smoothness help. They do not furnish the final estimate for the remaining centered interior.

51. Consolidated dictionary, closing inputs, and saved status

The original sections 1-35 are preserved. Sections 36-41 record the first unsaved six-route round, and sections 42-50 record the subsequent combined attack and its checked refinements. This section supplies the current reading order and status.

Symbol	Meaning in this continuation
μ, M	Ordinary Möbius function and its prefix sum
$I_M(T)$	Complete historical integral $\int_1^T M(t)^2 t^{-2} dt$
$\mathcal{A}, U$	Logarithmic prefix $\int_1^x M(t) dt/t$ and sliding mean $\mathcal{A}(2X) - \mathcal{A}(X)$
N, L	$N = 2X$ and $L = \lfloor \sqrt{N} \rfloor$
H, K in sections 42-43	Removed terminal width and retained cutoff $K = L - H$
Θ, e	Exact odd logarithmic count and its centered remainder
m_J	Raw harmonic sum $\sum_{a \leq J} \mu(a)/a$
b'_L	Endpoint-adjusted harmonic sum $m_L - M(L)/L$
ν	Zero-total endpoint-adjusted coefficients from section 23
$Q_J(N)$	Raw Möbius quadratic form with cutoff J
$\mathcal{S}_X$	Endpoint-adjusted reciprocal sum; $U = \mathcal{B}_X - \mathcal{S}_X$
$\mathcal{D}_X$	Negative part of the complete larger-product expression from section 25
$K_R(y)$	Reciprocal integral (S11), zero outside its support
Q_{sf}	Balanced squarefree child (S12), distinct from Q_J
G_K, B_N, D_N	Terminal-strip remainder, squarefree inherited correction, and real-to-integer cutoff correction
W_{ij}, E_i, b_i, V_i	Actual packet interaction, energy, reciprocal charge, and nonnegative saving

The same letter H denotes total interval width locally in section 47; it is not a packet height there. The counting remainder e is not the prime discrepancy. Record intervals, first-return packets, positive intervals of U , and Fibonacci prime blocks are different indexed families.

The combined direct target. With the cutoff and surviving set in section 43, a sufficient arithmetic statement is

$$\boxed{\begin{aligned} \frac{N}{2} m_K^2 - \frac{\log 2}{2} M(K)^2 + \sum_{(a,b) \in \mathcal{C}_{N,K}} \mu(a) \mu(b) e(N/(ab)) \\ \geq -C_\varepsilon N^{1/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \end{aligned}} \quad (\text{S34})$$

For each ε , the constant is independent of N ; finitely many small N may be absorbed into it. The complete strip, product rectangles, common-factor wedge, and prefix are already within that scale. Consequently (S34) implies the bound on U used in section 26.

The following are alternative sufficient inputs to the same closing deduction:

- The original all-scale deficit allowance in section 25.
- The equivalent raw centered lower estimate (S34).
- The fourth-power observations in section 31, including the adapted starts of section 40.
- The cumulative or weighted sparse deficit allowances in section 32.

- A sufficiently strong complete historical-energy estimate, transferred by section 33.

One need not prove every alternative. No further variational floor, Fourier-tail onset, or derivative-regularity premise is appended after one of these inputs has been proved.

What is established in this continuation. Exact raw hyperbola conversion; complete terminal-strip payment; the $N^{4/5}$ full-rectangle allowance; the common-factor wedge allowance; reflection and triangle credits with their boundary restrictions; short-interval aggregate payment; the actual historical projection and its mixed remainder; the squarefree-lift transfer and complete boundary estimate; finite-capacity long-interval transfer; and the exact signed Mellin average with its positive first channel.

What is not established. A bound of the strength (S34), or one of its sufficient averaged alternatives, for the remaining authentic signed interior. The squarefree child sum and the long-interval signed area are alternative representations or estimates toward that same problem, not additional proved closure theorems. The classical arbitrary-logarithmic savings recorded earlier do not supply a fixed power improvement.

Relation to existing research. The archived long/open-block and record allowances remain part of the argument with their original definitions. A targeted record check distinguished H_{record} (collective record debit), H_{terminal} (retained-record height), and H_{θ} (coupled theta positivity). A theorem assuming a full matrix-positivity premise for H_{θ} does not establish the retained-height estimate. The scoped audit did not recover a later proof of that retained arithmetic rate; it does not claim to exhaust every archive record.

Source and preservation record. This PDF was rebuilt from the saved version-6 source package and its 35-section manuscript. Both unsaved parallel rounds are now incorporated in the continuation. The source package includes the prior manuscript in its history directory, the two full recovered source notes already supplied with version 6, both analytical continuation files, and the structured two-round checkpoint. The companion remains distinct from the 12-page summary, the running proof manuscript, and the large research master. This save updates this companion and its rebuild package.

The analytical deductions are recorded with explicit hypotheses and endpoints. The final aggregate growth estimate remains an identified arithmetic premise, not a statement awaiting only a change in label.

52. Recovery of the unsaved continuation and its dictionary

This part incorporates the compression and fixed-cutoff positive allowance supplied in the conversation, together with the subsequent complete prime-discrepancy residual. It supersedes their historical “not saved” notices. Sections 1-51 are preserved. The estimate required at the end remains a one-sided estimate for the complete signed expression.

All logarithms are natural. Let

$$\mu(n) = \begin{cases} (-1)^r & n \text{ is a product of } r \text{ distinct primes,} \\ 0 & n \text{ has a squared prime factor,} \end{cases} \quad M(x) = \sum_{n \leq x} \mu(n), \quad \mathcal{A}(x) = \int_1^x \frac{M(t)}{t} dt.$$

Set $U(X) = \mathcal{A}(2X) - \mathcal{A}(X)$, and

$$\Theta(z) = \sum_{\substack{m \leq z \\ m \text{ modd}}} \log(z/m), \quad e(z) = \Theta(z) - z/2 + (\log 2)/2.$$

Here e is the odd-counting remainder, not the earlier prime-discrepancy function $\psi(x) - x$. For the cutoffs below, every argument of e is at least one. The earlier odd-counting estimate is $|e(z)| \leq 1/(4z)$.

For an integer $N = 2X$, let

$$L = \lfloor \sqrt{N} \rfloor, \quad 1 \leq H \leq L/4, \quad K = L - H, \quad m_K = \sum_{a \leq K} \frac{\mu(a)}{a},$$

and define

$$Q_K(N) = \sum_{a, b \leq K} \mu(a)\mu(b)\Theta\left(\frac{N}{ab}\right), \quad G_K(N) = \sum_{\substack{a, b > K \\ ab \leq N}} \mu(a)\mu(b) \log \frac{N}{ab}.$$

The already established reconstruction, including its cutoff tail, is

$$\boxed{U(N/2) = 2(\log 2)M(K) - Q_K(N) + G_K(N), \quad |G_K(N)| \leq 6H^3/L.} \quad (\text{C1})$$

For large L , choose $H = \lfloor L^{2/3} \rfloor$; finitely many smaller cutoffs can use any admissible H , and are absorbed in constants. Then $G_K = O(\sqrt{N})$. The full center is

$$Q_K(N) = \frac{N}{2} m_K^2 - \frac{\log 2}{2} M(K)^2 + \sum_{a, b \leq K} \mu(a)\mu(b) e\left(\frac{N}{ab}\right). \quad (\text{C2})$$

The second term has a minus sign and is retained throughout.

The comparison notation used later is distinct: $F(x, y)$ is an actual Möbius sum over integers with no prime factor at most y ; $\mathfrak{F}(x, y)$ is a separately defined continuous comparison. Their difference produces $\mathcal{Z}_N$. None of G_K , F , $\mathfrak{F}$, and $\mathcal{Z}_N$ denotes the historical energy I_M .

53. Reciprocal-preserving compression with both centers exact

Let $\eta = N^{-1/4}$, and choose integer nodes

$$t_0 = 1, \quad t_{j+1} = \min\{K, t_j + \max(1, \lfloor \eta t_j \rfloor)\}, \quad t_J = K. \quad (\text{C3})$$

For an integer a strictly between successive nodes $t < r$, put

$$\vartheta_a = \frac{1/t - 1/a}{1/t - 1/r}.$$

Assign $(1 - \vartheta_a)\mu(a)$ to t , and $\vartheta_a\mu(a)$ to r . At a node assign its coefficient to that node once. Let β_j be the resulting signed masses. These assignments give, exactly,

$$\sum_j \beta_j = M(K), \quad \sum_j \frac{\beta_j}{t_j} = m_K, \quad \sum_j \frac{|\beta_j|}{t_j} \leq \sum_{a \leq K} \frac{|\mu(a)|}{a} \leq 1 + \log K. \quad (\text{C4})$$

The masses need not have absolute value at most one. Define

$$\widehat{Q}_K(N) = \sum_{i,j} \beta_i \beta_j \Theta\left(\frac{N}{t_i t_j}\right).$$

Then

$$|Q_K(N) - \widehat{Q}_K(N)| \leq \eta^2 N (1 + \log K)^2 = \sqrt{N} (1 + \log K)^2. \quad (\text{C5})$$

In particular,

$$\widehat{Q}_K(N) = \frac{N}{2} m_K^2 - \frac{\log 2}{2} M(K)^2 + \sum_{i,j} \beta_i \beta_j e\left(\frac{N}{t_i t_j}\right). \quad (\text{C6})$$

No centering approximation occurs.

Proof of the error bound. Fix $b \leq K$, allowing b to be a node, and put $c = N/b$. In logarithmic coordinates,

$$k(x) = \Theta(c e^{-x}) = \sum_{m \text{ odd}} (\log(c/m) - x)_+.$$

Its second derivative as a measure is a sum of unit atoms at the hinge locations $x = \log(c/m)$. On a node cell $[t, r]$, write $g = r - t$, $d = \log(r/t)$, and $v = \log(a/t)$. A cell of width one has no interior integer to move. On every other cell, $g \leq \eta t$.

First use the log-linear weight v/d . A single hinge in the cell contributes at most $d/4$ to the interpolation error at any integer. Its total contribution over the cell is at most

$$gd/4 \leq g^2/(4t) \leq \eta^2 t/4 \leq \eta^2 c/(4m).$$

Each hinge belongs to one cell, with boundary hinges giving no error. The relevant odd indices lie between c/K and c . Since $c/K \geq 1$, their reciprocal sum is at most $1 + \log K$. Thus the sum of log-linear errors is at most $\eta^2 c(1 + \log K)/4$.

Next replace the log-linear weight by the reciprocal weight. Concavity of $1 - e^{-v}$ and its second derivative bound give

$$0 \leq \vartheta_a - v/d \leq \frac{d^2}{8(1 - e^{-d})}.$$

The endpoint kernel difference is at most $c(1 - e^{-d})/t$, because the number of odd integers up to a positive real z is at most z when it is nonzero. Consequently this change costs at most

$$\frac{cd^2}{8t} \leq \frac{(1 + \eta)\eta^2 c}{8a}.$$

Using $\eta \leq 1$, the complete one-coordinate error, summed over the original coefficients, is at most

$$\frac{\eta^2 N}{2b} (1 + \log K).$$

Telescope the two-coordinate replacement. In the first coordinate sum with the original weights $|\mu(b)|/b$; in the second use the compressed weights $|\beta_j|/t_j$. Formula (C4) bounds both by $1 + \log K$. This proves (C5), including every cross-term.

There are $O(\eta^{-1}(1 + \log K))$ nodes. Below $2/\eta$ there are at most $2/\eta$ integers. Above it, the node increment is at least $\eta t/2$, so geometric growth gives the claimed count.

54. The compressed coordinates are historical energy means

Define

$$\Delta_j = \frac{1}{t_j} - \frac{1}{t_{j+1}}, \quad c_j = \frac{1}{\Delta_j} \int_{t_j}^{t_{j+1}} \frac{M(u)}{u^2} du \quad (0 \leq j < J). \quad (C7)$$

Integration by parts on each cell yields

$$\beta_0 = c_0, \quad \beta_j = c_j - c_{j-1} \quad (0 < j < J), \quad \beta_J = M(K) - c_{J-1}. \quad (C8)$$

For example the contribution of an interior coefficient $\mu(a)$ to c_j is $(1/a - 1/t_{j+1})/\Delta_j = 1 - \vartheta_a$. The differences in (C8) therefore reproduce exactly the assignment in section 53. The final coefficient explicitly retains the inherited terminal.

Partial summation and the weighted variance identity give

$$m_K = \frac{M(K)}{K} + \sum_j \Delta_j c_j, \quad \boxed{I_M(K) = \sum_j \Delta_j c_j^2 + V, \quad 0 \leq V \leq \frac{1}{4}.} \quad (C9)$$

Here

$$V = \sum_j \int_{t_j}^{t_{j+1}} \frac{(M(u) - c_j)^2}{u^2} du.$$

A width-one cell has zero variance because M is constant almost everywhere there. On a cell of width $g \geq 2$, the range of M is at most g . A variable supported in an interval of length g has variance at most $g^2/4$: expand the nonnegative product $(Y - \min Y)(\max Y - Y)$, then complete the square in the mean. Hence that cell contributes at most

$$\Delta_j g^2/4 = \frac{g^3}{4t_j t_{j+1}} \leq \eta^2 g/4.$$

Summing gives $V \leq \eta^2(K-1)/4 \leq 1/4$. Thus the omitted within-cell energy has a fixed allowance; it is not another unbounded remainder.

Combining (C1) and (C5) gives the complete bridge

$$\boxed{|U(N/2) - [2(\log 2)M(K) - \widehat{Q}_K(N)]| \leq 6H^3/L + \sqrt{N}(1 + \log K)^2.} \quad (C10)$$

The sufficient arithmetic statement remains $\widehat{Q}_K(N) \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$, with constants independent of N .

55. Exact output grouping and its cutoff hypotheses

This records the arithmetic grouping from the supplied continuation without extending it to arbitrary cutoffs. Assume $N \geq 16$, $K \geq 2$, and

$$2N/3 \leq K^2 \leq N.$$

Let

$$T(z) = \log_+ z - 2\log_+(z/2) + \log_+(z/4), \quad P(N) = \sum_{p \text{ modd prime}} T(N/p). \quad (C11)$$

Thus $T(z) = \log z$ for $1 < z \leq 2$, $T(z) = \log(4/z)$ for $2 \leq z < 4$, and it is zero otherwise. In particular $T \geq 0$.

Write $Q_K(N) = \sum_{n \leq N} c_K(n) \log(N/n)$. Ordinary Dirichlet convolution gives

$$c_K(n) = 2\mu(n)\mathbf{1}_{n \leq K} - 2\mathbf{1}_{2|n}\mu(n/2)\mathbf{1}_{n \leq 2K} - [\mu(n) - \mathbf{1}_{2|n}\mu(n/2)] + g_K(n), \quad g_K(n) = \sum_{\substack{ab=n \\ a, b > K}} \mu(a)\mu(b). \quad (\text{C12})$$

Indeed $\mu * \mathbf{1}_{\text{odd}} = \delta_1 - \delta_2$. Expanding the two truncated factors gives (C12). The tail has no further odd multiplier: $a, b > K$ and an odd multiplier at least three would force $n > 3K^2 > N$.

In the tail, $\mu(a)\mu(b) \neq 0$ requires prime exponents at most two. Suppose the odd part of n has exactly one prime p appearing to the first power, with a nontrivial squared odd factor. In a contributing factorization, squared factors belong to both factors. The ratio of the unequal factors is at least p , or at least $p/2$ when the exponent of two is one. Thus that ratio is at least $3/2$. But both factors exceeding K would imply $n > (3/2)K^2 \geq N$. The tail vanishes; all the other terms in (C12) vanish because of the squared odd factor. Higher exponents vanish automatically.

If the odd part is a square, the same argument excludes an odd exponent of two in a tail term (ratio two). Actual square outputs contribute

$$S_K(N) = \sum_{K < a \leq \sqrt{N}} \mu(a)^2 \log(N/a^2) \geq 0.$$

For odd-prime outputs, grouping $p, 2p, 4p$ gives the tent credit $P(N)$. The inherited prefix in (C12) gives $-2(\log 2)\pi_{\text{odd}}(K/2, K]$ for those outputs. Therefore

$$\boxed{Q_K(N) = P(N) + S_K(N) - 2(\log 2)\pi_{\text{odd}}(K/2, K] + R_{\geq 2}(N).} \quad (\text{C13})$$

Here $R_{\geq 2}$ is defined to be the complete contribution of the other outputs whose odd part has at least two first-power primes. Their inherited terms remain included. This is an exact grouping, not positivity of $R_{\geq 2}$.

56. A proved positive allowance at each fixed roughness cutoff

Let $P^-(1) = \infty$ and

$$F(x, y) = \sum_{P^-(n) > y} \mu(n)T(x/n), \quad B_\theta(N) = -F(N, N^\theta), \quad 0 < \theta < 1. \quad (\text{C14})$$

The summation includes the unit $n = 1$. For every **fixed** θ ,

$$\boxed{B_\theta(N) = [\rho(1/\theta - 1) + o_\theta(1)] \frac{N}{4 \log N} > 0 \quad (N \text{ sufficiently large}),} \quad (\text{C15})$$

where

$$\rho(v) = 1 \quad (0 \leq v \leq 1), \quad v\rho'(v) = -\rho(v-1) \quad (v > 1). \quad (\text{C16})$$

All allowable numbers of distinct prime factors, with alternating signs, occur in (C15).

Here is the analytic derivation, including the cutoff qualification. Factoring each nonunit squarefree integer by its smallest prime gives the universally valid identity

$$\boxed{F(x, y) = T(x) - \sum_{p > y} F(x/p, p).} \quad (\text{C17})$$

An exact separated version is

$$F(x, y) = T(x) - \sum_{p > y} T(x/p) - \sum_{y < p \leq \sqrt{x}} \{F(x/p, p) - T(x/p)\}. \quad (\text{C18})$$

For $x \geq 16$, $2 \leq y < x/4$, this simplifies to

$$F(x, y) = -P(x) - \sum_{y < p \leq \sqrt{x}} F(x/p, p). \quad (C19)$$

The conditions in (C19) matter. For example one must not drop the unit term at a small child argument. Equation (C17) is the formula used in subsequent exact recursions.

The prime number theorem and partial summation imply

$$P(x) \sim \frac{x}{\log x} \int_{1/4}^1 T(1/t) dt = \frac{x}{4 \log x}. \quad (C20)$$

For completeness the integral equals $\int_{1/4}^{1/2} \log(4t) dt + \int_{1/2}^1 \log(1/t) dt = 1/4$.

An absolute estimate controls transition strips in the recursion. For fixed $V > 1$, uniformly in $y \geq x^{1/V}$,

$$\sum_{P^-(n) > y} |\mu(n)| T(x/n) \ll_V x / \log x \quad (x \text{ large}). \quad (C21)$$

A nonzero summand has $x/4 < n < x$ and at most $\lfloor V \rfloor$ prime factors. If there are j factors, choose the largest p , and write $n = dp$. Then $p \geq (x/4)^{1/j}$, while $p \leq x/d$; the elementary consequence $\pi(z) \ll z / \log z$ of PNT bounds the number of such choices by $O_j(x/(d \log x))$. The reciprocal sum for the remaining distinct prime factors is bounded by

$$\frac{1}{(j-1)!} \left(\sum_{y < p \leq x} \frac{1}{p} \right)^{j-1} \ll_V 1.$$

The last bound also follows by prime partial summation. Sum over the finitely many j .

Put $u = \log x / \log y$, and write the leading term as $-F(x, x^{1/u}) \sim f(u)x/(4 \log x)$. For $1 < u \leq 2$, only singleton primes contribute for large x , so $f(u) = 1$. Induct on bounded ranges of u . In (C19) the child ratio is $w = \log(x/p) / \log p \in [1, u-1]$. On $w \geq 1 + \gamma$ the inductive asymptotic is uniform on compact intervals. On the remaining transition strip (C21) and prime partial summation give $O_V(\gamma x / \log x)$. Applying prime partial summation on the main range, then taking $x \rightarrow \infty$ and $\gamma \downarrow 0$, gives

$$f(u) = 1 - \int_1^{u-1} \frac{f(w)}{w} dw. \quad (C22)$$

This is exactly the defining delay equation for $f(u) = \rho(u-1)$. The same induction gives the asserted compact uniformity away from $u = 1$. Finally

$$v\rho(v) = \int_{v-1}^v \rho(t) dt \quad (v \geq 1) \quad (C23)$$

follows by differentiation and its value at one. A first zero of ρ would contradict (C23), proving strict positivity. These steps establish (C15) for each fixed θ ; they do not assert an error estimate uniform as $\theta \rightarrow 0$.

For $\theta = 1/3$, no product of three permitted primes fits below N , and

$$B_{1/3}(N) = P(N) - \sum_{N^{1/3} < p < q} T(N/(pq)) \sim (1 - \log 2) \frac{N}{4 \log N}. \quad (C24)$$

The semiprime-to-prime ratio is $\int_{1/3}^{1/2} [\alpha(1-\alpha)]^{-1} d\alpha = \log 2 < 1$. Thus the prime allowance pays for that whole family and leaves a positive surplus.

57. The smaller-prime contribution is the same whole comparison

For $N \geq 4$ and $N^\theta \geq 2$, define

$$D_\theta(N) = - \sum_{2 < p \leq N^\theta} F(N/p, p).$$

The least-prime partition gives

$$D_\theta(N) - B_\theta(N) = F(N, 2).$$

The latter is exactly $U(N/2)$. To verify this directly, put $A_o(x) = \sum_{n \leq x, n \text{ modd}} \mu(n) \log(x/n)$. The even squarefree terms give $\mathcal{A}(x) = A_o(x) - A_o(x/2)$, and the tent gives

$$F(N, 2) = A_o(N) - 2A_o(N/2) + A_o(N/4) = \mathcal{A}(N) - \mathcal{A}(N/2). \quad (\text{C25})$$

Consequently

$$\boxed{Q_K(N) = 2(\log 2)M(K) + B_\theta(N) - D_\theta(N) + G_K(N).} \quad (\text{C26})$$

The lower bound for Q_K , or its compressed version with (C5), is therefore equivalent up to the paid errors to

$$D_\theta(N) \leq B_\theta(N) + C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{C27})$$

The positivity of B_θ proves a credit, not (C27). All smaller-prime branches must remain coupled to that credit. This is the exact unresolved comparison stated in the user's checkpoint.

58. A historical-coordinate identity that retains interactions

This auxiliary identity gives another way to inspect the compressed object; it is not an additional closing requirement. Set $c_J = M(K)$, so that (C8) expresses β as successive differences. Write $W_{ij} = \Theta(N/(t_i t_j))$. Summation by parts in both indices yields

$$\widehat{Q}_K = c^\top A c,$$

where, for $i, j < J$,

$$A_{ij} = W_{ij} - W_{i+1,j} - W_{i,j+1} + W_{i+1,j+1}, \quad A_{iJ} = W_{iJ} - W_{i+1,J}, \quad A_{JJ} = W_{JJ}.$$

All these entries are nonnegative: each log-coordinate hinge has a nonnegative mixed finite difference, and the endpoint differences are nonnegative. This is entrywise positivity, not positive semidefiniteness. The row sums are

$$r_i = \Theta(N/t_i) - \Theta(N/t_{i+1}) \quad (i < J), \quad r_J = \Theta(N/K).$$

Thus the exact polarization identity is

$$\boxed{\widehat{Q}_K = \sum_i r_i c_i^2 - \sum_{i < j} A_{ij} (c_i - c_j)^2.} \quad (\text{C28})$$

Both the positive term and the debit are determined by the same historical coordinates. Positivity of the coefficients alone cannot compare their sizes. Formula (C28) retains the full terminal and every off-diagonal interaction.

Background for the standard notation and PNT: NIST Digital Library of Mathematical Functions, section 27.2, <https://dlmf.nist.gov/27.2>. Background for the Dickman function: A. Hildebrand and G. Tenenbaum, Integers without large prime factors, Journal de Theorie des Nombres de Bordeaux 5 (1993), 411-484, section 2, https://jtnb.centre-mersenne.org/item/JTNB_1993__5_2_411_0.pdf. The interpolation, output, and comparison arguments above are supplied here explicitly; these background references are not claims of external validation of the new estimates.

59. The exact continuous comparison

Let ρ be the function defined by

$$\rho(v) = 1 \quad (0 \leq v \leq 1), \quad v\rho'(v) = -\rho(v-1) \quad (v > 1).$$

Define a DIFFERENT function, with a continuous prime-density measure,

$$\mathfrak{F}(x, y) = T(x) - \int_y^\infty T(x/r) \frac{\rho(\log r / \log y - 1)}{\log r} dr, \quad y \geq 2. \quad (\text{Z3})$$

The integral is finite, because $T(x/r) = 0$ when $r \geq x$. Formula (Z3) is a definition, not an approximation asserted for the actual primes.

This comparison satisfies the exact continuous recursion

$$\mathfrak{F}(x, y) = T(x) - \int_y^\infty \mathfrak{F}(x/t, t) \frac{dt}{\log t}. \quad (\text{Z4})$$

Here is a direct derivation. Iterating the integral operator gives ordered factors $y < t_1 < \dots < t_k$. Symmetry converts this to

$$\mathfrak{F}(x, y) = T(x) + \sum_{k \geq 1} \frac{(-1)^k}{k!} \int_{t_1, \dots, t_k > y} T\left(\frac{x}{t_1 \dots t_k}\right) \prod_{j=1}^k \frac{dt_j}{\log t_j}. \quad (\text{Z5})$$

At each fixed x, y , only finitely many k can contribute, since a nonzero term requires $t_1 \dots t_k < x$.

To evaluate the product density, write $t_j = y^{u_j}$, $u_j > 1$. Apart from the common factor $y^{u_1 + \dots + u_k}$, the measure is the product of du_j/u_j . Put

$$h(u) = \frac{\mathbf{1}_{\{u > 1\}}}{u}, \quad \sigma(u) = \sum_{k \geq 1} \frac{(-1)^k}{k!} h^{*k}(u).$$

The sum is locally finite. Multiplying a convolution by the sum of its arguments gives

$$u\sigma(u) = - \left[1 + \int_0^{u-1} \sigma(v) dv \right] \quad (u > 1),$$

while $\sigma = 0$ on $(0, 1)$. Therefore $1 + \int_0^u \sigma(v) dv = \rho(u)$, so $\sigma = \rho'$. Changing back from the product coordinate to r gives precisely (Z3). The ordered-factor construction proves (Z4).

The identity

$$v\rho(v) = \int_{v-1}^v \rho(w) dw \quad (v \geq 1) \quad (\text{Z6})$$

follows by differentiation and the value at $v = 1$. A first zero is impossible, so $\rho > 0$; its differential equation then gives monotonicity.

For each fixed $0 < \theta < 1$, partial rescaling of (Z3) gives

$$-\mathfrak{F}(N, N^\theta) \sim \rho(1/\theta - 1) \frac{N}{4 \log N}. \quad (\text{Z7})$$

Thus this comparison has exactly the leading coefficient from the earlier fixed- θ result. Extending the definition to $y = 2$ does not extend that asymptotic statement for the actual arithmetic function F ; their difference is retained explicitly next.

Bounded contribution at the full cutoff

For $N \geq 4$, $T(N) = 0$, hence

$$\boxed{\mathfrak{F}(N, 2) \leq 0.} \quad (\text{Z8})$$

It is also uniformly bounded. From (Z6) and monotonicity, $\rho(m) \leq \rho(m-1)/m$, so $\rho(v) \leq 1/\lfloor v \rfloor!$ for $v \geq 1$. For $N \geq 16$, set $v_N = \log N / \log 2 - 3$ and $m = \lfloor v_N \rfloor$. The tent restricts r to $(N/4, N)$, giving

$$\begin{aligned} 0 \leq -\mathfrak{F}(N, 2) &\leq (\log 2) \frac{N}{\log(N/4)} \rho(v_N) \\ &\leq \frac{N}{2m!} < 8 \frac{2^m}{m!} \leq 8e^2. \end{aligned}$$

For $4 \leq N < 16$, a direct integral majorant gives a bound of 16. In particular,

$$\boxed{0 \leq -\mathfrak{F}(N, 2) \leq 16e^2 \quad (N \geq 4).} \quad (\text{Z9})$$

This is an all-scale bound for the explicitly defined comparison function, not for U .

60. The complete prime-discrepancy residual

For a two-variable function H , write

$$(\mathsf{P}H)(x, y) = \sum_{p > y} H(x/p, p), \quad (\mathsf{C}H)(x, y) = \int_y^\infty H(x/t, t) \frac{dt}{\log t}.$$

Equations (C17) and (Z4) say $(I + \mathsf{P})F = T$ and $(I + \mathsf{C})\mathfrak{F} = T$. Subtracting gives

$$(I + \mathsf{P})(F - \mathfrak{F}) = -(\mathsf{P} - \mathsf{C})\mathfrak{F}. \quad (\text{Z10})$$

At each finite x, y , the inverse on the functions here is a finite alternating expansion: once the product of the successive factors exceeds x , the tent support makes the term zero. Thus no convergence assumption or infinite resolvent is being imposed.

Define the prime-counting discrepancy for $t \geq 2$ by

$$\Delta_\pi(t) = \pi(t) - 1 - \int_2^t \frac{dv}{\log v}.$$

Its Stieltjes measure on $(2, \infty)$ is the actual odd-prime counting measure minus $dt/\log t$. For $P^+(1) = 1$, put

$$\boxed{\mathcal{K}_N(t) = \sum_{\substack{d \leq N/t \text{ odd} \\ P^+(d) < t}} \mu(d) \mathfrak{F}\left(\frac{N}{dt}, t\right).} \quad (\text{Z11})$$

The strict inequality $P^+(d) < t$ matters at a prime atom: it includes only smaller actual primes in the inherited cofactor. The $d = 1$ term is included.

Then the complete error is

$$\boxed{\mathcal{Z}_N = \int_{(2, N]} \mathcal{K}_N(t) d\Delta_\pi(t), \quad F(N, 2) = \mathfrak{F}(N, 2) - \mathcal{Z}_N.} \quad (\text{Z12})$$

Proof. Let $R = (\mathsf{P} - \mathsf{C})\mathfrak{F}$. Expanding (Z10) gives

$$F - \mathfrak{F} = - \sum_{k \geq 0} (-\mathsf{P})^k R.$$

An increasing chain of actual primes has one squarefree product d and coefficient $\mu(d)$. In the R term, the next factor t exceeds every prime in d . Writing R as an integral against $d\Delta_\pi$ and interchanging the finite sum with that integral gives (Z11)-(Z12). Every inherited cofactor occurs once. The continuous factors remaining above t are already contained in $\mathfrak{F}$.

This is a hybrid ordering identity: the primes below the discrepancy insertion are actual, while the factors above it belong to the explicitly defined comparison. The sum accounts for every possible position of that insertion. It is not a claim that the actual primes are independently distributed.

Return to the original lower bound

Combining (C25) and (Z12),

$$\boxed{B_\theta(N) - D_\theta(N) = -\mathfrak{F}(N, 2) + \mathcal{Z}_N.} \quad (\text{Z13})$$

The entire θ -dependent partition has recombined. The prime contribution has been used once, and no subset has been separately discarded.

Since (Z9) is bounded independently of N , the original one-sided comparison is equivalent, up to constants, to

$$\boxed{\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon} \quad \text{for every } \varepsilon > 0.} \quad (\text{Z14})$$

For the raw quadratic form, the inherited identity now reads

$$Q_K(N) = 2(\log 2)M(K) - \mathfrak{F}(N, 2) + \mathcal{Z}_N + G_K(N). \quad (\text{Z15})$$

The already-established compression error transfers this exact expression to $\widehat{Q}_K$. The original centering is retained through this equality; it has not been independently estimated or assigned a favorable sign.

61. A fictional closing mechanism for the exact residual

Status of this section. The finite identities below are real identities. The growth law labeled F1 is an invented hypothesis for exploration, not an established property of the primes. The deduction from F1 is a rigorous conditional proof. The tests in sections 63-66 examine the proposed mechanism and state its limitations. F1 is a sufficient certificate, not a new necessary condition added to the original one-sided target.

Use the exact $\mathcal{K}_N$, Δ_π , and $\mathcal{Z}_N$ from section 60. For real $N \geq 4$, partition $(2, N]$ into the disjoint intervals

$$I_{N,j} = (\max\{2, N/2^{j+1}\}, N/2^j], \quad 0 \leq j < J_N, \quad J_N = \lceil \log_2(N/2) \rceil.$$

Define the complete signed contribution of each interval and its squared sum by

$$z_{N,j} = \int_{I_{N,j}} \mathcal{K}_N(t) d\Delta_\pi(t), \quad S_N = \sum_{j=0}^{J_N-1} z_{N,j}^2. \quad (\text{F0})$$

All inherited cofactors and continuous branches are combined inside $z_{N,j}$ before it is squared. End-point atoms belong to exactly one interval. In particular $\mathcal{Z}_N = \sum_j z_{N,j}$ exactly. This does not assert that contributions in different intervals are independent.

For an integer $r \geq 0$, put

$$E_r = \sup_{4 \leq N \leq 2^{r+2}} \frac{S_N}{N}.$$

This is a finite quantity for each r . On a bounded N -range there are finitely many arithmetic factors, the comparison integrals are bounded, and all kernels have bounded finite sums. The supremum includes real cutoffs and intermediate values, not only powers of two.

The fictional law is:

$$1 + E_{r+1} \leq \left(1 + \frac{A}{r+1}\right) (1 + E_r) \quad \text{for every } r \geq 0, \quad (\text{F1})$$

for one finite constant $A \geq 0$, independent of r . It says that, after dividing the squared contribution by the counting scale N , the additional fractional cost of another doubling is at most $A/(r+1)$. Raw energy need not decrease, primes need not be regularly spaced, and individual signed contributions may be negative. The decreasing quantity in this fictional rule is the permitted fractional increase of the normalized running budget.

Conditional theorem. If F1 holds for the actual coefficients in section 60, then

$$|\mathcal{Z}_N| \ll_A \sqrt{N} (1 + \log N)^{(A+1)/2}. \quad (\text{F2})$$

Consequently $\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$, and the selected analytic closing argument applies.

Proof. Multiplying F1 and using $\log(1+x) \leq x$ gives

$$1 + E_r \leq (1 + E_0) \prod_{k=1}^r (1 + A/k) \leq (1 + E_0) e^{A(1+\log r)} \ll_A (1+r)^A.$$

Choose r with $N \leq 2^{r+2}$ and $r \ll 1 + \log N$. Then $S_N \ll_A N(1 + \log N)^A$. Cauchy-Schwarz gives

$$\mathcal{Z}_N^2 = \left(\sum_j z_{N,j} \right)^2 \leq J_N S_N \ll_A N(1 + \log N)^{A+1}.$$

Taking square roots proves (F2). Every fixed power of $\log N$ is $O_\varepsilon(N^\varepsilon)$, which gives the required one-sided bound. No independence, positivity of each interval, or numerical extrapolation enters this deduction.

The sole invented arithmetic step in this proof is F1. The product calculation and Cauchy-Schwarz cannot prove F1. An argument that says “the continuous comparison decays, therefore F1 holds” would leave precisely that step unjustified.

62. What the fictional proof would complete

This section spells out the closing implications so that the arithmetic premise and the analytic conclusion have no missing link. It also applies if the original one-sided bound is proved directly, without F1.

Assume $\mathcal{Z}_N \geq -C_\varepsilon N^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Equations (Z9) and (Z12) give

$$U(N/2) = \mathfrak{F}(N, 2) - \mathcal{Z}_N \leq C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{F3})$$

This is the needed upper bound; a lower bound on U is not needed. If it is known initially only for integer N , it extends to real arguments. Indeed, almost everywhere,

$$U'(X) = \frac{M(2X) - M(X)}{X}, \quad |U'(X)| \leq 1 + 1/X \leq 2 \quad (X \geq 1).$$

Adjacent half-integer sample arguments therefore differ by at most a fixed constant.

Fix $q > 1/2$. Choose $\varepsilon < q - 1/2$, telescope the dyadic increments U , and absorb the bounded initial interval to obtain

$$\mathcal{A}(x) \leq C_q x^q \quad (x \geq 1). \quad (\text{F4})$$

Increase C_q if necessary so that $g_q(x) = C_q x^q - \mathcal{A}(x) \geq 0$. For $\Re s > \max(1, q)$, absolute convergence and two partial integrations give

$$\int_1^\infty \mathcal{A}(x) x^{-s-1} dx = \frac{1}{s^2 \zeta(s)}, \quad \int_1^\infty g_q(x) x^{-s-1} dx = \frac{C_q}{s - q} - \frac{1}{s^2 \zeta(s)}. \quad (\text{F5})$$

For clarity, the positivity principle needed here has an elementary analytic proof. A Mellin transform of a nonnegative locally integrable function with a finite real abscissa of convergence cannot be analytic at that abscissa. Otherwise take a real point just to its right inside an analytic disk. The Taylor derivatives are the moments of powers of $-\log x$. At a real point slightly to the left, their Taylor terms all have the same sign. Tonelli’s theorem identifies the convergent Taylor sum with the integral there, contradicting the definition of the abscissa. This is the usual Landau positivity argument.

Apply that principle to g_q , with $1/2 < q < 1$. The right side of (F5) is analytic in a neighborhood of every real $s > q$: $\zeta(s) > 0$ for real $s > 1$, and it is nonzero for $0 < s < 1$ because the positive alternating eta series equals $(1 - 2^{1-s})\zeta(s)$. At $s = 1$, $1/\zeta(s)$ is analytic with a zero. Therefore the abscissa of the nonnegative integral is at most q . Its integral is holomorphic on $\Re s > q$, and (F5) continues $1/\zeta(s)$ there holomorphically. A zero of ζ in that half-plane would produce a pole, a contradiction.

Letting $q \downarrow 1/2$ excludes zeros to the right of the critical line. The functional equation reflects non-trivial zeros under $s \mapsto 1 - s$, excluding those to the left as well. This is the conditional RH conclusion. The proof uses the original one-sided residual bound; it requires no separate historical-energy, Fourier-tail, or variational-regularity estimate afterward.

63. Analytical attack I: positivity fails, and the upper band is explicit

The fictional proof deliberately allows negative contributions. An unconditional claim $\mathcal{Z}_N \geq 0$ at every cutoff is already false.

At $N = 4$, the actual tent sum is $F(4, 2) = -\log(4/3)$. For $2 < r < 4$, the Dickman factor in (Z3) is one, so

$$\mathcal{Z}_4 = \log(4/3) - \int_2^4 \frac{\log(4/r)}{\log r} dr. \quad (\text{F6})$$

The integrand is $\log 4 / \log r - 1$, a convex function on $[2, 4]$. Jensen's inequality gives

$$\int_2^4 \frac{\log(4/r)}{\log r} dr \geq 2 \left(\frac{\log 4}{\log 3} - 1 \right) > \log(4/3),$$

the last inequality involving fixed elementary constants. Thus $\mathcal{Z}_4 < 0$ analytically. This refutes universal positivity; it does not contradict a negative allowance of square-root size.

There is also a useful exact simplification, valid for all $N \geq 4$:

$$\boxed{\mathcal{K}_N(t) = \log(N/t) \quad (N/2 < t \leq N).} \quad (\text{F7})$$

Only $d = 1$ can occur in that interval, and $N/t \leq 2 \leq t$, so the continuous child integral vanishes. Consequently

$$\boxed{z_{N,0} = \sum_{N/2 < p \leq N} \log(N/p) - \int_{N/2}^N \frac{\log(N/t)}{\log t} dt.} \quad (\text{F8})$$

The top interval already contains a definite smoothed prime-counting error. It has no remaining complicated cofactor to remove.

More generally, when $t > \sqrt{N}$, every allowed cofactor satisfies $d < t$ and every continuous child has argument below its cutoff. Hence

$$\boxed{\mathcal{K}_N(t) = F(N/t, 2) \quad (t > \sqrt{N}).} \quad (\text{F9})$$

This exposes an exact smaller-scale Möbius response inside the larger-prime discrepancy integral. It is a useful structural identity, not by itself an upper or lower bound.

64. Analytical attack II: how strong the invented law really is

F1 implies a square-root estimate, up to logarithms, for each individual $z_{N,j}$, because $|z_{N,j}| \leq \sqrt{S_N}$. In particular it would bound (F8) at that scale. The prime number theorem used for (C15) supplies only an asymptotic error $o(N/\log N)$ for this fixed smooth prime weight; that statement does not imply a square-root error. Thus F1 cannot be deduced merely by repeating the PNT partial summation already used for the positive allowance.

The square budget is stronger than the original objective in a second way. It bounds $|\mathcal{Z}_N|$, whereas the closing argument needs only its negative part. It also limits every interval before cancellations between intervals are used. For an abstract two-interval vector $(L, -L)$, the total is zero but its squared sum is $2L^2$. This is an algebraic illustration of the difference between the two criteria, not a proposed model of the actual primes.

Consequently a failure to prove F1 would not introduce a new gap in the established route. One can instead try to prove the original signed lower bound directly, allowing a large negative interval to be paid by an actual positive interval elsewhere. The current paper retains those cross-interval credits in $\mathcal{Z}_N$; none is thrown away in (Z12) or (Z15).

An absolute-value shortcut is also too costly with the elementary estimates available here. Since $0 \leq \rho \leq 1$,

$$|\mathfrak{F}(x, y)| \leq \log 2 + (\log 2)x / \log y.$$

Summing this bound over all allowed d , even after discarding zero coefficients, gives the crude estimate

$$|\mathcal{K}_N(t)| \ll \frac{N}{t}(1 + \log N), \quad 2 < t \leq N.$$

Integrating against the total variation of $d\Delta_\pi$, and using only the harmonic bound for the prime sum, yields

$$\int_{(2,N]} |\mathcal{K}_N(t)| d|\Delta_\pi|(t) \ll N(1 + \log N)^2. \quad (\text{F10})$$

This is an upper bound too large for the desired scale, not a proof that every possible absolute estimate must be large. It pinpoints what the straightforward triangle inequality fails to accomplish.

The remaining analytic safeguards are equally concrete. The fixed- θ coefficient in (C15) cannot be substituted at $\theta = \log 2 / \log N$ without a new uniform error estimate. The continuous comparison at $y = 2$ is defined exactly and bounded, but its arithmetic difference is still $\mathcal{Z}_N$. Finally the interval contributions share the same primes and inherited Möbius coefficients; calling them independent or orthogonal would be an extra unsupported assumption. The proof of F2 uses neither claim.

65. Numerical attacks on signs, normalization, and reconstruction

The following calculations are floating-point diagnostics. They check formulas, cutoff conventions, and plausible failure modes. They are not interval-certified estimates, a proof of F1, or evidence that a bound holds at untested scales. No fitted exponent is extrapolated.

The saved script constructs ρ by the method of steps for its differential equation and evaluates the continuous comparison by adaptive quadrature. It computes each dyadic interval contribution from its prime sum minus its continuous integral, retaining $P^+(d) < t$. A separate direct odd Möbius sum computes $F(N, 2)$. The check is the independent agreement of $\sum_j z_{N,j}$ with $\mathfrak{F}(N, 2) - F(N, 2)$.

N	$\mathcal{Z}_N$, numerical	S_N/N , numerical	Sum of negative interval parts
4	-0.377360	0.035600	0.377360
8	-0.198395	0.003114	0.198395
16	0.702913	0.098857	0.472937
32	1.207232	0.042517	0.247214
64	1.312628	0.030819	0.761474
128	1.486366	0.025758	0.673683
256	1.142616	0.031613	2.321994

The largest reconstruction discrepancy in this run was less than $2 \cdot 10^{-9}$. In particular there are negative interval contributions even when the total is positive. The sampled normalized square sum is not monotonically decreasing: it rises sharply from $N = 8$ to $N = 16$. Therefore the simplest “normalized energy always decreases” guess is contradicted by this diagnostic. F1 was formulated with a running supremum and an allowed fractional increase, not with monotone decay of S_N/N .

The table samples only seven endpoints. It does not evaluate the supremum over real N in E_r , and must not be described as a verification of F1.

A second saved script checks the compression and historical-coordinate formulas at $N = 256, 1024, 4096, 16384$:

N	K	Nodes	$ Q_K - \widehat{Q}_K $	Within-cell variance V
256	12	10	0.048493	0.004132
1024	24	17	0.151295	0.010078
4096	49	27	0.181035	0.013893
16384	103	43	0.123568	0.010196

The mass and historical-coordinate identities agreed to approximately 10^{-15} ; the largest raw reconstruction discrepancy was below $5 \cdot 10^{-11}$. The stated error allowances held at these samples. Their proof is the analytic argument in sections 53-54, not this computation.

The accompanying source package's diagnostics/ folder contains:

- residual_diagnostics.py and residual_diagnostics.json;
- compression_diagnostics.py and compression_diagnostics.json.

The scripts record the numerical method and preserve the individual interval contributions.

66. What survived the attacks, and the precise next arithmetic object

The fictional mechanism survives as a valid conditional implication. Its arithmetic law has not been derived. The attacks establish the following distinctions.

Proposed step	Finding
Every residual is nonnegative	False: section 63 proves $\mathcal{Z}_4 < 0$
The normalized square sum always decreases	The diagnostic has a large increase between 8 and 16
F1 implies the desired rate	Proved conditionally in section 61
The continuous comparison bounds the actual residual automatically	No: (Z12) keeps their difference explicitly
Fixed-cutoff PNT asymptotics imply F1	No such implication follows; even the top interval needs a stronger error estimate
The exact formula loses a terminal or prime atom	The finite expansion retains both; independent numerical checks agree
Failure of the square-budget route creates another mandatory gate	No: the original one-sided signed target remains sufficient

The strongest exact new simplification from this audit is (F9), together with its explicit top interval (F8). At a dyadic cutoff, the lower-bound comparison can now be inspected as

$$\mathcal{Z}_N = \underbrace{\int_{(N/2, N]} \log(N/t) d\Delta_\pi(t)}_{\text{explicit top contribution}} + \underbrace{\int_{(2, N/2]} \mathcal{K}_N(t) d\Delta_\pi(t)}_{\text{all inherited remaining contributions}} . \quad (\text{F11})$$

This equality is valid at every real $N \geq 4$, not only dyadic cutoffs. No estimate for one term has been subtracted from another budget. The desired lower bound concerns their combined signed value. A proof can use compensation between them, instead of proving the stronger individual estimates that F1 would entail.

The unresolved real statement is still exactly

$$\boxed{\int_{(2, N]} \mathcal{K}_N(t) d\Delta_\pi(t) \geq -C_\varepsilon N^{1/2+\varepsilon} \text{ for every } \varepsilon > 0.} \quad (\text{F12})$$

Here all coefficients, cutoffs, and continuous comparison terms are defined in sections 52 and 59-60. The fictional growth law explains one sufficient shape of a proof. It has not supplied the arithmetic justification of that law.

67. Documentation map and status ledger

This edition records every displayed link in the supplied continuation and the subsequent exact residual. The former disconnected-workspace statements are historical notices, now superseded for this material.

Supplied or subsequent result	Location in this edition	Status
Complete strip removal and inherited terminal	(C1), section 52; original proof in section 42	Proved identity and allowance
Reciprocal-preserving masses and both centers	(C3)-(C6), section 53	Proved
Compression error with all cross-terms	(C5), section 53	Proved
Historical means, terminal, and $V \leq 1/4$	(C7)-(C10), section 54	Proved
Squarefull cancellation, square credit, prime credit	(C12)-(C13), section 55	Proved under displayed cutoff conditions
Positive fixed- θ allowance and $1 - \log 2$ case	(C15), (C24), section 56	Proved for fixed θ
Complete smaller-prime comparison	(C25)-(C27), section 57	Exact; its growth estimate remains open here
Historical-coordinate polarization	(C28), section 58	Exact; no PSD assertion
Continuous comparison and full-cutoff bound	(Z3)-(Z9), section 59	Proved for the comparison function
Complete actual-prime residual	(Z10)-(Z15), section 60	Exact
Fictional shrinking fractional budget	(F1), section 61	Invented hypothesis
Deduction from that budget to closure	(F2)-(F5), sections 61-62	Proved conditional implication
Negative residual and explicit larger-prime response	(F6)-(F9), section 63	Proved
Finite computational checks	Section 65 and saved scripts/results	Diagnostics only
Required all-scale signed lower bound	(F12), section 66	Not proved by this continuation

The original sections and their historical indexing remain available. The source package preserves the previous edition, both previously saved residual notes, the new continuation files, numerical diagnostics, the build scripts, and a file manifest with checksums. The lifetime companion is separate from the 12-page summary, the running proof manuscript, and the larger research master; this update does not silently replace those other documents.

68. Joint transfer: regularity, traces, and the common boundary

This continuation performs the joint integration by parts proposed after section 67. It uses the actual residual (Z12). The fictional growth law F1 is not assumed. The result is an exact finite expansion, an unconditional allowance for all sufficiently small discrepancy insertions, and explicit positive prime-square contributions. The complete remaining signed lower bound is kept as the same arithmetic target.

To keep the dictionary unambiguous, write

$$\nu(t) = \pi(t) - 1, \quad \ell(t) = \int_2^t \frac{dv}{\log v}, \quad D(t) = \nu(t) - \ell(t) = \Delta_\pi(t) \quad (t \geq 2). \quad (\text{J1})$$

Thus ν counts odd primes and $D(2) = 0$. The symbol $P(N)$ used for the earlier weighted prime allowance is unchanged; it is not used for this counting function.

Define the logarithmic derivative of the tent almost everywhere by

$$h(x) = xT'(x) = \mathbf{1}_{1 < x < 2} - \mathbf{1}_{2 < x < 4}.$$

Values at its three endpoints can be assigned arbitrarily in ordinary integrals. Let

$$\mathfrak{H}(x, y) = x \partial_x \mathfrak{F}(x, y) = h(x) - \int_y^\infty h(x/v) \frac{\rho(\log v / \log y - 1)}{\log v} dv. \quad (\text{J2})$$

For each integer $r \geq 0$, define a left-continuous kernel and its response:

$$\begin{aligned} \mathcal{K}_{N,r}^-(t) &= \sum_{\substack{d \text{ modd} \\ P^+(d) < t}} \mu(d) \mathfrak{F}\left(\frac{N}{dt^{r+1}}, t\right), \\ \mathcal{H}_{N,r}(t) &= \sum_{\substack{d \text{ modd} \\ P^+(d) < t}} \mu(d) \mathfrak{H}\left(\frac{N}{dt^{r+1}}, t\right). \end{aligned} \quad (\text{J3})$$

Every sum is finite: its summands vanish when $d \geq N/t^{r+1}$. The unit cofactor $d = 1$ is included. Let $\mathcal{K}_{N,r}^+$ use $P^+(d) \leq t$ instead; it is the right-continuous version. In particular $\mathcal{K}_{N,0}^- = \mathcal{K}_N$, and the prime atoms in the original residual use the left version.

These kernels are of bounded variation on each bounded interval in t . Here are the relevant regularity facts. The tent is Lipschitz in $\log x$. Its integrals in the finite ordered-factor formula (Z5) are locally absolutely continuous in their parameters. Differentiation in x under those finite integrals gives (J2), and differentiating the lower endpoint in (Z4) gives

$$\partial_y \mathfrak{F}(x, y) = \frac{\mathfrak{F}(x/y, y)}{\log y}. \quad (\text{J4})$$

On a bounded N, t -range only finitely many squarefree cofactors occur. A summand entering or leaving at $N/(dt^{r+1}) = 1$ has value zero. The tent corners cause derivative discontinuities, not jumps of the kernel. The only kernel jumps are therefore the changes in the finite cofactor set at primes. These facts justify the Stieltjes product rules used below without a further regularity premise.

At $R = \sqrt{N}$, both kernel traces agree and

$$\mathcal{K}_{N,0}^-(R) = \mathcal{K}_{N,0}^+(R) = F(R, 2). \quad (\text{J5})$$

Indeed $\mathfrak{F}(R/d, R) = T(R/d)$. Any cofactor excluded by strictness at $d = R$ has zero tent weight. For $t > R$, the already established identity is $\mathcal{K}_{N,0}^-(t) = F(N/t, 2)$, with

$$\frac{d}{dt} F(N/t, 2) = -\frac{M(N/t) - M(N/(2t))}{t} \quad \text{almost everywhere.} \quad (\text{J6})$$

Here $M(x) = 0$ for $0 < x < 1$.

69. The boundary cancels and every prime jump descends exactly

Factor a cofactor newly admitted at an odd prime p as $d = pa$, with $P^+(a) < p$. The nonzero coefficients are squarefree, so $\mu(pa) = -\mu(a)$. This gives

$$\boxed{\mathcal{K}_{N,r}^+(p) - \mathcal{K}_{N,r}^-(p) = -\mathcal{K}_{N,r+1}^-(p).} \quad (\text{J7})$$

For $r = 0$, the right side is $-\mathcal{K}_{N/p}^-(p)$. The jump vanishes when $p^{r+2} \geq N$. Thus the first jumps stop at the square-root boundary, the next family stops at the cube-root boundary, and subsequent families stop at the corresponding prime-power boundary.

Between primes, (J2)-(J4) give

$$\frac{d}{dt}\mathcal{K}_{N,r}(t) = -\frac{r+1}{t}\mathcal{H}_{N,r}(t) + \frac{\mathcal{K}_{N,r+1}^-(t)}{\log t}.$$

Combining this with the jumps proves the exact signed-measure identity

$$\boxed{d\mathcal{K}_{N,r}^+(t) = -(r+1)\mathcal{H}_{N,r}(t)\frac{dt}{t} - \mathcal{K}_{N,r+1}^-(t)dD(t).} \quad (\text{J8})$$

All terms on the right use the same actual prime-discrepancy measure.

For right-continuous bounded-variation functions f, g , the product rule in the convention needed here is

$$d(fg) = f_- dg + g df, \quad (\text{J9})$$

where g is evaluated after a jump in the second integral. At a jump this identity is simply $f_- \Delta g + g \Delta f = fg - f_- g_-$. This convention accounts for the interaction at a shared prime atom exactly once.

Apply (J9) below and above $R = \sqrt{N}$. The low piece is

$$\int_{(2,R]} \mathcal{K}_{N,0}^- dD = F(R, 2)D(R) - \int_{(2,R]} D d\mathcal{K}_{N,0}^+.$$

The high piece is

$$\int_{(R,N]} \mathcal{K}_{N,0}^- dD = -F(R, 2)D(R) + \int_R^N D(t) \frac{M(N/t) - M(N/(2t))}{t} dt.$$

The displayed boundary terms cancel, including when R itself is prime, because (J7) gives a zero jump there. Using (J8), the resulting whole identity is

$$\boxed{\mathcal{Z}_N = \int_2^N D(t)\mathcal{H}_{N,0}(t)\frac{dt}{t} + \int_{(2,N]} D(t)\mathcal{K}_{N,1}^-(t)dD(t).} \quad (\text{J10})$$

At a prime in the last integral, $D(t)$ is the post-jump value and the kernel is the pre-jump value. The inherited correction is not omitted. It is the term that will generate the squared discrepancy together with the actual prime count.

70. An exact finite expansion retaining coincident prime interactions

The expansion can be based at any Y with $2 \leq Y < N$. Define

$$\nu_Y(t) = \pi(t) - \pi(Y), \quad \ell_Y(t) = \int_Y^t \frac{dv}{\log v}, \quad D_Y(t) = D(t) - D(Y) = \nu_Y(t) - \ell_Y(t).$$

Introduce coefficients by the convergent generating function near $z = 0$:

$$\boxed{\sum_{k \geq 0} \mathcal{B}_{k,Y}(t) z^k = e^{-\ell_Y(t)z} (1-z)^{-\nu_Y(t)}} \quad (\text{J11})$$

These coefficients are distinct from the earlier fixed- θ allowance B_θ . They are explicit polynomials in the actual count and its smooth centering. The first terms are

$$\begin{aligned} \mathcal{B}_{0,Y} &= 1, & \mathcal{B}_{1,Y} &= D_Y, \\ \mathcal{B}_{2,Y} &= \frac{1}{2}(D_Y^2 + \nu_Y), \\ \mathcal{B}_{3,Y} &= \frac{1}{6}(D_Y^3 + 3\nu_Y D_Y + 2\nu_Y^2). \end{aligned} \quad (\text{J12})$$

They can also be calculated by

$$k\mathcal{B}_{k,Y} = D_Y \mathcal{B}_{k-1,Y} + \nu_Y \sum_{j=0}^{k-2} \mathcal{B}_{j,Y},$$

with an empty sum for $k = 1$.

Exact differential identity. With post-jump values on the right,

$$\boxed{d\mathcal{B}_{k+1,Y}(t) = \mathcal{B}_{k,Y}(t) dD(t)}. \quad (\text{J13})$$

Between primes the generating function differentiates to $-z/\log t$ times itself. At a prime it is multiplied by $(1-z)^{-1}$, so the jump of its $(k+1)$ -st coefficient equals the post-jump k -th coefficient. This proves both parts of (J13). In particular replacing $\mathcal{B}_{2,Y}$ by $D_Y^2/2$ would lose the $\nu_Y/2$ contribution from coincident jumps.

For $q_Y = \lceil \log N / \log Y \rceil$, define

$$J_r(Y) = \int_{(Y,N]} \mathcal{B}_{r,Y}(t) \mathcal{K}_{N,r}^-(t) dD(t).$$

Using (J9), (J13), and then (J8), we obtain

$$J_r(Y) = (r+1) \int_Y^N \mathcal{B}_{r+1,Y}(t) \mathcal{H}_{N,r}(t) \frac{dt}{t} + J_{r+1}(Y). \quad (\text{J14})$$

There is no omitted boundary term: $\mathcal{B}_{r+1,Y}(Y) = 0$, while $\mathcal{K}_{N,r}^+(N) = 0$. When $r = q_Y - 1$, $N/t^{r+1} < 1$ on $(Y, N]$, so the kernel and $J_r(Y)$ vanish. Iteration is finite, yielding

$$\boxed{\int_{(Y,N]} \mathcal{K}_N(t) dD(t) = \int_Y^N \sum_{k=1}^{q_Y-1} k\mathcal{B}_{k,Y}(t) \mathcal{H}_{N,k-1}(t) \frac{dt}{t}}. \quad (\text{J15})$$

Every inherited prime-power interaction appears once. The terms remain in one signed sum and share the measure dt/t . This is a finite identity for each N, Y , without a limiting exchange or a growth assumption.

71. An unconditional allowance for the complete small-insertion range

Let

$$\mathfrak{e}_{N,Y} = \int_{(2,Y]} \mathcal{K}_N(t) dD(t), \quad 2 \leq Y \leq N.$$

This interval is indexed by the location of the discrepancy insertion. All actual cofactors below that insertion and all continuous factors above it remain present. It is a different partition from the integer-output family $C_y(N)$ in the preserved small-prime-output note.

Uniform finite bound. For every such N, Y ,

$$|\mathfrak{e}_{N,Y}| \leq [\nu(Y) + \ell(Y)] 2^{\nu(Y)} (\log 2 + Y^2 e^Y). \quad (\text{J16})$$

In particular, for sufficiently large N , set

$$Y_N^{(0)} = \frac{1}{2} \log N.$$

Then

$$\mathfrak{e}_{N,Y_N^{(0)}} = O_\varepsilon(N^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (\text{J17})$$

Equivalently its absolute value is at most $N^{1/2+o(1)}$. Smaller N can use a cutoff of two, with finitely many initial values absorbed in the constants. A sharper cutoff is obtained below.

Proof. The factorial estimate for ρ from section 59 gives, for $y \geq 2$,

$$\begin{aligned} \int_y^\infty \frac{\rho(\log v / \log y - 1)}{\log v} dv &\leq \sum_{m \geq 0} \frac{y^{m+2}}{m! \log y} \\ &= \frac{y^2 e^y}{\log y}. \end{aligned}$$

Here the m -th interval is $[y^{m+1}, y^{m+2}]$, and $\rho \leq 1/m!$ there. Since $|T| \leq \log 2$,

$$|\mathfrak{F}(x, y)| \leq (\log 2) \left(1 + \frac{y^2 e^y}{\log y} \right) \quad \text{uniformly for every } x > 0. \quad (\text{J18})$$

There are exactly $2^{\nu(t-)}$ odd squarefree products of the primes below t , before imposing any size cutoff. Every nonzero coefficient in $\mathcal{K}_N(t)$ belongs to this finite family. Therefore, for $2 < t \leq Y$,

$$|\mathcal{K}_N(t)| \leq 2^{\nu(Y)} (\log 2 + Y^2 e^Y).$$

The total variation of dD on $(2, Y]$ is $\nu(Y) + \ell(Y)$: its positive part consists of the prime atoms and its negative part is the continuous density. This proves (J16).

Only the elementary upper estimate $\pi(Y) \ll Y/\log Y$ is needed for its rate. One proof uses the product of the primes in $(n, 2n]$, which divides $\binom{2n}{n} < 4^n$. Thus their count is at most $n \log 4 / \log n$. Split $(\sqrt{Y}, Y]$ into dyadic intervals and bound primes below $\sqrt{Y}$ trivially; this gives the stated upper estimate. Splitting the integral for $\ell(Y)$ at $\sqrt{Y}$ similarly gives $\ell(Y) \ll Y/\log Y$. The logarithm of the right side of (J16) is consequently

$$Y + O\left(\frac{Y}{\log Y} + \log Y\right).$$

Substituting $Y = \frac{1}{2} \log N$ proves (J17). This estimate holds for the entire insertion interval, including arbitrarily large inherited products; it is not a termwise truncation of those products.

Sharper complete allowance. The factorial bound can be improved by retaining the exact Laplace moment of ρ . Put

$$L_\rho(a) = \int_0^\infty \rho(u) e^{au} du, \quad J(a) = \int_0^a \frac{e^v - 1}{v} dv \quad (a \geq 0).$$

The singularity of the integrand at zero is removable. The factorial estimate just used proves convergence of all these moments and justifies differentiation and integration by parts. Extend $\rho(u) = 0$ for $u < 0$. The delay equation gives $(u\rho(u))' = \rho(u) - \rho(u-1)$. Consequently

$$aL'_\rho(a) = (e^a - 1)L_\rho(a), \quad L_\rho(a) = L_\rho(0)e^{J(a)} \leq e e^{J(a)}.$$

The last inequality follows from $L_\rho(0) \leq \sum_{m \geq 0} 1/m! = e$. Change variables $v = y^{u+1}$ in the comparison integral to obtain the stronger uniform bound

$$\begin{aligned} \int_y^\infty \frac{\rho(\log v / \log y - 1)}{\log v} dv &= y \int_0^\infty \frac{\rho(u)y^u}{u+1} du \\ &\leq yL_\rho(\log y). \end{aligned}$$

Thus another explicit, all- N bound for the same complete insertion interval is

$$|\epsilon_{N,Y}| \leq [\nu(Y) + \ell(Y)] 2^{\nu(Y)} (\log 2) [1 + eY e^{J(\log Y)}]. \quad (\text{J18a})$$

Elementary l'Hopital comparison gives $J(\log Y) \sim Y / \log Y$. The prime number theorem, already used in section 56, gives $\nu(Y) \sim Y / \log Y$. Taking logarithms in (J18a) therefore shows

$$\log(\text{right side of (J18a)}) \leq (1 + \log 2 + o(1)) \frac{Y}{\log Y}.$$

For the remainder of this continuation choose, for large N ,

$$Y_N = \frac{1}{4} \log N \log \log N.$$

Then

$$|\epsilon_{N,Y_N}| \leq N^{(1+\log 2)/4+o(1)} = O(\sqrt{N}). \quad (\text{J18b})$$

The exponent $(1+\log 2)/4$ is strictly less than $1/2$, so the last big-O assertion follows for all sufficiently large N . Set $Y_N = 2$ on the finitely many smaller values if needed. This sharper allowance is unconditional and uses no hypothesis on $D(t)$.

Combine (J15) with this paid interval, using the coefficients based at the same Y_N :

$$\mathcal{Z}_N = \epsilon_{N,Y_N} + \int_{Y_N}^N \sum_{k=1}^{q_{Y_N}-1} k \mathcal{B}_{k,Y_N}(t) \mathcal{H}_{N,k-1}(t) \frac{dt}{t}. \quad (\text{J19})$$

Here $q_{Y_N} = O(\log N / \log \log N)$. Starting the coefficient recursion at Y_N is essential: all $\mathcal{B}_{k,Y_N}(Y_N)$, $k \geq 1$, vanish. One must not simply delete the lower interval from the expansion based at two, because that would leave its nonzero boundary terms unpaid.

72. Positive even coefficients and complete prime-square credits

The coefficients in (J11) contain exact positivity. For every $j \geq 0$,

$$\mathcal{B}_{2j,Y}(t) \geq 0 \quad (t \geq Y). \quad (\text{J20})$$

To prove this, put $m = \nu_Y(t)$, $a = \ell_Y(t)$. If $m = 0$, the coefficient is $(-a)^k/k!$. If $m \geq 1$, the elementary integral for $(1-z)^{-m}$ gives

$$\mathcal{B}_{k,Y}(t) = \frac{1}{k!(m-1)!} \int_0^\infty (u-a)^k u^{m-1} e^{-u} du. \quad (\text{J21})$$

Expansion of $e^{z(u-a)}$, or differentiation at zero, proves the identity. For even k the integrand is nonnegative. This is a deterministic representation of coefficients of the exact counting function; it assumes no randomness or independence of primes.

For the second level, the complete coefficient is

$$2\mathcal{B}_{2,Y} = D_Y^2 + \nu_Y. \quad (\text{J22})$$

On the terminal interval $\sqrt{N/2} < t < \sqrt{N}$, with $t \geq 2$, only the unit cofactor contributes to $\mathcal{H}_{N,1}$, its continuous child integral vanishes, and its tent response is one. Thus $\mathcal{H}_{N,1}(t) = 1$ there. If $Y \leq \sqrt{N/2}$, its exact nonnegative contribution is

$$\mathcal{C}_{2,Y}(N) = \int_{\sqrt{N/2}}^{\sqrt{N}} ([D(t) - D(Y)]^2 + \pi(t) - \pi(Y)) \frac{dt}{t} \geq 0. \quad (\text{J23})$$

It contains the actual prime-counting discrepancy inside the same square. For the cutoff Y_N , prime partial summation gives the explicit lower allowance

$$\mathcal{C}_{2,Y_N}(N) \geq [2(1 - 1/\sqrt{2}) + o(1)] \frac{\sqrt{N}}{\log N}. \quad (\text{J24})$$

Indeed one can omit the nonnegative square and apply the prime number theorem to the remaining count integral; $\pi(Y_N)$ gives a negligible subtraction.

There is a second useful grouping that pays both sides of the unit-cofactor tent. Decompose $\mathcal{H}_{N,1}$ into its unit term $h(N/t^2)$ and all remaining terms from (J2)-(J3). Isolate only the $\nu_Y(t)h(N/t^2)$ part of (J22). If $Y \leq \sqrt{N/2}$, integration by parts gives

$$\begin{aligned} \int_Y^N \nu_Y(t) h(N/t^2) \frac{dt}{t} &= \frac{1}{2} \int_{(Y,N]} T(N/t^2) d\pi(t) \\ &= \left[\frac{1}{2} \sum_{p>Y} T(N/p^2) \geq 0. \right] \end{aligned} \quad (\text{J25})$$

There are no end terms because the tent vanishes at both ends of its support. This combines its positive and negative derivative portions before assigning the credit. The sum already has the allowed size:

$$0 \leq \frac{1}{2} \sum_{p>Y_N} T(N/p^2) \ll \frac{\sqrt{N}}{\log N},$$

and more precisely

$$\frac{1}{2} \sum_{p>Y_N} T(N/p^2) \sim (3 - 2\sqrt{2}) \frac{\sqrt{N}}{\log N}. \quad (\text{J26})$$

For the constant, scale $p = \sqrt{N}u$ and calculate $\int_{1/2}^1 T(1/u^2) du = 3 - 2\sqrt{2}$. Prime partial summation then proves (J26). The upper estimate follows already from the elementary prime-counting bound used in section 71.

Equations (J23) and (J25) are two different groupings of overlapping parts of the second level. Their credits must not be added as if they came from disjoint terms. In (J25), the remaining part of the same level is exactly

$$D_Y^2 \mathcal{H}_{N,1} + \nu_Y [\mathcal{H}_{N,1} - h(N/t^2)],$$

integrated against dt/t . In (J23), the remaining part is the full second level outside the displayed terminal interval. Both descriptions preserve the entire signed expression.

73. Checks, the surviving signed comparison, and updated status

The derivative responses $\mathcal{H}_{N,r}$ retain signs. Even-coefficient positivity therefore does not make every level positive. This can be checked analytically: at $N = 64$, for

$$8/\sqrt{3} < t < 4\sqrt{2},$$

the argument $64/t^2$ lies between two and three, only $d = 1$ occurs, and its continuous child vanishes. Hence $\mathcal{H}_{64,1}(t) = -1$ on this whole interval. The matching positive coefficient must remain coupled to the other intervals and levels, exactly as in (J19).

The new diagnostic script evaluates the derivative integrals and the original odd Möbius tent sum separately. It includes square boundaries $N = 9, 25, 49, 81$, a cube boundary $N = 27$, and ordinary dyadic cutoffs. These computations check conventions and identities; the proofs above are analytical.

N	Direct residual	First level	Second level	All higher levels combined
8	-0.198395	-0.305781	0.107386	0.000000
16	0.702913	0.390018	0.320717	-0.007822
32	1.207232	1.062697	0.159632	-0.015096
64	1.312628	1.916836	-0.658812	0.054604
128	1.486366	2.336627	-0.549572	-0.300690

In this run the full unanchored reconstruction discrepancy was below $1.3 \cdot 10^{-8}$. The trace identity at $\sqrt{N}$ and prime-jump identity agreed to floating-point precision. Additional tests reset the baseline at $Y = 3$, $Y = 5$, $Y = \frac{1}{2} \log 128$, and the sharper moving cutoff at $N = 512$, including baselines that are themselves primes. The largest discrepancy for the full reset formula, with the small interval added back, was below $7 \cdot 10^{-8}$. These are non-certified floating-point diagnostics and imply no estimate at untested scales.

The source package's diagnostics/ folder contains:

- `joint_transfer_diagnostics.py`, the editable calculation;
- `joint_transfer_diagnostics.json`, the complete recorded results.

The direct attempt has established:

Result	Location	Status
Regularity and matching square-root traces	(J2)-(J6)	Proved
Prime jumps equal inherited smaller responses	(J7)-(J8)	Proved
Shared boundary cancellation with post-jump correction	(J9)-(J10)	Proved
Complete finite expansion at any baseline	(J11)-(J15)	Proved
Full small-insertion interval fits the permitted rate	(J16)-(J19)	Proved
All even coefficient polynomials are nonnegative	(J20)-(J22)	Proved
Terminal square credit and whole unit-tent prime-square credit	(J23)-(J26)	Proved, with distinct grouping rules
Original all-scale lower bound for the complete residual	(Z14), equivalently the signed integral below	Remains unproved here

After the paid interval, the exact sufficient comparison is

$$\int_{Y_N}^N \left[\sum_{k=1}^{q_{Y_N}-1} k \mathcal{B}_{k,Y_N}(t) \mathcal{H}_{N,k-1}(t) \right] \frac{dt}{t} \geq -C_\varepsilon N^{1/2+\varepsilon}. \quad (\text{J27})$$

This is the same original one-sided bound with an explicitly bounded part removed. No square-budget law, separate energy condition, or additional analytic closing premise has been introduced. Equation (J19) returns it directly to $\mathcal{Z}_N$, and section 62 supplies the conditional zero-exclusion argument.

The next signed comparison must retain the discrepancy powers, prime counts, and inherited derivative responses together. The formulas above identify actual positive credits and pay a complete family, but they do not establish that these credits dominate all negative contributions in (J27).